



AL RAJHI PAYMENT GATEWAY

MERCHANT INTEGRATION GUIDE - REST API'S

Version 1.24

This document explains the step-by-step procedures to integrate Al Rajhi payment gateway in merchants' application.

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Chapter 1 INTRODUCTION

This chapter provides the purpose of this document and its details.

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Purpose of the Document

The Payment Gateway follows industry standards and norms as prescribed by Master-Card and Visa International as well in conformity with Payment Card Industry – Data Security Standards commonly referred to as PCI – DSS.

In order that the AL Rajhi Bank merchants are integrated in a secure and mandated manner, this reference document is being shared. The expectation being that the merchant's system integrator or auditor can refer to a document while performing integration as well as post integration. It contains the technical integration details including message formats to be used in communicating to the Payment Gateway irrespective of the merchant platform being used. The document also shares the best practices and recommendations the merchant should follow during the integration with Payment Gateway.

Target Audience

This document is intended for Partner integration teams, project managers, developers, and testers.

Chapter 2 GETTING STARTED

This chapter contains detailed instructions of building merchant application with ARB Payment Gateway.

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Start Building with ARB Payment Gateway

Alrajhi Payment Gateway offers a secure, PCI-DSS compliant approach to save Debit card, Credit card, Net-Banking, UPI and Wallet payments from the merchants.

1. Once the Merchant on-boarded on ARB payment gateway, merchant would be provided with Payment gateway application URL and login credentials.

Notes: Generate your staging account credentials. These are required to explore ARB Payment Gateway integration solutions.

2. Merchant to logs in to ARB payment gateway merchant portal and download the following:
 - Tranportal ID & Tranportal password
 - **Tranportal ID** and **Tranportal password** are unique values provided by ARB Payment Gateway via registered mail as well.
 - The details are required for integration with ARB payment gateway.
 - Resource Key
 - **Resource key** is a unique secret key used for secure encryption and decryption of every request.
 - **This should not be shared with anyone.**

Basic understanding of ARB Payment Gateway

The ARB payment gateway's merchant application is a well secured interface that captures payment related data between the merchant and the customer:

Some of the features are :

- View payments received from your customers
- View transactions ARB PG makes into your account
- Initiate and track the refund
- Generate and download all kinds of transaction reports

Basics of collecting payments with ARB Payment Gateway

1. A user adds goods & services in cart and clicks the **Alrajhi Payment Gateway** button in your application (the merchant's application).
2. A checkout form is shown where the user fills in the payment details and authorizes the payment.

There are two ways of processing the payment with ARB payment gateway with API integration:

- Bank Hosted setup
- Merchant Hosted setup

3. After completion of a transaction, Alrajhi Payment Gateway posts the response (success or failed) on a Callback URL defined by you (the merchant).
4. Based on the response received, you display the order status to the customer.
5. See a real-time summary of payments received and other detail in the merchant portal URL:

<https://digitalpayments.alrajhibank.com.sa/mrchptl/merchant.htm>

Navigation to transaction detail report in merchant portal is as below :

[Merchant Portal](#) -> [Reports](#) -> [Transaction Reports](#) -> [Transaction Detail Report](#)

6. Receive payments collected from customers in your bank account on the next business day.

Integration Guidelines

Mandatory steps to be followed by Merchant Integration Team:

1. Verify the endpoints by sending request in postman application.
2. Frame the Json request of **Plain Trandata** with request parameters as specified in the document
3. Encrypt the plain json request with the **Resource Key** received in the mail.
4. Frame the encrypted request and send it to ARB Payment Gateway endpoint URL shared via mail.
5. After receiving the response from ARB Payment Gateway, decrypt the encrypted trandata using the resource key received in the mail.
6. Process the plain trandata - response message.

Test Instruments

ARB PG APIs are complete RESTFUL API's; Merchants can test the API's on sandbox setup before testing on Live environment.

Test Card details.

Card type	Card Number	Expiry Date	CVV
Credit	4012001037141112	12/2027	212
Credit	5297412201764352	01/2022	999

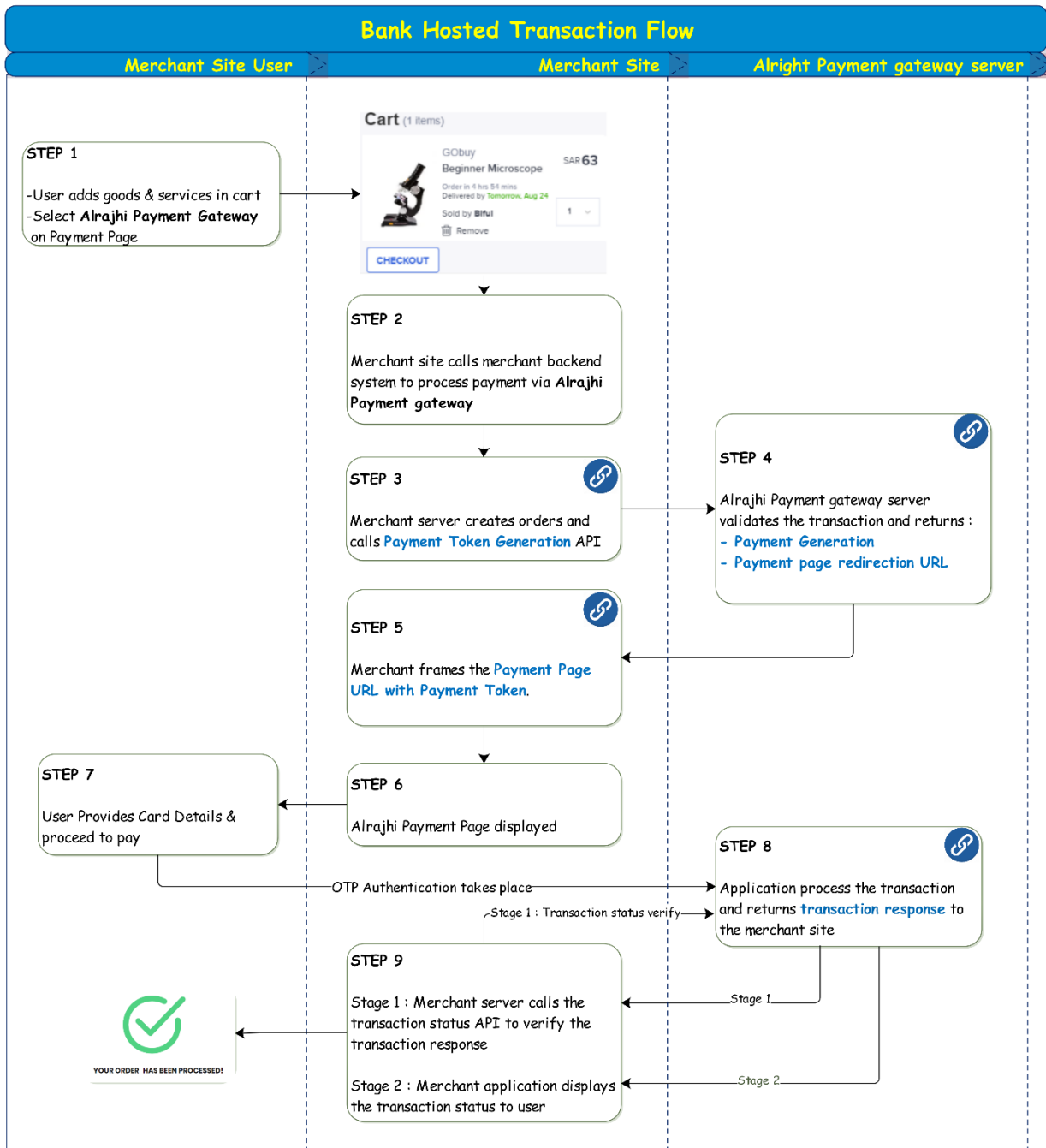
Chapter 3 TRANSACTION FLOW

This chapter illustrates various transaction flows of payment gateway integrate on merchants' application.

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Bank Hosted Integration

This section illustrates how you can integrate the bank hosted flow on your website application.



1. User visits the merchant application and creates order.
2. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
3. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID](#) and [Payment Page URL](#) in the response.
 - Merchant needs to [frame the payment page URL with Payment ID](#), Hence the ARB payment page is displayed.
 - In case of failure, ARB PG provides **Error Code** and **Description**.

Note: If merchant notification is disabled, then ARB Payment gateway will provide the [final response in URL redirection](#).

4. User enters the payment details and authorizes the transaction on the ARB PG bank's page.
5. The ARB PG application process the transaction and returns the transaction response to the merchant site.
6. Merchant server calls the transaction status API to verify the transaction response.
7. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization: 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	response URL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while processing the transaction.
14	langid	O	Alphabetic	Language ID . Based on language ID arabic language will be displayed on payment page. Value should be 'ar' or 'AR' for arabic language.

Sample JSON request - Request from Merchant to ARB PG

```
[{  
  //Mandatory Parameters  
  "id":"IPAYlCR6qZF7q6w",  
  "trandata":"<encrypted trandata> ",  
  "responseURL":"https://merchantpage/PaymentResult.jsp",  
  "errorURL":"https://merchantpage/PaymentResult.jsp"  
}]
```

Plain Trandata:

Trandata will contain below parameters encrypted with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value PGKEYENCDECIVSPC under Resource Key.

```
[{  
  //Mandatory Parameters  
  "amt":"12.00",  
  "action":"1",      // 1 - Purchase , 4 - Authorization  
  "password":"q@a680$27@JLkcK",  
  "id":"IPAYlCR6qZF7q6w",  
  "currencyCode":"682",  
  "trackId":"12345656789",  
  "responseURL":"https://merchantpage/PaymentResult.jsp",  
  "errorURL":"https://merchantpage/PaymentResult.jsp",  
  //Optional Parameters  
  "udf1":"udf1text",
```

```

"udf2":"udf2text",
"udf3":"udf3text",
"udf4":"udf4text",
"udf5":"udf5text",
"langid":"ar",
}]

```

MADA Mandatory Parameters

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and

				then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanumeric	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.

Sample JSON request - Request from Merchant to ARB PG

```
{
  //MADA Mandatory Parameters
  "amt": "12.00",
  "action": "1", // 1 - Purchase , 2 - Authorization
  "password": "q@a680$27@JLkcK",
  "id": "IPAY1CR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "12345656789",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp",
  //Optional Parameters
  "udf1": "udf1text",
  "udf2": "udf2text",
  "udf3": "udf3text",
  "udf4": "udf4text",
}
```

```
"udf5": "udf5text",
"langid": "ar",
}]
```

Split Payment or Payout.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanumeric	Bank Identification Code Min - 8 Max - 12
3	iBanNum	C	Alphanumeric	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//Conditional if Merchant Opted for Split Payment or Payout.

"accountDetails": [
  {
    "bankIdCode": "12345d6f",
    "iBanNum": "567896743281926354276254",
    "beneficiaryName": "AlRajhi Bank Services",
    "serviceAmount": "200.00",
    "valueDate": "20201231"
  },
  {
    "bankIdCode": "1234ret3",
    "iBanNum": "987656743281926354276254",
```

```
"beneficiaryName":"DIGITAL CO",
"serviceAmount":"300.00","valueDate":"20201231" }],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

"billingDetails" :

{"IDType":"01","IDNumber":"1072587916","billNumber":"00100100018","billType":"POSTPAID","billerID":"169","billAmount":5,"transactionType":"ADVANCE"}
,
```

Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg
1.1.1.1.7	destinationArrivalTime	C	Alphanumeric	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanumeric	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanumeric	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanumeric	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanumeric	
1.1.1.3	originCountry	C	Alphanumeric	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanumeric	The passenger first name
1.1.2.2	lastName	C	Alphanumeric	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanumeric	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanumeric	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanumeric	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanumeric	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare

1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - for airline merchant
```

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ { "carrierCode":
"MH", "departureAirport": "MNL", "departureDate": "2021-05-11", "departureTime":
"06:50:00Z", "destinationAirport": "KUL", "destinationArrivalDate": "2021-05-11",
"destinationArrivalTime": "10:35:00Z", "fareBasis": "BOWMPH6", "flightNumber":
"0805", "travelClass": "B" }, { "carrierCode": "UL", "departureAirport": "KUL",
"departureDate": "2021-05-11", "departureTime": "15:00:00Z", "destinationAirport":
"CMB", "destinationArrivalDate": "2021-05-11", "destinationArrivalTime":
"16:05:00Z", "fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName": "KAI
MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees": "59518.00",
"totalTaxes": "4600.00" } } }
}]
```

Initial Response - Payment ID and Payment Page URL

Attributes - Initial Response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Sample JSON Response - Initial Response from ARB PG to Merchant

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based status and result fields as mentioned below.

Success:

```
[{
  "status": "1",
  "result": "100201931620827468:https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm", //Payment ID:Paymentpage URL
  "error": null,
  "errorText": null }]
```

Failure:

```
[{  
  "status": "2",  
  "error": " IPAY0100124",  
  "errorText": "Problem occurred while validating transaction data",  
  "result": null  
}]
```

Framing Payment URL

After Initial Response from ARB PG, merchant needs to frame the payment page URL like the below sample.

<https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm?PaymentID=100201931620827468>

Final Response – Transaction Status

Attribute - Final URL redirection response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique payment Id generated by PG and merchant can use this ID to match the response from PG
2	trandata	C	Alphanum	All the below response parameters encrypted and send the encrypted value in trandata
3	error	C	Alphanum	If any error, PG will send the error code
4	errorText	C	Alphanum	If any error, PG will send the error description

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along

S. No	Fields	M/C/O	Field Type	Description
				with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values

S. No	Fields	M/C/O	Field Type	Description
				1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{
  //Redirection Parameters
  "paymentId":"100201935166676976",
  "trandata":"<encrypted trandata>",
  "error":"","
  "errorText":""
}]
```

Plain Trandata:

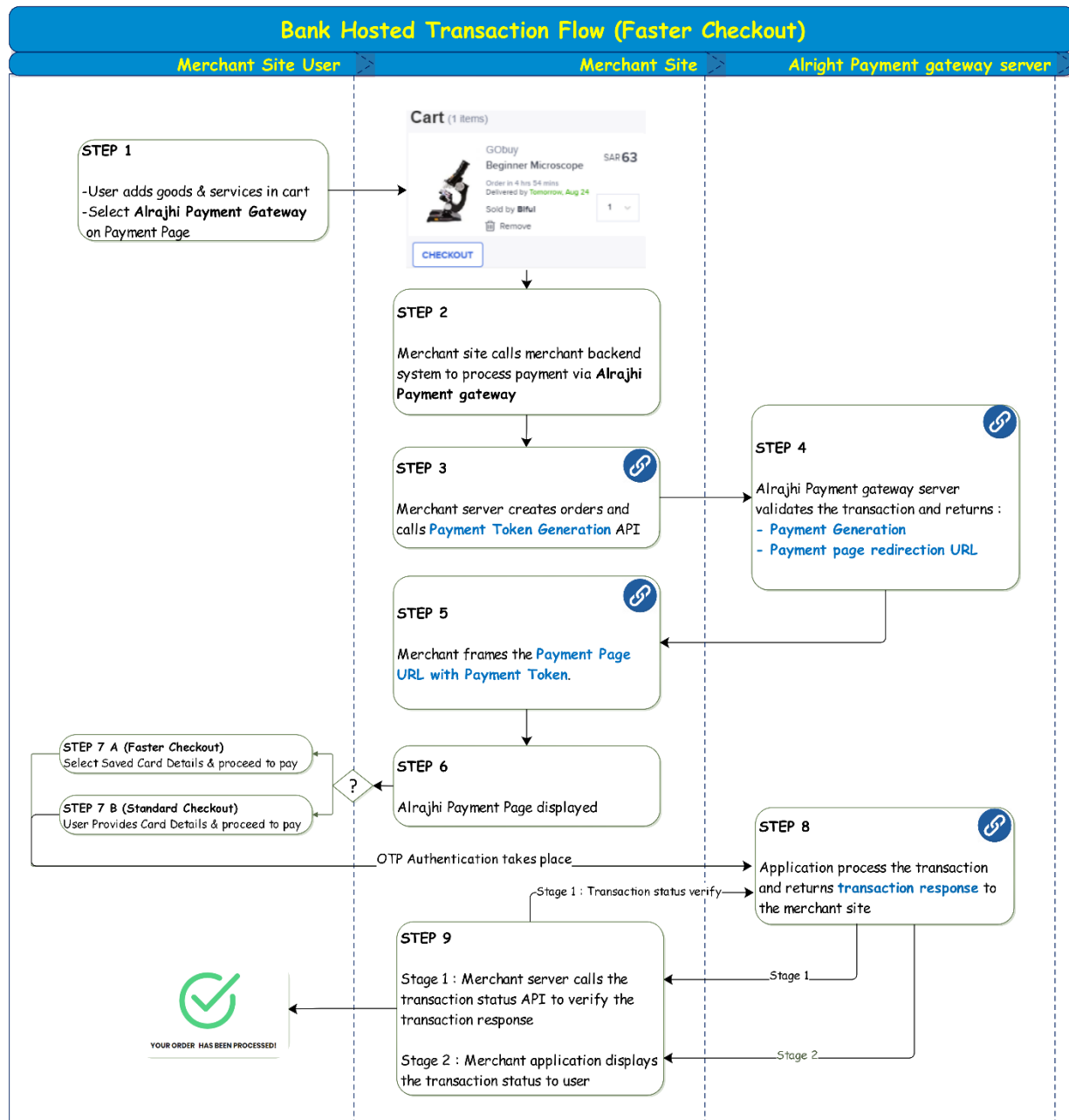
Trandata will contain below parameters encrypted.

```
[{
  "paymentId":"100201935166676976",
  "result": "CAPTURED",
  "transId":201935166561122,
  "ref":"935110000001",
  "date":"1217",
```

```
"trackId":"1003383844",  
  
"udf1":"","  
  
"udf2":"","  
  
"udf3":"8870091137",  
  
"udf4":"FC",  
  
"udf5":"Tidal5",  
  
"amt":"70.0",  
  
"authRespCode","00",  
  
"authCode":"000000",  
  
"cardType":"Visa",  
  
"actionCode":"1"  
  
}]
```

Bank Hosted Integration (Faster Checkout)

This section illustrates how you can integrate the bank hosted (Faster checkout) flow on your website application. Faster Checkout functionality is supported for the merchants only when Faster Checkout flag is enabled at the terminal level.



1. User visits the merchant application and creates order.
2. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
3. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID and Payment Page URL](#) in the response.
 - Merchant needs to [frame the payment page URL with Payment ID](#), Hence the ARB payment page is displayed.
 - In case of failure, ARB PG provides **Error Code** and **Description**.

Note: If merchant notification is disabled, then ARB Payment gateway will provide the [final response in URL redirection](#).

4. Select the payment details already stored in the bank's database. (Faster checkout)
Or
User enters the payment details and authorizes the transaction on the ARB PG bank's page. (Standard checkout)
5. The ARB PG application process the transaction and returns the transaction response to the merchant site.
6. Merchant server calls the transaction status API to verify the transaction response.
7. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	response URL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while processing the transaction.
14	langid	O	Alphabetic	Language ID . Based on language ID arabic language will be displayed on payment page. Value should be 'ar' or 'AR' for arabic language .
15	custid	M	Numeric	Unique Customer ID generated on faster checkout registration.
16	cust_card HolderName	O	Alphabetic	Cardholder Name

S. No	Fields	M/C/O	Field Type	Description
17	cust_mobile_number	O	Numeric	Customer Mobile Number
18	cust_emailId	O	Alphanum	Customer E-mail ID

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata> ",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata:

Trandata will contain below parameters **encrypted** with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource Key.

```
[{
  //Mandatory Parameters
  "amt": "12.00",
  "action": "1", // 1 - Purchase , 4 - Authorization
  "password": "q@a680$27@JLkcK",
  "id": "IPAY1CR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "12345656789",
}]
```

```

"responseURL": "https://merchantpage/PaymentResult.jsp",
"errorURL": "https://merchantpage/PaymentResult.jsp",
//Optional Parameters
"udf1": "udf1text",
"udf2": "udf2text",
"udf3": "udf3text",
"udf4": "udf4text",
"udf5": "udf5text",
"langid": "ar",
//conditional if Merchant opted for Faster Checkout.
"custid": "201936122890007",
"cust_cardHolderName": "Test",
"cust_mobile_number": "7788667755",
"cust_emailId": "Test@gmail.com"
}]

```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanum	Bank Identification Code Min - 8 Max -12

3	iBanNum	C	Alphanu m	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabe tic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails": [

  { "bankIdCode": "12345d6f", "iBanNum": "567896743281926354276254", "beneficiaryName": "AlRajhi Bank Services", "serviceAmount": "200.00", "valueDate": "20201231" },

  { "bankIdCode": "1234ret3", "iBanNum": "987656743281926354276254", "beneficiaryName": "DIGITAL CO", "serviceAmount": "300.00", "valueDate": "20201231" } ],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanu m	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanu m	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanu m	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end

5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
    }
```

Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg

1.1.1.1.7	destinationArrivalTime	C	Alphanum	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanum	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanum	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanum	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanum	
1.1.1.3	originCountry	C	Alphanum	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanum	The passenger first name
1.1.2.2	lastName	C	Alphanum	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanum	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanum	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanum	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanum	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare
1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

//conditional - for airline merchant

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ {
"carrierCode": "MH", "departureAirport": "MNL", "departureDate": "2021-05-11",
"departureTime": "06:50:00Z", "destinationAirport": "KUL",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "10:35:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0805", "travelClass": "B" }, {
"carrierCode": "UL", "departureAirport": "KUL", "departureDate": "2021-05-11",
"departureTime": "15:00:00Z", "destinationAirport": "CMB",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "16:05:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName":
"KAI MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees":
"59518.00", "totalTaxes": "4600.00" } } }
```


Initial Response - Payment ID and Payment Page URL

Attributes - Initial Response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Sample JSON Response - Initial Response from ARB PG to Merchant

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based status and result fields as mentioned below.

Success:

```
[{
  "status": "1",
  "result": "100201931620827468:https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm", //Payment ID:Paymentpage URL
  "error": null,
  "errorText": null }]
```

Failure:

```
[{
  "status": "2",
```

```
"error": " IPAY0100124",  
"errorText": "Problem occurred while validating transaction data",  
"result": null  
}]
```

Framing Payment URL

After Initial Response from ARB PG, merchant needs to frame the payment page URL like the below sample.

<https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm?PaymentID=100201931620827468>

Final Response – Transaction Status

Attribute - Final URL redirection response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique payment Id generated by PG and merchant can use this ID to match the response from PG
2	trandata	C	Alphanum	All the below response parameters encrypted and send the encrypted value in trandata
3	error	C	Alphanum	If any error, PG will send the error code
4	errorText	C	Alphanum	If any error, PG will send the error description

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	fcCustId	M	Numeric	Unique Customer ID generated on faster checkout registration.
5	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
6	date	M	Numeric	Transaction date and time
7	trackId	M	Numeric	Merchant unique reference no

S. No	Fields	M/C/O	Field Type	Description
8	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
13	amt	M	Numeric	Transaction amount
14	authRespCode	M	Numeric	Auth response code provided by PG
15	authCode	M	Numeric	6 digit authorization code received from switch
16	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".

S. No	Fields	M/C/O	Field Type	Description
17	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{
  //Redirection Parameters
  "paymentId":"100201935166676976",
  "trandata":"<encrypted trandata>",
  "error":"","
  "errorText":""
}]
```

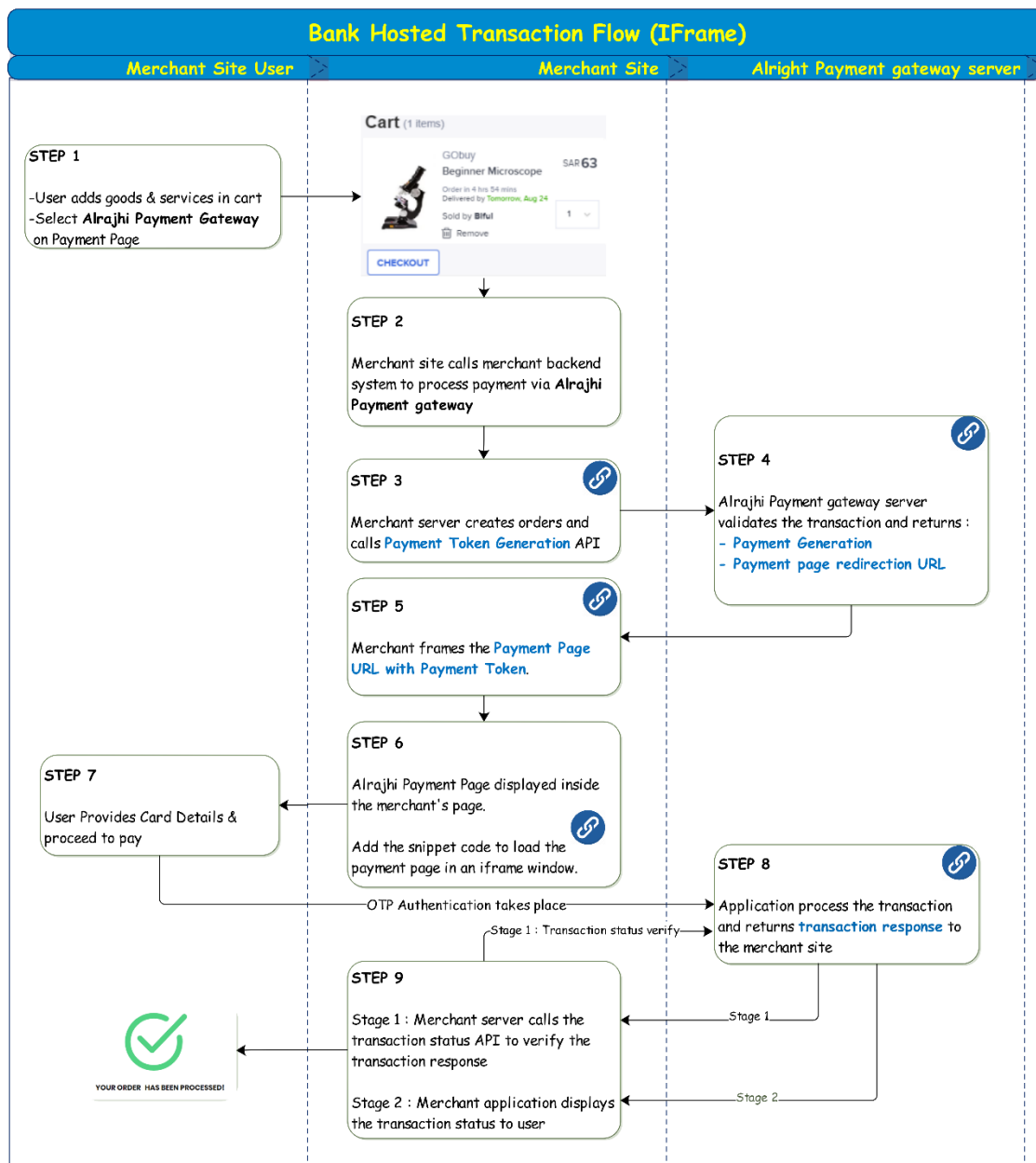
Plain Trandata:

```
[{
  "paymentId":"100201935166676976",
  "result":"CAPTURED",
  "transId":"201935166561122",
  "fcCustId":"201936122890007",
  "ref":"935110000001",
  "date":"1217",
```

```
"trackId":"1003383844",  
  
"udf1":"","  
  
"udf2":"","  
  
"udf3":"8870091137",  
  
"udf4":"FC",  
  
"udf5":"Tidal5",  
  
"amt":"70.0,  
  
"authRespCode","00",  
  
"authCode":"000000",  
  
"actionCode":"1"  
  
}]
```

Bank Hosted Integration (Iframe)

Payment gateway shall support Iframe integration to enable merchants to accept online payments on their own checkout page without having to be a PCI compliant. Instead of being redirected to ARB Payment page, customers will remain on merchant website to make payments. However, at the back end, ARB PG will securely collect the payment information and get it authorized.



1. User visits the merchant application and creates order.
2. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
3. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID and Payment Page URL](#) in the response.
 - Merchant needs to [frame the payment page URL with Payment ID](#), Hence the ARB payment page is displayed.
 - Merchant needs to add the [code snippet](#) to load the payment page in an iframe window.
 - In case of failure, ARB PG provides **Error Code** and **Description**.

Note: If merchant notification is disabled, then ARB Payment gateway will provide the [final response in URL redirection](#).

4. Select the payment details already stored in the bank's database. (For faster checkout)
Or
User enters the payment details and authorizes the transaction on the ARB PG bank's page.
5. The ARB PG application process the transaction and returns the transaction response to the merchant site.
6. Merchant server calls the transaction status API to verify the transaction response.
7. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	M	Alphanum	Merchant needs to send value 'iframe' in the UDF3 field.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	response URL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
14	langid	O	Alphabetic	Language ID . Based on language ID arabic language will be displayed on payment page. Value should be 'ar' or 'AR' for arabic language .
15	custid	M	Numeric	Unique Customer ID generated on faster checkout registration.
16	cust_card HolderName	O	Alphabetic	Cardholder Name
17	cust_mobile_number:	O	Numeric	Customer Mobile Number

S. No	Fields	M/C/O	Field Type	Description
18	cust_emailId	O	Alphanumeric	Customer E-mail ID

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata> ",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata:

Trandata will contain below parameters **encrypted** with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource Key.

```
[{
  //Mandatory Parameters
  "amt": "12.00",
  "action": "1",
  "password": "q@a680$27@JLkcK",
  "id": "IPAY1CR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "12345656789",
  "udf3": "iframe",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
}]
```

```

"errorURL": "https://merchantpage/PaymentResult.jsp",

//Optional Parameters

"udf1":"udf1text",

"udf2":"udf2text",

"udf4":"udf4text",

"udf5":"udf5text",

"langid":"ar",

}]

```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanumeric	Bank Identification Code Min - 8 Max -12
3	iBanNum	C	Alphanumeric	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails":[

{"bankIdCode": "12345d6f", "iBanNum":
"567896743281926354276254", "beneficiaryName": "AlRajhi Bank Services",
"serviceAmount": "200.00", "valueDate": "20201231" },

{"bankIdCode": "1234ret3", "iBanNum":
"987656743281926354276254", "beneficiaryName": "DIGITAL CO",
"serviceAmount": "300.00", "valueDate": "20201231" }],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
    }
```

Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg
1.1.1.1.7	destinationArrivalTime	C	Alphanumeric	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanumeric	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanumeric	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanumeric	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanumeric	
1.1.1.3	originCountry	C	Alphanumeric	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanumeric	The passenger first name

1.1.2.2	lastName	C	Alphanum	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanum	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanum	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanum	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanum	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare
1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

//conditional - for airline merchant

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ {
"carrierCode": "MH", "departureAirport": "MNL", "departureDate": "2021-05-11",
"departureTime": "06:50:00Z", "destinationAirport": "KUL",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "10:35:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0805", "travelClass": "B" }, {
"carrierCode": "UL", "departureAirport": "KUL", "departureDate": "2021-05-11",
"departureTime": "15:00:00Z", "destinationAirport": "CMB",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "16:05:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName":
"KAI MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees":
"59518.00", "totalTaxes": "4600.00" } } }
}]
```

Initial Response - Payment ID and Payment Page URL

Attributes - Initial Response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Sample JSON Response - Initial Response from ARB PG to Merchant

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based status and result fields as mentioned below.

Success:

```
[{
  "status": "1",
  "result": "100201931620827468:https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm", //Payment ID:Paymentpage URL
  "error": null,
  "errorText": null }]
```

Failure:

```
[{
  "status": "2",
```



```
"error": " IPAY0100124",  
"errorText": "Problem occurred while validating transaction data",  
"result": null  
}]
```

Framing Payment URL

After Initial Response from ARB PG, merchant needs to frame the payment page URL like the below sample.

<https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm?PaymentID=100201931620827468>

Merchant needs to add the following code snippet to load the payment page in an iframe window:

Code Snippet

```
if ("iframe".equals(pipe.getUdf3()))
{
%>
<script>

    if (window.parent.document.getElementById("iframe") != null)
    {

        var division=document.createElement("div");

        division.setAttribute("id", "payframe");

        division.setAttribute("style", "min-height: 100%; position:
fixed; top: 0px; left: 0px; width: 100%; height: 100%; background: rgba(0,
0, 0, 0); padding-right: 0px; padding-left: 0px;padding-top: 0px;");

        division.innerHTML = '<div style="position: absolute;right:
0px;top: 0px;cursor: pointer;font-size: 24px;opacity: .6;width:
100%;text-align: center;line-height: 0px;z-index: 1;" class="close"
id="F" onclick="javascript:
window.parent.document.getElementById(\'iframe\').parentNode.removeChil
d(window.parent.document.getElementById(\'iframe\'));window.parent.docu
ment.getElementById(\'payframe\').parentNode.removeChild(window.parent.
document.getElementById(\'payframe\'));">x</div><iframe
id="iframe"
style="opacity: 7; height: 100%; position: relative; background: none;
display: block; border: 0px none transparent; margin-left: 0%; padding:
0px; z-index: 2; width: 100%; margin-top: 0%" allowtransparency="true"
frameborder="0" allowpaymentrequest="true"
src="<%=pipe.getWebAddress() %>"></iframe>';
```

```
        document.body.appendChild(division);

    }

    else

    {

        var division=document.createElement("div");

        division.setAttribute("id", "payframe");

        division.setAttribute("style", "min-height: 100%;
transition: all 0.3s ease-out 0s; position: fixed; top: 0px; left: 0px;
width: 100%; height: 100%; background: rgba(0, 0, 0, 0.4); padding-right:
10px; padding-left: 250px;padding-top: 0px;");

        division.innerHTML ='<div style="position: absolute;right:
0px;top: 0px;cursor: pointer;font-size: 24px;opacity: .6;width:
24px;text-align: center;line-height: 0px;z-index: 1;" class="close"
id="F" onclick="javascript:
window.parent.document.getElementById(\'iframe\').parentNode.removeChil
d(window.parent.document.getElementById(\'iframe\'));window.parent.docu
ment.getElementById(\'payframe\').parentNode.removeChild(window.parent.
document.getElementById(\'payframe\'));">x</div><iframe id="iframe"
style="opacity: 7; height: 100%; position: relative; background: none;
display: block; border: 0px none transparent; margin-left: 7%; padding:
0px; z-index: 2; width: 65%; margin-top: 0%" allowtransparency="true"
frameborder="0" allowpaymentrequest="true"
src="<%=pipe.getWebAddress() %>"></iframe>';

        document.body.appendChild(division);

    }

</script>

}

else

{

    response.sendRedirect(pipe.getWebAddress());

}
```

Final Response – Transaction Status

Attribute - Final URL redirection response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique payment Id generated by PG and merchant can use this ID to match the response from PG
2	trandata	C	Alphanum	All the below response parameters encrypted and send the encrypted value in trandata
3	error	C	Alphanum	If any error, PG will send the error code
4	errorText	C	Alphanum	If any error, PG will send the error description

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	fcCustId	M	Numeric	Unique Customer ID generated on faster checkout registration.
5	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
6	date	M	Numeric	Transaction date and time
7	trackId	M	Numeric	Merchant unique reference no

S. No	Fields	M/C/O	Field Type	Description
8	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
13	amt	M	Numeric	Transaction amount
14	authRespCode	M	Numeric	Auth response code provided by PG
15	authCode	M	Numeric	6 digit authorization code received from switch
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values

S. No	Fields	M/C/O	Field Type	Description
				1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{
  //Redirection Parameters
  "paymentId":"100201935166676976",
  "trandata":"<encrypted trandata>",
  "error":"","
  "errorText":""
}]
```

Plain Trandata:

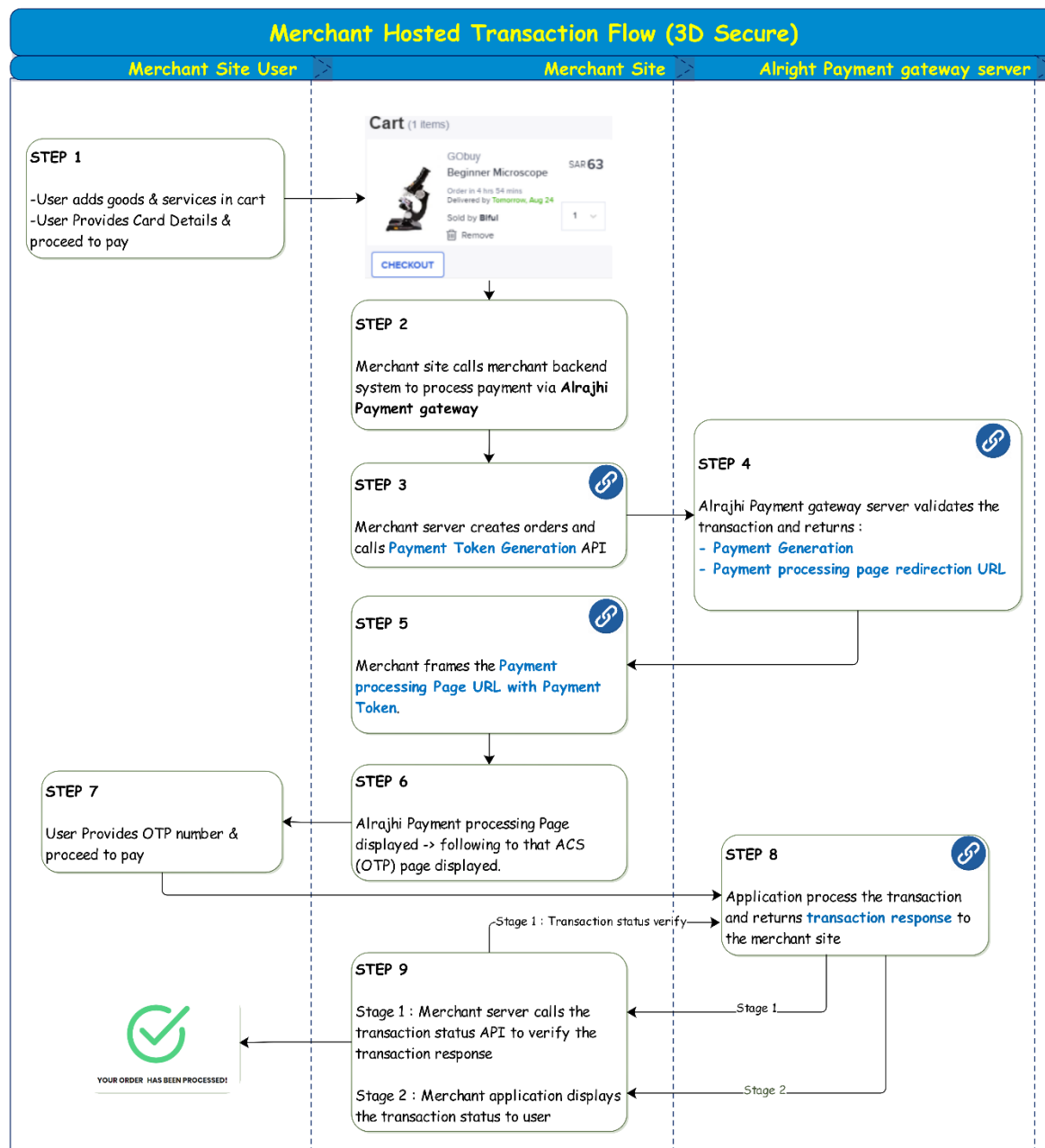
Trandata will contain below parameters encrypted.

```
[{
  "paymentId":"100201935166676976",
  "result":"CAPTURED",
  "transId":"201935166561122",
  "fcCustId":"201936122890007",
  "ref":"935110000001",
```

```
"date": "1217",  
  
"trackId": "1003383844",  
  
"udf1": "",  
  
"udf2": "",  
  
"udf3": "8870091137",  
  
"udf4": "FC",  
  
"udf5": "Tidal5",  
  
"amt": "70.0",  
  
"authRespCode": "00",  
  
"authCode": "000000",  
  
"actionCode": "1"  
}]
```

Merchant Hosted Transaction (3D Secure)

The merchant will collect the customer's card details on their own website hence, the customer will not be redirected to the ARB payment gateway payment page, as the payment option and card details are already received by merchant. The card details are later posted to ARB payment gateway.



This section illustrates how you can integrate the merchant hosted flow on your website application.

1. User visits the merchant application and creates order.
2. User enters the payment details.
3. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
4. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID and Payment Processing Page URL](#) in the response.
 - Merchant needs to [frame the payment processing page URL with Payment ID](#), Hence the ARB payment processing page is displayed.
 - In case of failure, ARB PG provides **Error Code** and **Description**.
5. Upon authorization, the customer redirects to ARB Payment gateway.

The flow takes the user to the login ACS page of the bank, where the user needs to complete the transaction by using the OTP sent by the bank to the registered mobile number. PG then process for authorization with the respective schemes. Once payment response received from respective scheme, then ARB Payment gateway returns the response to merchant. This is URL redirection.

6. After authorization, the ARB PG application process the transaction and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
7. Merchant server calls the transaction status API to verify the transaction response.
8. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
9	udf4	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
10	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
11	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	response URL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanumeric	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
14	expYear	M	Numeric	Expiry year of card
15	expMonth	M	Numeric	Expiry month of card
16	member	M	Alphanumeric	Card holder name
17	cvv2	M	Numeric	CVV of the card
18	cardNo	M	Numeric	Cardholders card number
19	cardType	M	Alphanumeric	Card type Ex : Credit card – C, Debit Card – D

S. No	Fields	M/C/O	Field Type	Description
20	browserLanguage	M	Alphanumeric	Value representing the browser language Returned from "navigator.language" property. Length 1 to 8 characters.
21	browserColorDepth	M	Alphanumeric	Value representing the bit depth of the colour palette for displaying images, in bits per pixel. Obtained from Cardholder browser using the "screen.colorDepth" property. Length 1 to 2 characters. Values Accepted : 1 = 1 bit 4 = 4 bits 8 = 8 bits 15 = 15 bits 16 = 16 bits 24 = 24 bits 32 = 32 bits 48 = 48 bits
22	browserScreenHeight	M	Alphanumeric	Total height of the Cardholder's screen in pixels. Value is returned from the screen.height property. Length 1 to 6 characters.
23	browserScreenWidth	M	Alphanumeric	Total width of the cardholder's screen in pixels. Value is returned from the screen.width property. Length 1 to 6 characters.
24	browserJavaEnabled	M	Alphanumeric	Value is returned from the navigator.javaEnabled property. Boolean value.
25	browserTZ	M	Alphanumeric	Time difference between UTC time and the Cardholder browser local time, in minutes. Value is returned from the getTimezoneOffset() method. Length 1 to 5 characters.
26	jsEnabled	M	Alphanumeric	Value whether the java script is enabled in browser or not.

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
```

```
"id":"IPAY1CR6qZF7q6w",  
  
"trandata":"<encrypted trandata>",  
  
"responseURL":"https://merchantpage/PaymentResult.jsp",  
  
"errorURL":"https://merchantpage/PaymentResult.jsp"  
}]
```

Plain Trandata:

Trandata will contain below parameters **encrypted** with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource key.

```
[{  
  
//Mandatory Parameters  
  
"amt":"12.00",  
  
"action":"1",          // 1 - Purchase , 4 - Authorization  
  
"password":"q@a680$27@JLkcK",  
  
"id":"IPAY1CR6qZF7q6w",  
  
"currencyCode":"682",  
  
"trackId":"123456",  
  
"expYear":"2022",  
  
"expMonth":"12",  
  
"member":"cardholdername",  
  
"cvv2":"212",  
  
"cardNo","5453*****5539",  
  
"cardType":"C",  
  
"responseURL":"https://merchantpage/PaymentResult.jsp",  
  
"errorURL":"https://merchantpage/PaymentResult.jsp",  
}]
```

```

"browserJavaEnabled":"true",

"browserLanguage":"en",

"browserColorDepth":"48",

"browserScreenHeight":"400",

"browserScreenWidth":"600",

"browserTZ":"0",

"jsEnabled":"true",

//Optional Parameters

"udf1":"udf1text",

"udf2":"udf2text",

"udf3":"udf3text",

"udf4":"udf4text",

"udf5":"udf5text",

}]

```

Credit Card Installment Transactions

Detailed description of Plain Trandata request parameters

S.No	Fields	M/C	Field Type	Description
		/O		
1	udf6	C	Alphanum	Credit Card Installment Short Code
2	udf7	C	Alphanum	Credit Card Installment Type

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - Credit Card Installment Transactions
```

```
"udf6": "12345",
"udf7": "C",
```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanumeric	Bank Identification Code Min - 8 Max - 12
3	iBanNum	C	Alphanumeric	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails": [
  {
    "bankIdCode": "12345d6f",
    "567896743281926354276254",
    "beneficiaryName": "AlRajhi Bank Services",
    "serviceAmount": "200.00",
    "valueDate": "20201231"
  },
  {
    "bankIdCode": "1234ret3",
    "987656743281926354276254",
    "beneficiaryName": "DIGITAL CO",
    "serviceAmount": "300.00",
    "valueDate": "20201231"
  }
]
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
    }
```


Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg
1.1.1.1.7	destinationArrivalTime	C	Alphanumeric	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanumeric	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanumeric	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanumeric	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanumeric	
1.1.1.3	originCountry	C	Alphanumeric	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanumeric	The passenger first name
1.1.2.2	lastName	C	Alphanumeric	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanumeric	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanumeric	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanumeric	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanumeric	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare

1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - for airline merchant
```

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ {
"carrierCode": "MH", "departureAirport": "MNL", "departureDate": "2021-05-11",
"departureTime": "06:50:00Z", "destinationAirport": "KUL",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "10:35:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0805", "travelClass": "B" }, {
"carrierCode": "UL", "departureAirport": "KUL", "departureDate": "2021-05-11",
"departureTime": "15:00:00Z", "destinationAirport": "CMB",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "16:05:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName":
"KAI MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees":
"59518.00", "totalTaxes": "4600.00" } } }
```

Initial Response - Payment ID and Payment Processing Page URL

Attributes - Initial Response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Sample JSON Response - Initial Response from ARB PG to Merchant

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided.

Success:

```
[{
  "status": "1",
  "result": "700212030953264091:https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700212030953264091&id=r9Ht8R4U6g9dYtY",
  //Payment ID:Payment URL
  "error": null,
  "errorText": null
}]
```

Failure:

```
[{  
  "status": "2",  
  "error": " IPAY0100124",  
  "errorText": "Problem occurred while validating transaction data",  
  "result": null  
}]
```

Framing Payment URL

After Initial Response from ARB PG, merchant needs to frame the payment page URL like the below sample.

<https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700112030953264091&id=r9Ht8R4U6g9dYtYg>

Final Response – Transaction Status

Merchant needs to redirect the customer to ARB Payment gateway.

The flow takes the user to the login ACS page of the bank, where the user needs to complete the transaction by using the OTP sent by the bank to the registered mobile number. PG then process for authorization with the respective schemes. Once payment response received from respective scheme, then ARB Payment gateway returns the response to merchant. This is URL redirection.

Attribute - Final URL redirection response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by Payment gateway. Merchant can store the payment ID to match the final URL redirection response
2	trandata	C	Alphanum	All the below response parameters encrypted and send the encrypted value in trandata Ex: <pre>[{"paymentId":1002019351666769,"result":"CAPTURED","ref":"935110000001","transId":201935166561122,"date":1217,"trackId":"1003383844","udf1":"","udf2":"","udf3":"8870091137","udf4":"FC","udf5":"Tidal5","amt":"70.0","authRespCode","00"}]</pre>
3	Error	C	Numeric	If any error, PG will provide the error code
4	ErrorText	C	Alphanum	PG will provide the error description if any transaction declined.

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{  
  
  "paymentId": "100201935044735860",  
  
  "trandata": "<encrypted trandata>",  
  
  "Error": "",  
  
  "ErrorText": ""  
  
}]
```

Plain Trandata:

Trandata will contain below parameters encrypted.

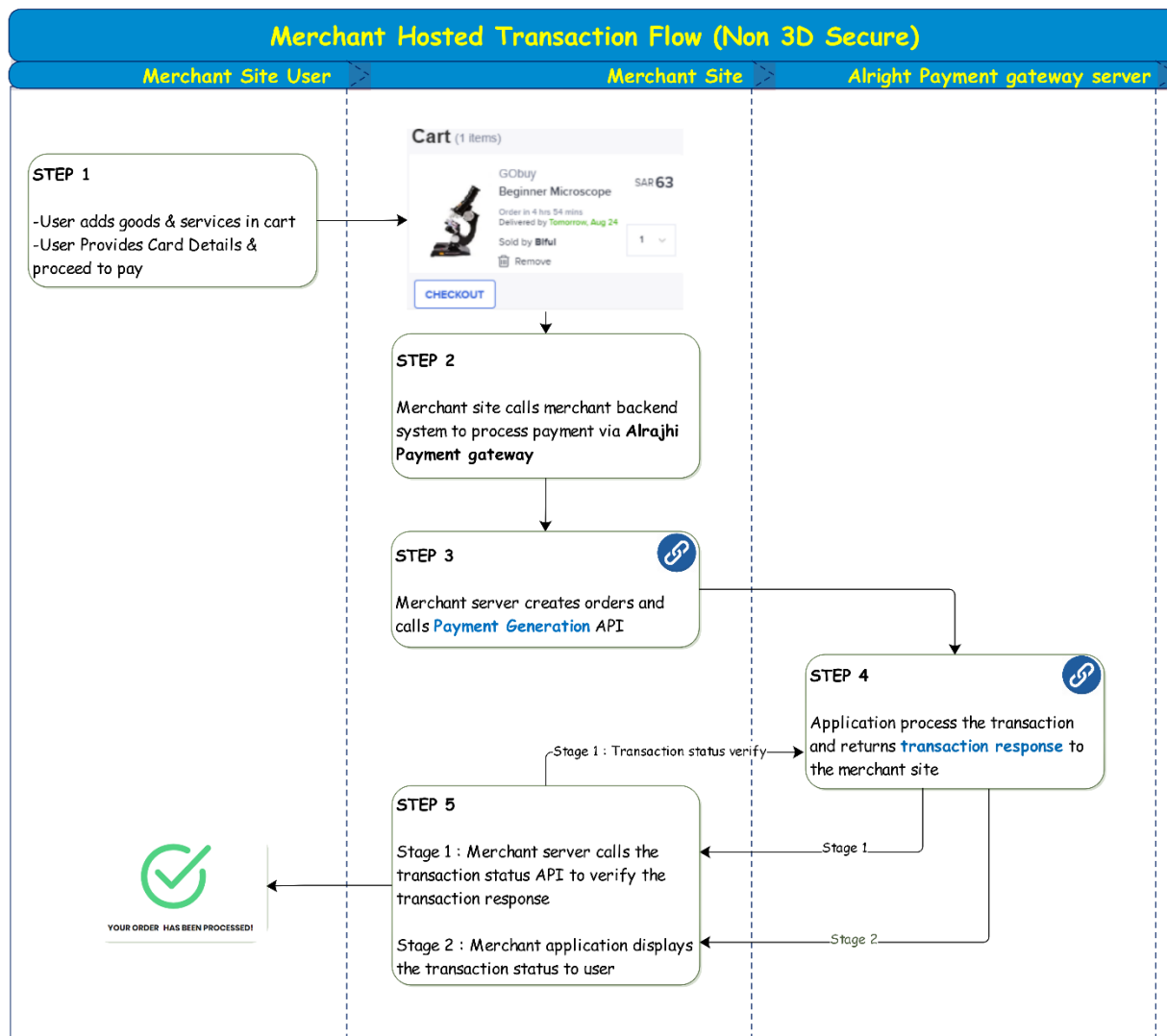
```
[{  
  
  "paymentId": "100201935166676976",  
  
  "result": "CAPTURED",  
  
  "transId": 201935166561122,  
  
  "ref": "935110000001",  
  
  "date": "1217",  
  
  "trackId": "1003383844",  
  
  "udf1": "",  
  
  "udf2": "",  
  
  "udf3": "8870091137",  
  
  "udf4": "FC",  
  
  "udf5": "Tidal5",  
  
  "amt": "70.0",  
  
  "authRespCode": "00",  
  
  "authCode": "000000",  
  
  "cardType": "Visa",  
  
}]
```



```
"actionCode": "1"  
}]
```

Merchant Hosted Transaction (Non 3D Secure)

This section illustrates how you can integrate the merchant hosted flow (Non 3D Secure) on your website application.



1. User visits the merchant application and creates order.
2. User enters the payment details.
3. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token and to process payment via Alrajhi Payment gateway
4. After authorization, the ARB PG application process the transaction and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
5. Merchant server calls the transaction status API to verify the transaction response.
6. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanum	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanum	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanum	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanum	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a

				transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	expYear	M	Numeric	Expiry year of card
13	expMonth	M	Numeric	Expiry month of card
14	member	M	Alphanum	Card holder name
15	cvv2	M	Numeric	CVV of the card
16	cardNo	M	Numeric	Cardholders card number
17	cardType	M	Alphanum	Card type Ex : Credit card – C, Debit Card – D

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>"
}]
```

Plain Trandata:

```
[{
```

```
//Mandatory Parameters

"amt":"12.00",

"action":"1",      // 1 - Purchase , 4 - Authorization

"password":"q@a68O$27@JLkcK",

"id":"IPAYlCR6qZF7q6w",

"currencyCode":"682",

"trackId":"123456",

"expYear":"2022",

"expMonth":"12",

"member":"cardholdername",

"cvv2":"212",

"cardNo","5453*****5539",

"cardType":"C",

//Optional Parameters

"udf1":"udf1text",

"udf2":"udf2text",

"udf3":"udf3text",

"udf4":"udf4text",

"udf5":"udf5text",

}]
```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanumeric	Bank Identification Code Min - 8 Max -12
3	iBanNum	C	Alphanumeric	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails": [

  { "bankIdCode": "12345d6f", "iBanNum": "567896743281926354276254", "beneficiaryName": "AlRajhi Bank Services", "serviceAmount": "200.00", "valueDate": "20201231" },

  { "bankIdCode": "1234ret3", "iBanNum": "987656743281926354276254", "beneficiaryName": "DIGITAL CO", "serviceAmount": "300.00", "valueDate": "20201231" } ],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
  }
```


Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg
1.1.1.1.7	destinationArrivalTime	C	Alphanumeric	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanumeric	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanumeric	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanumeric	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanumeric	
1.1.1.3	originCountry	C	Alphanumeric	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanumeric	The passenger first name
1.1.2.2	lastName	C	Alphanumeric	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanumeric	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanumeric	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanumeric	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanumeric	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare

1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - for airline merchant
```

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ {
"carrierCode": "MH", "departureAirport": "MNL", "departureDate": "2021-05-11",
"departureTime": "06:50:00Z", "destinationAirport": "KUL",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "10:35:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0805", "travelClass": "B" }, {
"carrierCode": "UL", "departureAirport": "KUL", "departureDate": "2021-05-11",
"departureTime": "15:00:00Z", "destinationAirport": "CMB",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "16:05:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName":
"KAI MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees":
"59518.00", "totalTaxes": "4600.00" } } }
```

Final Response – Transaction Status

The ARB payment gateway verifies the transaction and returns the response to the same request.

Attribute - Final response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	Alphanum	If any error during processing, PG will provide the error code
3	errorText	C	Alphanum	If any error during processing, PG will provide the error description
4	status	M	Alphanum	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

The ARB payment gateway verifies the transaction and returns the response to the same request.

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
  "error": null,
  "errorText": null
}]
```

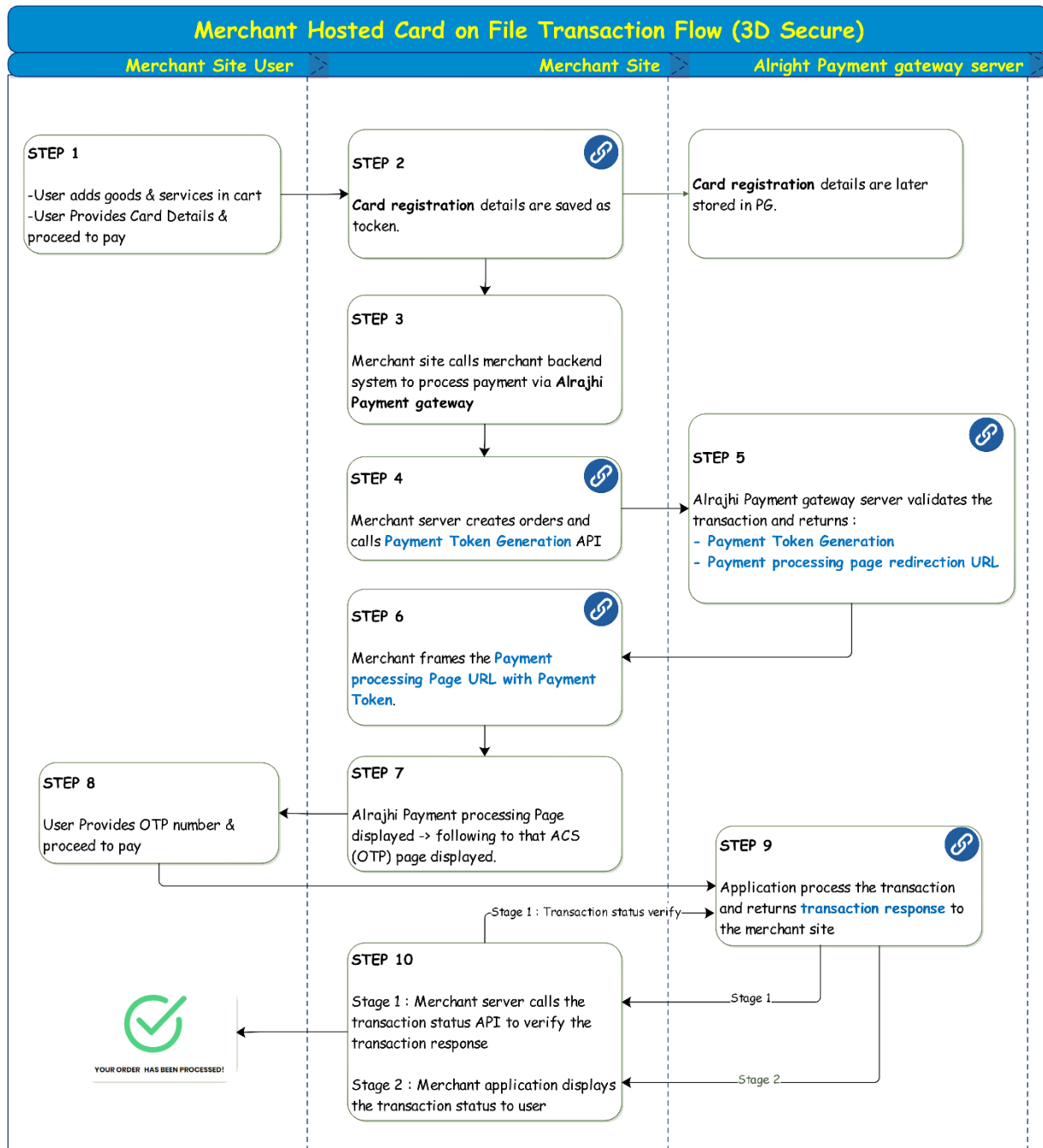
Plain Trandata:

Trandata will contain below parameters encrypted.

```
[{
  "paymentId": -1,
  "result": "CAPTURED",
  "amt": "10.55",
  "date": "1221",
  "ref": "935110000001",
  "udf1": "udf1text",
  "udf2": "udf2text",
  "udf3": "udf3text",
  "udf4": "udf4text",
  "udf5": "udf5text",
  "trackId": "3423423",
  "transId": "1242345345234",
}
```

```
"authRespCode": "00",  
"authCode": "000000",  
"cardType": "Visa",  
"actionCode": "1"  
}]
```

Merchant Hosted Card on File Transactions (3D secure)



1. User visits the merchant application and creates order.
2. User enters the payment card details.
3. The card details are saved as token with the merchant as well the [card registration](#) information is later posted to PG.
4. The Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
5. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID and Payment Processing Page URL](#) in the response.
 - Merchant needs to [frame the payment processing page URL with Payment ID](#), Hence the ARB payment processing page is displayed.
 - In case of failure, ARB PG provides **Error Code** and **Description**.
6. Upon authorization, the customer redirects to ARB Payment gateway.

The flow takes the user to the login ACS page of the bank, where the user needs to complete the transaction by using the OTP sent by the bank to the registered mobile number. PG then process for authorization with the respective schemes. Once payment response received from respective scheme, then ARB Payment gateway returns the response to merchant. This is URL redirection.

7. After authorization, the ARB PG application process the transaction and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
8. Merchant server calls the transaction status API to verify the transaction response.
9. Finally, the merchant application displays the transaction status to user.

Saving Cards During Transaction (Card Registration)

When the merchant hosted transaction is in progress, the sensitive card information entered by the customer is saved as token in the PCI compliant merchant's database as well the information is later posted to PG. The next time the customer makes any transactions, the customer can pay directly by entering the CVV of the card and OTP received.

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanum	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanum	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanum	Merchant error URL

Detailed description of Plain trandata request parameters

S.No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanum	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanum	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.

S.No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
10	udf5	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
13	expYear	M	Numeric	Expiry year of card
14	expMonth	M	Numeric	Expiry month of card
15	member	M	Alphanum	Card holder name
16	cvv2	M	Numeric	CVV of the card
17	cardNo	M	Numeric	Cardholders card number
18	cardType	M	Alphanum	Card type Ex : Credit card – C, Debit Card – D
19	cardOnFileAction	M	Alphanum	Card on File action .Mandatory field for Card On File. Value should be "transaction" for Card On File transactions.
20	cardOnFileToken	C	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card . Merchant needs to send this field only for saving subsequent cards for the customer and for transaction using saved cards.
21	browserLanguage	M	Alphanum	Value representing the browser language Returned from "navigator.language" property. Length 1 to 8 characters.
22	browserColorDepth	M	Alphanum	Value representing the bit depth of the colour palette for displaying images, in bits per pixel. Obtained from Cardholder browser using the "screen.colorDepth" property. Length 1 to 2 characters. Values Accepted :

S.No	Fields	M/C/O	Field Type	Description
				1 = 1 bit 4 = 4 bits 8 = 8 bits 15 = 15 bits 16 = 16 bits 24 = 24 bits 32 = 32 bits 48 = 48 bits
23	browserScreenHeight	M	Alphanumeric	Total height of the Cardholder's screen in pixels. Value is returned from the screen.height property. Length 1 to 6 characters.
24	browserScreenWidth	M	Alphanumeric	Total width of the cardholder's screen in pixels. Value is returned from the screen.width property. Length 1 to 6 characters.
25	browserJavaScriptEnabled	M	Alphanumeric	Value is returned from the navigator.javaEnabled property. Boolean value.
26	browserTZ	M	Alphanumeric	Time difference between UTC time and the Cardholder browser local time, in minutes. Value is returned from the getTimezoneOffset() method. Length 1 to 5 characters.
27	jsEnabled	M	Alphanumeric	Value whether the java script is enabled in browser or not.

Request from Merchant to ARB Payment gateway:

```
[{
  //Mandatory Parameters
  "id":"IPAY1CR6qZF7q6w",
  "trandata":"<encrypted trandata>",
  "responseURL":"https://merchantpage/PaymentResult.jsp",
  "errorURL":"https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata

Trandata will contain below parameters encrypted with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource key.

```
[{  
  
  //Mandatory Parameters  
  
  "amt": "12.00",  
  
  "action": "1",          // 1 - Purchase , 4 - Authorization  
  
  "password": "q@a680$27@JLkcK",  
  
  "id": "IPAYlCR6qZF7q6w",  
  
  "currencyCode": "682",  
  
  "trackId": "123456",  
  
  "expYear": "2022",  
  
  "expMonth": "12",  
  
  "member": "cardholdername",  
  
  "cvv2": "212",  
  
  "cardNo": "5453*****5539",  
  
  "cardType": "C",  
  
  "responseURL": "https://merchantpage/PaymentResult.jsp",  
  
  "errorURL": "https://merchantpage/PaymentResult.jsp",  
  
  "cardOnFileAction": "transaction",  
  
  "browserJavaEnabled": "true",  
  
  "browserLanguage": "en",  
  
  "browserColorDepth": "48",  
  
  "browserScreenHeight": "400",  
  
  "browserScreenWidth": "600",  
  
  "browserTZ": "0",  
}
```

```
"jsEnabled":"true",

//Conditional - To be excluded when a customer saves first card. Required
for saving subsequent cards for a customer.

"cardOnFileToken":"201936122890007",

//Optional Parameters

"udf1":"udf1text",

"udf2":"udf2text",

"udf3":"udf3text",

"udf4":"udf4text",

"udf5":"udf5text"

}]
```

Request - Payment Token Generation API

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanum	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanum	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanum	Merchant error URL

Detailed description of Plain trandata request parameters

S.No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanum	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanum	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.

S.No	Fields	M/C/O	Field Type	Description
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanumeric	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
13	member	M	Alphanumeric	Card holder name
14	cvv2	M	Numeric	CVV of the card
15	cardType	M	Alphanumeric	Card type Ex : Credit card – C, Debit Card – D
16	cardOnFileAction	M	Alphanumeric	Card on File action .Mandatory field for Card On File. Value should be "transaction" for Card On File transactions
17	cardOnFileToken	M	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card .
18	maskedCardNo	M	Alphanumeric	Masked card number for card on file . First 6 and last 4 digits visible.
19	browserLanguage	M	Alphanumeric	Value representing the browser language Returned from "navigator.language" property. Length 1 to 8 characters.
20	browserColorDepth	M	Alphanumeric	Value representing the bit depth of the colour palette for displaying images, in bits per pixel. Obtained from Cardholder browser using the "screen.colorDepth" property. Length 1 to 2 characters. Values Accepted : 1 = 1 bit 4 = 4 bits 8 = 8 bits 15 = 15 bits 16 = 16 bits 24 = 24 bits 32 = 32 bits 48 = 48 bits

S.No	Fields	M/C/O	Field Type	Description
21	browserScreenHeight	M	Alphanumeric	Total height of the Cardholder's screen in pixels. Value is returned from the screen.height property. Length 1 to 6 characters.
22	browserScreenWidth	M	Alphanumeric	Total width of the cardholder's screen in pixels. Value is returned from the screen.width property. Length 1 to 6 characters.
23	browserJavaScriptEnabled	M	Alphanumeric	Value is returned from the navigator.javaEnabled property. Boolean value.
24	browserTZ	M	Alphanumeric	Time difference between UTC time and the Cardholder browser local time, in minutes. Value is returned from the getTimezoneOffset() method. Length 1 to 5 characters.
25	jsEnabled	M	Alphanumeric	Value whether the java script is enabled in browser or not.

Request from Merchant to ARB Payment gateway:

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata

Trandata will contain below parameters encrypted with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource key.

```
[{ //Mandatory Parameters
  "amt": "12.00",
```



```
"action": "1",          // 1 - Purchase , 4 - Authorization

"password": "q@a680$27@JLkcK",

"id": "IPAY1CR6qZF7q6w",

"currencyCode": "682",

"trackId": "123456",

"member": "cardholdername",

"cvv2": "212",

"cardType": "C",

"responseURL": "https://merchantpage/PaymentResult.jsp",

"errorURL": "https://merchantpage/PaymentResult.jsp",

"cardOnFileAction": "transaction",

"cardOnFileToken": "201936122890007",

"maskedCardNo": "545301*****5539",

"browserJavaEnabled": "true",

"browserLanguage": "en",

"browserColorDepth": "48",

"browserScreenHeight": "400",

"browserScreenWidth": "600",

"browserTZ": "0",

"jsEnabled": "true",

//Optional Parameters

"udf1": "udf1text",

"udf2": "udf2text",

"udf3": "udf3text",
```

```
"udf4": "udf4text",  
"udf5": "udf5text"  
}]
```

Initial Response - Payment ID and Payment Page URL

Initial Response from PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanu m	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error description

Plain Response:

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response if the validation success. If failure then, Error code and description will be provided.

```

Success:

[ {
  "status": "1",
  "result": "700212030953264091:https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700212030953264091&id=r9Ht8R4U6g9dYtY",
  //Payment ID:Payment URL
  "error": null,
  "errorText": null
} ]

Failure:

[ {
  "status": "2",
  "error": " IPAY0100124",

```

```
"errorText": " Problem occurred while validating transaction data",  
"result": null  
}]
```

Framing Payment URL

Merchant needs to frame the URL like the below sample

<https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700112030953264091&id=r9Ht8R4U6g9dYtYg>

Final Response – Transaction Status

Merchant needs to redirect the customer to ARB Payment gateway.

Customer browser will redirect to ACS page and will complete the authentication. PG then process for authorization with the respective schemes. Once payment response received from respective scheme, then ARB Payment gateway returns the response to merchant. This is URL redirection. Below is sample response from ARB PG to merchant,

Final Response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by Payment gateway. Merchant can store the payment ID to match the final URL redirection response
2	trandata	C	Alphanumeric	All the response parameters encrypted and sent in encrypted value in trandata
3	Error	C	Numeric	If any error, PG will provide the error code
4	ErrorText	C	Alphanumeric	PG will provide the error description if any transaction declined.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanumeric	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	cardOnFileToken	M	Numeric	Unique token ID (customer ID) generated by PG per customer when customer saves the first card . This should be sent in merchant request when the customer saves the subsequent cards next time.
16	maskedCardNo	M	Alphanumeric	Masked card number for card on file transactions. First 6 digits and last 4 digits will be visible.
17	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

Redirection Parameters

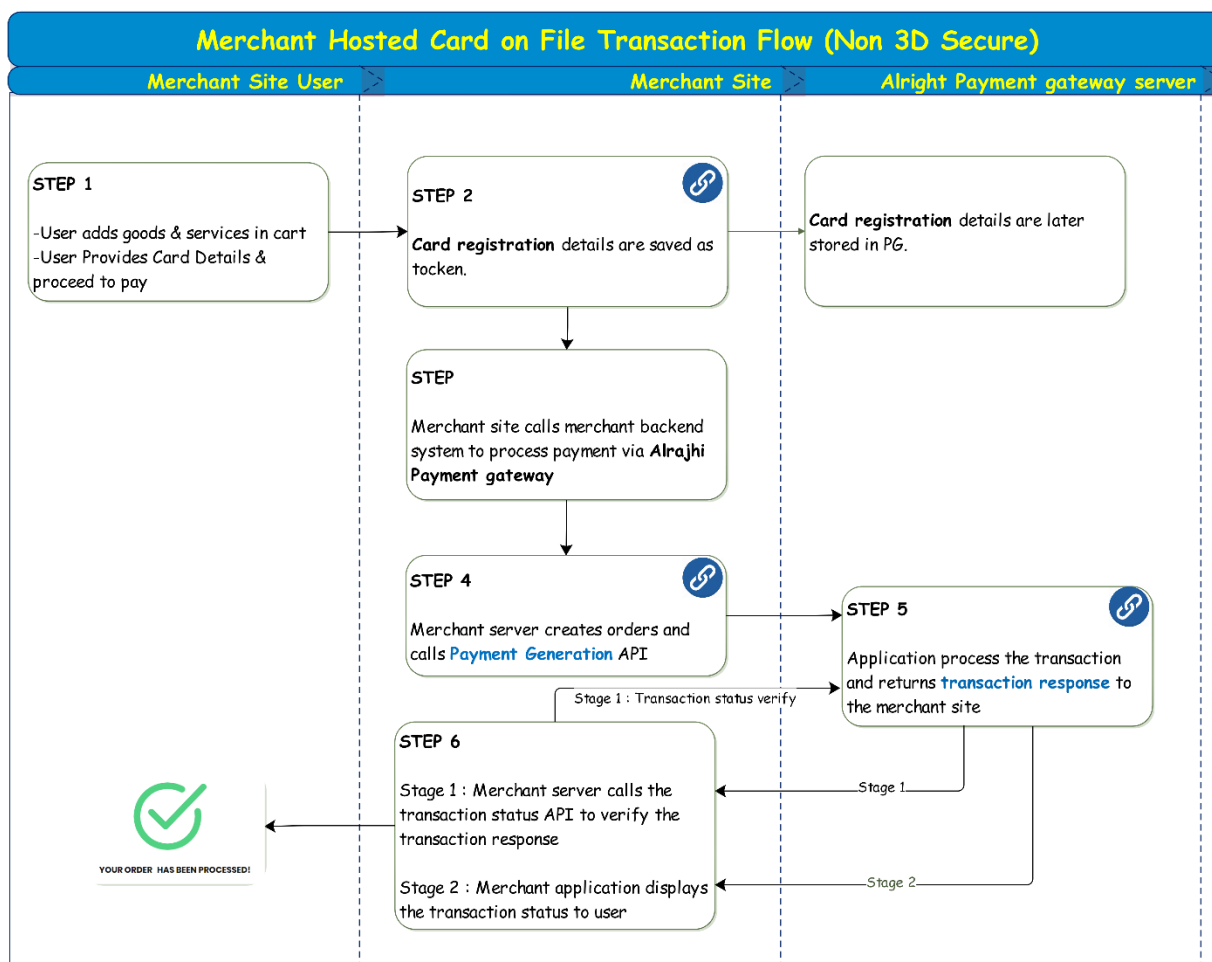
```
[{  
  
  "paymentId": "100201935044735860",  
  
  "trandata": "<encrypted trandata>",  
  
  "Error": "",  
  
  "ErrorText": ""  
  
}]
```

Plain Trandata

```
[{"paymentId": "100201935044735860",  
  
  "result": "CAPTURED",  
  
  "amt": "10.55",  
  
  "date": 1221,  
  
  "ref": "935110000001",  
  
  "udf1": "udf1text",  
  
  "udf2": "udf2text",  
  
  "udf3": "udf3text",  
  
  "udf4": "udf4text",  
  
  "udf5": "udf5text",  
  
  "trackId": "3423423",  
  
  "transId": "1242345345234",  
  
  "authRespCode": "00",  
  
  "authCode": "000000",  
  
  "cardType": "Visa",  
  
  "cardOnFileToken": "201936122890007",  
  
  "maskedCardNo": "545301*****5539"
```

```
"actionCode": "1"  
}]
```


Merchant Hosted Card on File Transactions (Non-3D secure)



1. User visits the merchant application and creates order.
2. User enters the payment card details.
3. The card details are saved as [card registration](#) token and posted to PG.
4. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token and to process payment via Alrajhi Payment gateway.
5. After authorization, the ARB PG application processes the transaction and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
6. Merchant server calls the transaction status API to verify the transaction response.
7. Finally, the merchant application displays the transaction status to user.

Saving Cards During Transaction (Card Registration)

When the merchant hosted transaction is in progress, the sensitive card information entered by the customer is saved as token in the PCI compliant merchant's database as well the information is later posted to PG. The next time the customer makes any transactions, the customer can pay directly by entering the CVV of the card.

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanu m	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanu m	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanu m	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanu m	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	expYear	M	Numeric	Expiry year of card
12	expMonth	M	Numeric	Expiry month of card
13	member	M	Alphanu m	Card holder name
14	cvv2	M	Numeric	CVV of the card
15	cardNo	M	Numeric	Cardholders card number
16	cardType	M	Alphanu m	Card type Ex : Credit card – C, Debit Card - D
17	cardOnFileAct ion	M	Alphanu m	Card on File action .Mandatory field for Card On File. Value should be "transaction" for Card On File transactions
18	cardOnFileTok en	C	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card . Merchant needs to send this field only for saving subsequent cards for the customer and for transaction using saved cards.

Below is the sample encrypted request from Merchant to PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>"
}]
```

Below are the plain Trandata request

```
{
  //Mandatory Parameters
  "amt":"12.00",

  "action":"1",      // 1 - Purchase , 4 - Authorization
  "password":"q@a680$27@JLkcK",

  "id":"IPAYlCR6qZF7q6w",

  "currencyCode":"682",

  "trackId":"123456",

  "expYear":"2022",

  "expMonth":"12",

  "member":"cardholdername",

  "cvv2":"212",

  "cardNo","5453*****5539",

  "cardType":"C",

  "cardOnFileAction":"transaction",

  //Conditional - To be excluded for saving first card for a customer .
  Required for saving subsequent cards for the customer.
  "cardOnFileToken":"201936122890007",

  //Optional Parameters

  "udf1":"udf1text",

  "udf2":"udf2text",

  "udf3":"udf3text",

  "udf4":"udf4text",

  "udf5":"udf5text"
}
```

Request - Payment Token Generation API

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	member	M	Alphanumeric	Card holder name
12	cvv2	M	Numeric	CVV of the card
13	cardType	M	Alphanumeric	Card type Ex : Credit card – C, Debit Card - D
14	cardOnFileAction	M	Alphanumeric	Card on File action .Mandatory field for Card On File. Value should be "transaction" for Card On File transactions
15	cardOnFileToken	M	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card .
16	maskedCardNo	M	Alphanumeric	Masked card number for card on file. First 6 and last 4 digits visible.

Below is the sample encrypted request from Merchant to PG

```
[{
  //Mandatory Parameters
  "id":"IPAY1CR6qZF7q6w",
  "trandata":"<encrypted trandata>"
}]
```

Below are the plain Trandata request

```
[{
  //Mandatory Parameters
  "amt":"12.00",
  "action":"1",          // 1 - Purchase , 4 - Authorization
  "password":"q@a68O$27@JLkcK",
```

```
"id": "IPAY1CR6qZF7q6w",  
  
"currencyCode": "682",  
  
"trackId": "123456",  
  
"member": "cardholdername",  
  
"cvv2": "212",  
  
"cardType": "C",  
  
"cardOnFileAction": "transaction",  
  
"cardOnFileToken": "201936122890007",  
  
"maskedCardNo": "545301*****5539",  
  
  
//Optional Parameters  
  
"udf1": "udf1text",  
  
"udf2": "udf2text",  
  
"udf3": "udf3text",  
  
"udf4": "udf4text",  
  
"udf5": "udf5text"  
}]
```

Final Response – Transaction Status

Once ARB payment gateway verifies the transaction and returns the response to the same request.

Response from ARB Payment Gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	Alphanu m	All the response parameters will be provided in trandata field
2	error	C	Alphanu m	If any error during processing, PG will provide the error code
3	errorText	C	Alphanu m	If any error during processing, PG will provide the error description
4	status	M	Alphanu m	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanu m	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	cardOnFileToken	M	Numeric	Unique token ID (customer ID) generated by PG per customer when customer saves the first card . This should be sent in merchant request when the customer saves the subsequent cards next time.
16	maskedCardNo	M	Alphanumeric	Masked card number for card on file transactions. First 6 digits and last 4 digits will be visible.
17	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Below is the sample encrypted response from PG to Merchant

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
```

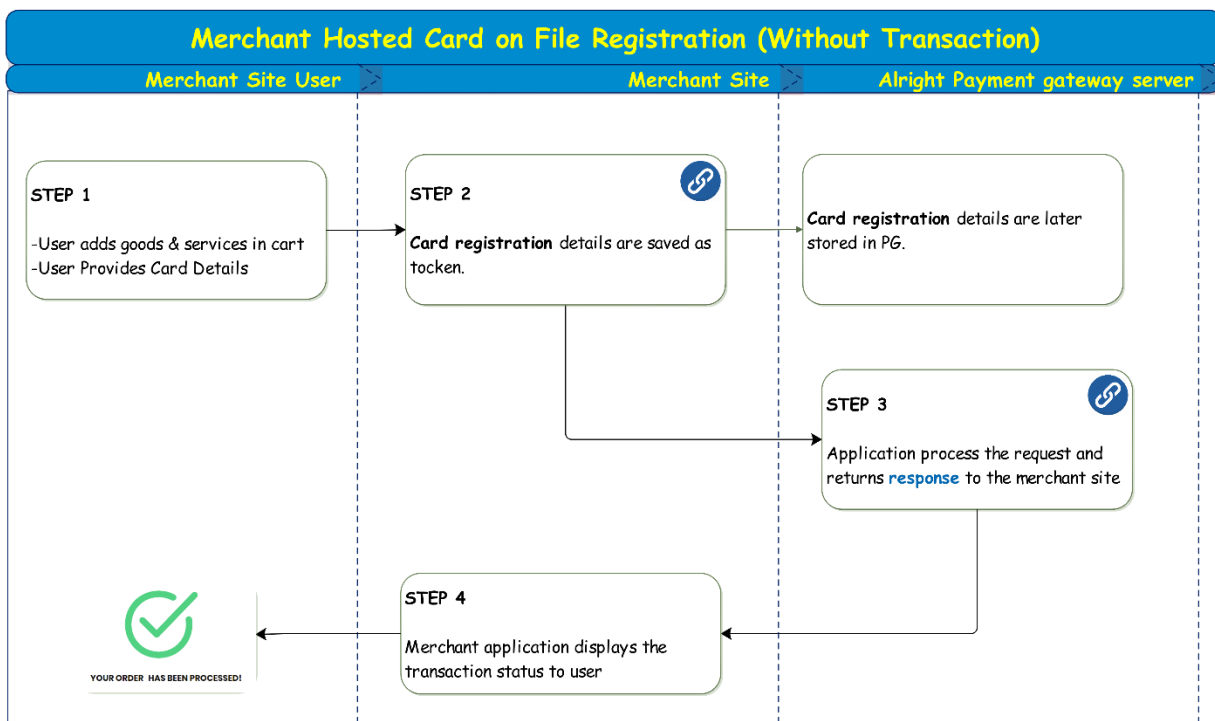
```
"error":null,  
  
"errorText": null  
  
}]
```

Plain trandata in response

```
[{  
  
  "paymentId":-1,  
  
  "result":"CAPTURED",  
  
  "amt":"10.55",  
  
  "date":"1221",  
  
  "ref":"935110000001",  
  
  "udf1":"udf1text",  
  
  "udf2":"udf2text",  
  
  "udf3":"udf3text",  
  
  "udf4":"udf4text",  
  
  "udf5":"udf5text",  
  
  "trackId","3423423",  
  
  "transId":"1242345345234",  
  
  "authRespCode":"00",  
  
  "authCode":"000000",  
  
  "cardType":"Visa",  
  
  "cardOnFileToken":"201936122890007",  
}
```

```
"maskedCardNo": "545301*****5539"  
  
"actionCode": "1"  
  
}]
```

Merchant Hosted Card on File Registrations (without transaction)



1. User visits the merchant application and click card registration option.
2. Merchant application backend server calls [Card Registration Token Generation API](#) to get the token and to register the card details in Alrajhi Payment gateway
3. After authorization, the ARB PG application saves the card details and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
4. Finally, the merchant application displays the transaction status to user.

Request – Card Registration Token Generation API

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
2	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
3	expYear	M	Numeric	Expiry year of card
4	expMonth	M	Numeric	Expiry month of card
5	cardNo	M	Numeric	Cardholders card number
6	cardOnFileAction	M	Alphanumeric	Card on File action . Value should be "registration" for registration without transaction.
7	cardOnFileToken	C	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card . Merchant needs to send this field only for saving subsequent cards for the customer.

Below is the sample encrypted request from Merchant to PG

```
[{  
  
  //Mandatory Parameters  
  
  "id":"IPAY1CR6qZF7q6w",  
  
  "trandata":"<encrypted trandata>"  
  
}]
```

Below are the plain Trandata request

```
[{  
  
  //Mandatory Parameters  
  
  "password":"q@a68O$27@JLkcK",  
  
  "id":"IPAY1CR6qZF7q6w",  
  
  "expYear":"2022",  
  
  "expMonth":"12",  
  
  "cardNo","5453*****5539",  
  
  "cardOnFileAction":"registration",  
  
  //conditional Parameter - To be excluded for saving first card for a  
  customer. Required for saving subsequent cards for the customer.  
  
  "cardOnFileToken":"201936122890007"  
  
}]
```

Response – Card Registration Status

Response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	C	Alphanu m	All the below response parameters encrypted and send the encrypted value in trandata
2	error	C	Numeric	If any error, PG will provide the error code
3	errorText	C	Alphanu m	PG will provide the error description if registration fails.
4	status	M	Numeric	1 for registration success case, 2 for failure case

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	cardOnFileToken	M	Numeric	Card on File Token . Unique token ID (customer ID) generated by PG per customer when customer saves the first card .
2	maskedCardNo	M	Alphanu m	Masked card number for card on file.

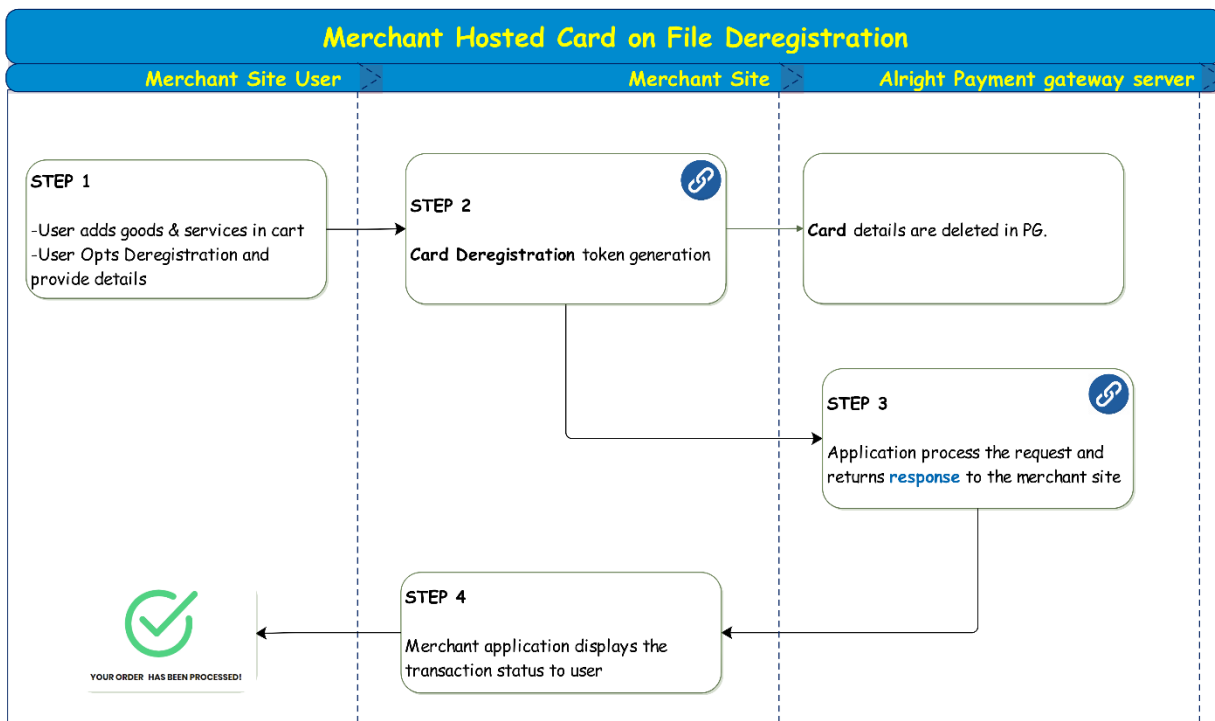
Below is the sample encrypted response from PG to Merchant

```
[{ "trandata": "<encrypted trandata>",
  "status": "1", //1 for registration success case, 2 for failure case
  "error": "",
  "errorText": ""
}]
```

Plain Trandata for PG response to Merchant

```
[{
  "cardOnFileToken": "201936122890007",
  "maskedCardNo": "545301*****5539"
}]
```

Merchant Hosted Card on File Deregistration



1. User visits the merchant application and click card deregistration option.
2. Merchant application backend server calls [Card Deregistration Token Generation API](#) to get the token and to deregister the card details in Alrajhi Payment gateway
3. After authorization, the ARB PG application deletes the card details and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
4. Finally, the merchant application displays the transaction status to user.

Request – Card Deregistration Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
2	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
3	cardOnFileAction	M	Alphanumeric	Card on File action. Value should be "deregistration" for deregistration of Card on File.
4	cardOnFileToken	M	Numeric	Card on File Token. Unique token ID (customer ID) generated by PG per customer.
5	maskedCardNo	M	Alphanumeric	Card on File masked card number. First 6 digits and last 4 digits to be visible.

Below is the sample encrypted request from Merchant to PG

```
[{
  //Mandatory Parameters
  "id":"IPAY1CR6qZF7q6w",
  "trandata":"<encrypted trandata>"
}]
```

Below are the plain Trandata request

```
{  
  //Mandatory Parameters  
  "password":"q@a68O$27@JLkcK",  
  "id":"IPAYlCR6qZF7q6w",  
  "cardOnFileAction":"deregistration",  
  "cardOnFileToken":"201936122890007",  
  "maskedCardNo":"545301*****5539"  
}
```

Response – Card Deregistration Status

Response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	error	C	Numeric	If any error, PG will provide the error code
2	errorText	C	Alphanu m	PG will provide the error description if deregistration fails.
3	status	M	Numeric	1 for deregistration success case, 2 for failure case

Below is the sample response from PG to Merchant

```
[{
  "status": "1", //1 for deregistration success case, 2 for failure case
  "error": "",
  "errorText": ""
}]
```

Merchant Hosted Transaction Flow (Inquiry, Void, Refund, Capture Transactions)

- **Inquiry Transactions:** As an added security measure ARB Payment Gateway allows merchants to do an inquiry of already completed transaction by passing certain details of the payment message, ARB Payment Gateway provides response to this request with appropriate fields in the response; the merchant is expected to verify the relevant fields like Transaction amount, transaction status and other transaction fields.
- **Refund Transactions:** Merchant collects the cancellation request and initiate the transaction using Refund API. Merchant may initiate refund for full amount or partial amount; Payment Gateway will allow amount until it reached to original transaction amount.
- **Authorization Extension Transactions (MADA):** As per the current functionality, only MADA is supporting Authorization Extension. Authorization Extension is nothing but extending the Authorization period not the Authorized amount.

Transaction Type		Action Code
Inquiry		8
void		3
refund		2
capture		5
Void Authorization Transactions (MADA)		9
Authorization Extension transactions (MADA)		14

Note: Capture transactions need to be performed within 7 days.
Auth Extension and refund needs to be performed within 14 days

Request

Request from Merchant to ARB Payment Gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanu m	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandat a	M	Alphanu m	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Tran data request parameters

S. No	Fields	M/C/O	Field Type	Description
1	id	M	AlphaNu m	Unique Tranportal ID.
2	password	M	Alphanu m	Tranportal password.
3	action	M	Numeric	Transaction action Ex: "2" for Refund "3" for Void purchase "8" for Inquiry "5" for Capture "9" for Void Auth – only MADA "14" for Auth Extension – only MADA
4	amt	M	Numeric	Transaction amount
5	currencyCode	M	Numeric	Currency code of merchant
6	trackId	O	Numeric	A unique tracking id issued by the merchant's system
7	udf5	M	Alphanu m	When merchant want to perform refund/Void/Inquiry/ Void Auth / Auth Extension based on PaymentID / TransID / TrackID then merchant need to pass, a word PaymentID / TRANID/ TrackID in Udf5 field.
8	Customer ID	O	Alphanu m	When Merchant required to perform Preatuh for unschedule transaction
9	transId	M	Numeric	When merchant want to perform refund/Void/Inquiry/ Void Auth/ Auth Extension based on PaymentID / TransID / TrackID then merchant need to pass the value of, PaymentID / TRANID/ TrackID in this field

10	udf10	O	Alphanum	Applicable only for MADA : Need to mention whether the transaction is partial capture or Full Capture in case of MADA transactions. Udf10 fields needs to be set as below, PARTIALCAPTURE for Partial Capture FINALCAPTURE for Final Capture
----	-------	---	----------	---

Below is the sample encrypted request from Merchant to PG

```
[{
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>"
}]
```

Below are the plain trandata request

Based on TransID (Payment Gateway Transaction ID):

```
[{
  "amt": "70.00",
  "action": "<action code>",
  "password": "q@a680$27@JLkcK",
  "id": "IPAY1CR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "696921377",
  "udf5": "TRANID",
  "transId": "201931951332346",
  "udf10": "FULLCAPTURE" //Applicable only for MADA capture transactions
}]
```

Based on Payment ID (Payment ID):

```
[{  
  "amt": "70.00",  
  "action": "<action code>",  
  "password": "q@a680$27@JLkcK",  
  "id": "IPAY1CR6qZF7q6w",  
  "currencyCode": "682",  
  "trackId": "696921377",  
  "udf5": "PaymentID",  
  "transId": "100201931948635783",  
  "udf10": "FULLCAPTURE" //Applicable only for MADA capture transactions  
}]
```

Based on TrackID (Merchant Transaction ID/Track ID):

```
[{  
  "amt": "70.00",  
  "action": "<action code>",  
  "password": "q@a680$27@JLkcK",  
  "id": "IPAY1CR6qZF7q6w",  
  "currencyCode": "682",  
  "trackId": "696921377",  
  "udf5": "TrackID",  
  "transId": "696921377",  
  "udf10": "FULLCAPTURE" //Applicable only for MADA capture transactions  
}]
```

Response

Response from ARB Payment Gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	Alphanumeric	If any error during processing, PG will provide the error code
3	errorText	C	Alphanumeric	If any error during processing, PG will provide the error description
4	Status	M	Alphanumeric	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanumeric	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in

S. No	Fields	M/C/O	Field Type	Description
				the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)
17	origTransactionID	C	Alphanumeric	It is applicable only for Supporting Transactions action code like 2,3,5,9 and 14. It refers to the "transId" of the Source/Original transaction

Below is the sample encrypted response from PG to Merchant

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
  "error": null,
  "errorText": null
}]
```

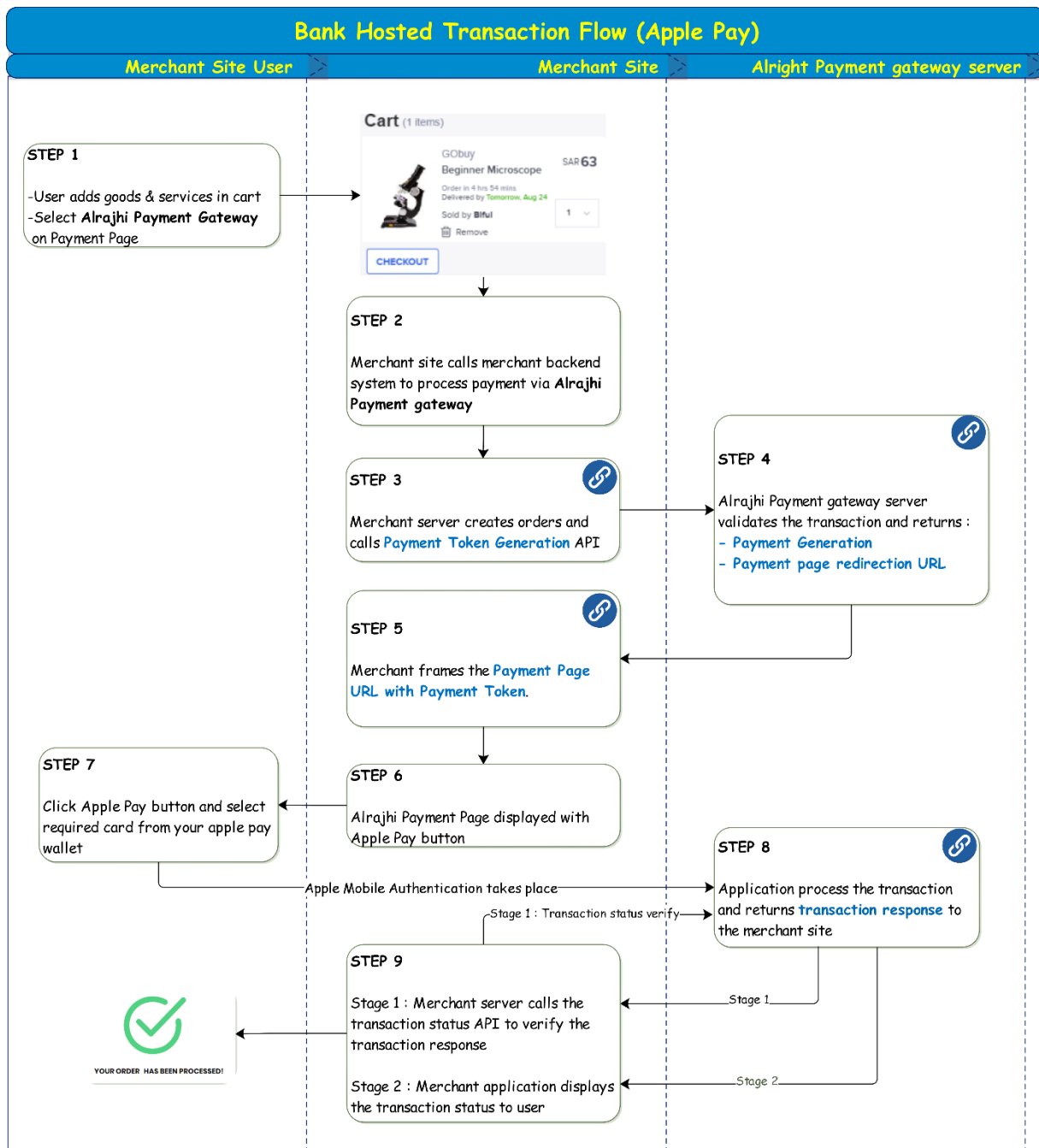
Plain trandata in response

```
[{
  "paymentId": 100201934525118923,
  "result": "success",
  "date": "1221",
  "udf1": "udf1text",
  "udf2": "udf2text",
  "udf3": "udf3text",
  "udf4": "udf4text",
  "udf5": "PaymentID",
  "trackId": "3423423",
  "transId": "201931951332346",
  "ref": "935110000001",
  "authRespCode": "00",
  "authCode": "000000",
}]
```

```
"cardType": "Visa"  
}]
```

Bank Hosted Integration Flow (Apple Pay)

This section illustrates how you can integrate the bank hosted flow on your website application.



1. User visits the merchant application and creates order.
2. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token.
3. ARB Payment gateway internally validates the request.
 - In case of successful validation, ARB PG provides [Payment ID and Payment Page URL](#) in the response.
 - Merchant needs to [frame the payment page URL with Payment ID](#), Hence the ARB payment page is displayed with Apple Pay button.
 - In case of failure, ARB PG provides **Error Code** and **Description**.

Note: If merchant notification is disabled, then ARB Payment gateway will provide the [final response in URL redirection](#).

4. User click the Apple Pay button and selects the required card details from apple wallet and performs apple authorization.
5. The ARB PG application process the transaction and returns the transaction response to the merchant site.
6. Merchant server calls the transaction status API to verify the transaction response.
7. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	response URL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
14	langid	O	Alphabetic	Language ID . Based on language ID arabic language will be displayed on payment page. Value should be 'ar' or 'AR' for arabic language .

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
```

```
"id": "IPAY1CR6qZF7q6w",  
  
"trandata": "<encrypted trandata> ",  
  
"responseURL": "https://merchantpage/PaymentResult.jsp",  
  
"errorURL": "https://merchantpage/PaymentResult.jsp"  
}]
```

Trandata will contain below parameters **encrypted** with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource Key.

Plain Trandata:

```
[{  
  
  //Mandatory Parameters  
  
  "amt": "12.00",  
  
  "action": "1",          // 1 - Purchase , 4 - Authorization  
  
  "password": "q@a68O$27@JLkcK",  
  
  "id": "IPAY1CR6qZF7q6w",  
  
  "currencyCode": "682",  
  
  "trackId": "12345656789",  
  
  "responseURL": "https://merchantpage/PaymentResult.jsp",  
  
  "errorURL": "https://merchantpage/PaymentResult.jsp",  
  
  //Optional Parameters  
  
  "udf1": "udf1text",  
  
  "udf2": "udf2text",  
  
  "udf3": "udf3text",  
  
  "udf4": "udf4text",  
  
  "udf5": "udf5text",  
}]
```



```
"langid":"ar",  
}]
```

MADA Mandatory Parameters

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response.

				Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanumeric	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.

Sample JSON request - Request from Merchant to ARB PG

```
{
  //MADA Mandatory Parameters
  "amt": "12.00",
  "action": "1", // 1 - Purchase , 2 - Authorization
  "password": "q@a680$27@JLkcK",
  "id": "IPAYlCR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "12345656789",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp",
  //Optional Parameters
  "udf1": "udf1text",
  "udf2": "udf2text",
  "udf3": "udf3text",
  "udf4": "udf4text",
  "udf5": "udf5text",
  "langid": "ar",
}]
```

Split Payment or Payout.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanum	Bank Identification Code Min - 8 Max -12
3	iBanNum	C	Alphanum	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//Conditional if Merchant Opted for Split Payment or Payout.

"accountDetails": [

{"bankIdCode": "12345d6f", "iBanNum": "567896743281926354276254",

"beneficiaryName": "AlRajhi Bank Services",

"serviceAmount": "200.00", "valueDate": "20201231" },

{"bankIdCode": "1234ret3", "iBanNum": "987656743281926354276254",

"beneficiaryName": "DIGITAL CO",

"serviceAmount": "300.00", "valueDate": "20201231" }] ,
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
  },
```

Airline

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	airline	C	JSON Object	Conditional - for Airline Merchant
1.1	bookingReference	C	Alphanumeric	The booking reference number
1.1.1	itinerary	C	JSON Object	Conditional - for Airline Merchant
1.1.1.1	leg	C	JSON Array	Conditional - for Airline Merchant
1.1.1.1.1	carrierCode	C	Alphanumeric	The carrier code for the leg
1.1.1.1.2	departureAirport	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.3	departureDate	C	Alphanumeric	The departure date for the leg
1.1.1.1.4	departureTime	C	Alphanumeric	The departure airport for the passenger
1.1.1.1.5	destinationAirport	C	Alphanumeric	The destination airport for the leg
1.1.1.1.6	destinationArrivalDate	C	Alphanumeric	The arrival date for the leg
1.1.1.1.7	destinationArrivalTime	C	Alphanumeric	The arrival time for the leg
1.1.1.1.8	fareBasis	C	Alphanumeric	The fare basis for the leg
1.1.1.1.9	flightNumber	C	Alphanumeric	The flight number for the leg
1.1.1.1.10	travelClass	C	Alphanumeric	The class of service for the leg
1.1.1.2	numberInParty	C	Alphanumeric	
1.1.1.3	originCountry	C	Alphanumeric	The origin Country of the itinerary
1.1.2	passenger	C	JSON Array	Conditional - for airline merchant
1.1.2.1	firstName	C	Alphanumeric	The passenger first name
1.1.2.2	lastName	C	Alphanumeric	The passenger last name
1.1.3	ticket	C	JSON Object	Conditional - for airline merchant
1.1.3.1	issue	C	JSON Object	Conditional - for airline merchant
1.1.3.1.1	carrierCode	C	Alphanumeric	Code of the airline that issuing the ticket
1.1.3.1.2	carrierName	C	Alphanumeric	Name of the airline that is issuing the ticket.
1.1.3.1.3	travelAgentCode	C	Alphanumeric	Code of the Travel Agent that issuing the ticket
1.1.3.1.4	travelAgentName	C	Alphanumeric	Name of the Travel Agent that issuing the ticket
1.1.3.2	totalFare	C	Numeric	Ticket Total Fare

1.1.3.3	totalFees	C	Numeric	Total fee for the ticket.
1.1.3.4	totalTaxes	C	Numeric	Tax portion of the order amount.

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - for airline merchant
```

```
{ "airline": { "bookingReference": "5WPU68", "itinerary": { "leg": [ {
"carrierCode": "MH", "departureAirport": "MNL", "departureDate": "2021-05-11",
"departureTime": "06:50:00Z", "destinationAirport": "KUL",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "10:35:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0805", "travelClass": "B" }, {
"carrierCode": "UL", "departureAirport": "KUL", "departureDate": "2021-05-11",
"departureTime": "15:00:00Z", "destinationAirport": "CMB",
"destinationArrivalDate": "2021-05-11", "destinationArrivalTime": "16:05:00Z",
"fareBasis": "BOWMPH6", "flightNumber": "0315", "travelClass": "B" } ],
"numberInParty": "1", "originCountry": "PHL" }, "passenger": [ { "firstName":
"KAI MR", "lastName": "QIAN" } ], "ticket": { "issue": { "carrierCode": "UL",
"carrierName": "SRILANKANAIRLINES", "travelAgentCode": "91401483",
"travelAgentName": "MANUL08AE" }, "totalFare": "54918.00", "totalFees":
"59518.00", "totalTaxes": "4600.00" } } }
```

Initial Response - Payment ID and Payment Page URL

Attributes - Initial Response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Sample JSON Response - Initial Response from ARB PG to Merchant

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based status and result fields as mentioned below.

Success:

```
[{
  "status": "1",
  "result": "100201931620827468:https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm", //Payment ID:Paymentpage URL
  "error": null,
  "errorText": null }]
```

Failure:

```
[{
  "status": "2",
```

```
"error": " IPAY0100124",  
"errorText": "Problem occurred while validating transaction data",  
"result": null  
}]
```

Framing Payment URL

After Initial Response from ARB PG, merchant needs to frame the payment page URL like the below sample.

<https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm?PaymentID=100201931620827468>

Final Response – Transaction Status

Attribute - Final URL redirection response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique payment Id generated by PG and merchant can use this ID to match the response from PG
2	trandata	C	Alphanum	All the below response parameters encrypted and send the encrypted value in trandata
3	error	C	Alphanum	If any error, PG will send the error code
4	errorText	C	Alphanum	If any error, PG will send the error description

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along

S. No	Fields	M/C/O	Field Type	Description
				with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values

S. No	Fields	M/C/O	Field Type	Description
				1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{
  //Redirection Parameters
  "paymentId":"100201935166676976",
  "trandata":"<encrypted trandata>",
  "error":"","
  "errorText":""
}]
```

Plain Trandata:

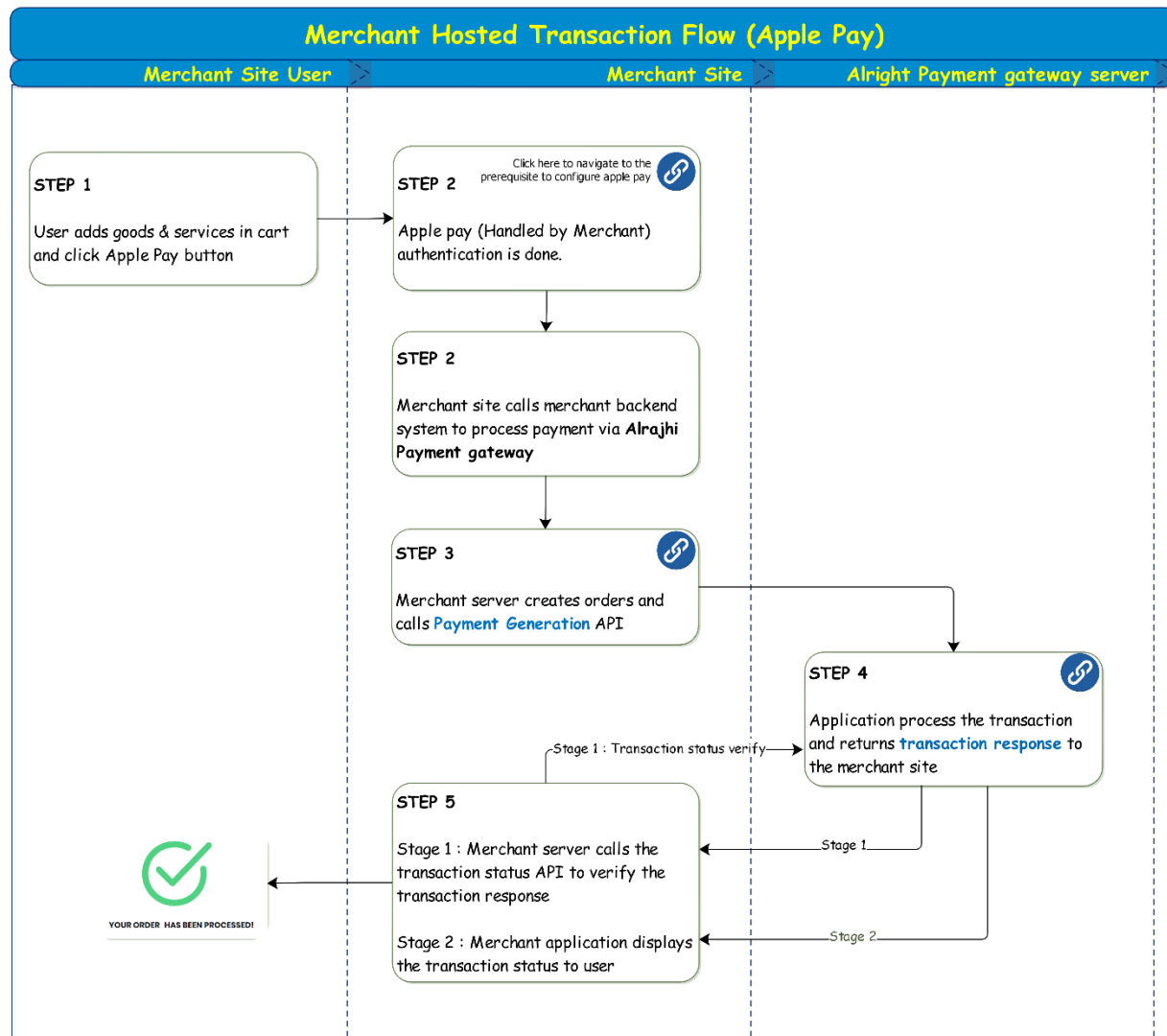
Trandata will contain below parameters encrypted.

```
[{
  "paymentId":"100201935166676976",
  "result": "CAPTURED",
  "transId":201935166561122,
  "ref":"935110000001",
  "date":"1217",
```

```
"trackId":"1003383844",  
  
"udf1":"","  
  
"udf2":"","  
  
"udf3":"8870091137",  
  
"udf4":"FC",  
  
"udf5":"Tidal5",  
  
"amt":"70.0",  
  
"authRespCode","00",  
  
"authCode":"000000",  
  
"cardType":"Visa",  
  
"actionCode":"1"  
  
}]
```

Merchant Hosted Transaction Flow (Apple Pay)

This section illustrates how you can integrate the merchant hosted flow (Apple Pay) on your website application.



1. User visits the merchant application and creates order.
2. In the merchant page, the **Apple Pay** button is displayed. User clicks the **Apple Pay** button and select required card from apple wallet. (The **Apple Pay** button in merchant page is handled by the merchants by following the [prerequisites](#) steps)

3. Merchant application backend server calls [Payment Token Generation API](#) to get the transaction token and to process payment via Alrajhi Payment gateway
4. After authorization, the ARB PG application process the transaction and returns the transaction response to the merchant site. The ARB Payment gateway will provide the [final response in URL redirection](#).
5. Merchant server calls the transaction status API to verify the transaction response.
6. Finally, the merchant application displays the transaction status to user.

Request - Payment Token Generation API

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanum	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanum	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanum	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanum	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a

				transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	expYear	M	Numeric	Expiry year of card
13	expMonth	M	Numeric	Expiry month of card
14	member	M	Alphanum	Card holder name
15	cvv2	M	Numeric	CVV of the card
16	cardNo	M	Numeric	Cardholders card number
17	cardType	M	Alphanum	Card type Ex : Credit card – C, Debit Card – D

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>"
}]
```


Plain Trandata:

```
[{  
  
  //Mandatory Parameters  
  
  "amt": "12.00",  
  
  "action": "1",          // 1 - Purchase , 4 - Authorization  
  
  "password": "q@a680$27@JLkcK",  
  
  "id": "IPAY1CR6qZF7q6w",  
  
  "currencyCode": "682",  
  
  "trackId": "123456",  
  
  "expYear": "2022",  
  
  "expMonth": "12",  
  
  "member": "cardholdername",  
  
  "cvv2": "212",  
  
  "cardNo": "5453*****5539",  
  
  "cardType": "C",  
  
  //Optional Parameters  
  
  "udf1": "udf1text",  
  
  "udf2": "udf2text",  
  
  "udf3": "udf3text",  
  
  "udf4": "udf4text",  
  
  "udf5": "udf5text",  
  
}]
```

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Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
  },
```

Final Response – Transaction Status

The ARB payment gateway verifies the transaction and returns the response to the same request.

Attribute - Final response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	Alphanum	If any error during processing, PG will provide the error code
3	errorText	C	Alphanum	If any error during processing, PG will provide the error description
4	status	M	Alphanum	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf4	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	udf5	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
12	amt	M	Numeric	Transaction amount
13	authRespCode	M	Numeric	Auth response code provided by PG
14	authCode	M	Numeric	6 digit authorization code received from switch
15	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

The ARB payment gateway verifies the transaction and returns the response to the same request.

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
  "error": null,
  "errorText": null
}]
```

Plain Trandata:

Trandata will contain below parameters encrypted.

```
[{
  "paymentId": -1,
  "result": "CAPTURED",
  "amt": "10.55",
  "date": "1221",
  "ref": "935110000001",
  "udf1": "udf1text",
  "udf2": "udf2text",
  "udf3": "udf3text",
  "udf4": "udf4text",
  "udf5": "udf5text",
  "trackId": "3423423",
  "transId": "1242345345234",
}
```

```
"authRespCode": "00",  
"authCode": "000000",  
"cardType": "Visa",  
"actionCode": "1"  
}]
```

Pre-Authentication Transactions (Airline)

Pre-authorization (pre-auth) is a process of authenticating and temporary blocking of the certain amount available on the card, based on the card details provided at the time of booking.

1. User visits the merchant application and creates order.
2. User enters the payment card details.
3. Call the Pre-auth API to block amount from the user's account.
4. Merchant proceed with order/service completion.
5. Once the fulfilment is complete, call Capture API with the final amount to capture the amount from user's card and then notify the payment status to the user.

Request

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	expYear	M	Numeric	Expiry year of card
12	expMonth	M	Numeric	Expiry month of card
13	member	M	Alphanumeric	Card holder name
14	cvv2	M	Numeric	CVV of the card
15	cardNo	M	Numeric	Cardholders card number
16	cardType	M	Alphanumeric	Card type Ex : Credit card – C, Debit Card – D
17	eci	M	Alphanumeric	Electronic commerce indicator received from third Party MPI
18	ucaf	M	Alphanumeric	UCAF received from third Party MPI
19	cavv	M	Alphanumeric	CAVV received from third Party MPI

Below is the sample encrypted request from Merchant to PG

```
[{
  //Mandatory Parameters
  "id":"IPAY1CR6qZF7q6w",
  "trandata":"<encrypted trandata>"
}]
```

Below are the plain Trandata request

```
[{
  //Mandatory Parameters
  "amt":"12.00",
```

```
"action": "1",  
  
"password": "q@a680$27@JLkcK",  
  
"id": "IPAY1CR6qZF7q6w",  
  
"currencyCode": "682",  
  
"trackId": "123456",  
  
"expYear": "2022",  
  
"expMonth": "12",  
  
"member": "cardholdername",  
  
"cvv2": "212",  
  
"cardNo": "5453*****5539",  
  
"cardType": "C",  
  
"eci": "7",  
  
"ucaf": "1",  
  
"cavv": "JcboZXndOg40CB ECC2BGbheAAAA=",  
  
  
//Optional Parameters  
  
"udf1": "udf1text",  
  
"udf2": "udf2text",  
  
"udf3": "udf3text",  
  
"udf4": "udf4text",  
  
"udf5": "udf5text",
```

Credit Card Installment Transactions

Detailed description of Plain Trandata request parameters

S.No	Fields	M/C /O	Field Type	Description
1	udf6	C	Alphanum	Credit Card Installment Short Code
2	udf7	C	Alphanum	Credit Card Installment Type

Sample JSON request - Request from Merchant to ARB PG

```
//conditional - Credit Card Installment Transactions

"udf6": "12345",

"udf7": "C",
```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanum	Bank Identification Code Min - 8 Max - 12
3	iBanNum	C	Alphanum	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount

6	valueDate	C	Numeric	Value Date Format: YYYYMMDD
---	-----------	---	---------	-----------------------------

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails":[
  {
    "bankIdCode": "12345d6f",
    "iBanNum": "567896743281926354276254",
    "beneficiaryName": "AlRajhi Bank Services",
    "serviceAmount": "200.00",
    "valueDate": "20201231"
  },
  {
    "bankIdCode": "1234ret3",
    "iBanNum": "987656743281926354276254",
    "beneficiaryName": "DIGITAL CO",
    "serviceAmount": "300.00",
    "valueDate": "20201231"
  }
],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    {"IDType":"01","IDNumber":"1072587916","billNumber":"00100100018","billType":"P
OSTPAID","billerID":"169","billAmount":5,"transactionType":"ADVANCE"
}
```

Response

Once ARB payment gateway verifies the transaction and returns the response to the same request.

Response from ARB Payment Gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	Alphanumeric	If any error during processing, PG will provide the error code
3	errorText	C	Alphanumeric	If any error during processing, PG will provide the error description
4	status	M	Alphanumeric	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanumeric	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Below is the sample encrypted response from PG to Merchant

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
  "error": null,
  "errorText": null
}]
```

```
}]
```

Plain trandata in response

```
[{  
  "paymentId":-1,  
  "result":"CAPTURED",  
  "amt":"10.55",  
  "date":"1221",  
  "ref":"935110000001",  
  "udf1":"udf1text",  
  "udf2":"udf2text",  
  "udf3":"udf3text",  
  "udf4":"udf4text",  
  "udf5":"udf5text",  
  "trackId","3423423",  
  "transId":"1242345345234",  
  "authRespCode":"00",  
  "authCode":"000000",  
  "cardType":"Visa",  
  "actionCode":"1"  
}]
```


Invoice Payment Transaction Flow

1. Customer visits the merchant site and selects the merchandise and confirms to pay using the payment gateway.
2. Merchant redirects the customer with the Invoice transaction data to the payment gateway.
3. Payment Gateway verifies the invoice transaction request from the merchant and generates the invoice Payment URL based on the invoice type.
4. Payment gateway will provide the response to merchant with invoice Payment URL.
5. Merchant to redirect the Payment URL to Payment Gateway.
 - There are two types of invoice transactions namely Dedicated and Open invoice.
 - a) If the initiated transaction type is dedicated, Merchant need to send the buyer details in the request.
PG presents the hosted payment page with buyer details to customer and prompts the customer to enter the card credentials.
 - b) If the initiated transaction type is open then merchant no need to send buyer details.
6. Payment gateway will get the buyer detail and process the transaction with the respective schemes.
7. Once the transaction complete, PG will send the response to merchant and customer via mail & SMS.

Request - Payment Token Generation API

Request from Merchant to ARB Payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.

Dedicated Invoice detailed description of Plain Trandata request parameters.

S. No	Fields	M/C/O	Field Type	Description
1	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
2	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
3	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
4	invoiceId	M	Alphanumeric	50-digit unique Id.
5	itemDesc	M	Alphanumeric	500-digit Item Description.
5	invoiceType	M	Alphabetic	1-digit value "D" or "O" (D-dedicated invoice / O – Open Invoice)
6	buyerName	M	Alphabetic	50-digit Buyer Name.
7	amt	M	Numeric	Transaction amount.
8	email	M	alphanumeric	Buyer mail Id.
9	mobile	M	Numeric	Buyer Mobile Number.
10	expiryDate	O	Alphanumeric	Expiry Date of the payment Link. Date Format: "dd-MM-yyyy hh:mm:ss"

Open Invoice detailed description of Plain Trandata request parameters.

S. No	Fields	M/C/O	Field Type	Description
1	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
2	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
3	currencyCode	M	Numeric	3-digit currency code of KSA. Ex: "682"
4	invoiceId	M	Alphanumeric	50-digit unique Id.
5	itemDesc	M	Alphanumeric	500-digit Item Description
5	invoiceType	M	Alphabetic	1-digit value "D" or "O" (D-dedicated invoice / O – Open Invoice)
7	amt	O	numeric	Transaction amount. Ex: "100.00"
10	expiryDate	O	Alphanumeric	Expiry Date of the payment Link. Date Format: "dd-MM-yyyy hh:mm:ss"

Initial Response - Payment ID and Processing Page URL

ARB Payment gateway internally validates the request and gives invoice payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based on the status and result fields as mentioned below.

Initial Response from PG to Merchant for Dedicated and Open Invoice

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanu m	It contains invoice Payment URL if the validation success
3	error	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error code
4	errorT ext	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error description

```

Success:

[ {
  "status": "1",
  "result": "https://securepayments.alrajhibank.com.sa/mrchpt1/iP.htm?Id=Y3FT9"
//Invoice Payment Page URL} ]

Failure:

[ {
  "status": "2",
  "error": " IPAY0100124",
  "errorText": " Problem occurred while validating transaction data"
} ]

```

- If success, Merchant needs to redirect the invoice payment page URL like the below sample
<https://securepayments.alrajhibank.com.sa/mrchptl/iP.htm?Id=Y3FT9>
- Once merchant redirects the link, ARB Payment Gateway shows the invoice Payment Page to customer based on the invoice type.

Final Response

After validating the customer card details then ARB Payment gateway will provide the final response to merchant and customer via mail & SMS.

Webhook Merchant Notification Flow

For Webhook transactions, ARB PG will send risk based, authentication based or transaction/payment based notification request to merchant as below :

Detailed Notification Request from PG to Merchant for Webhook

S. No	Fields	M/C/O	Field Type	Description
1	type	M	Alphanum(20)	Type of Notification to the Merchant
2	payLoad	M	JSON Object	Payment Status Info.All the Payment Gateway related information's available in the payload fields (Further table below)
3	result	M	JSON Object	Transaction Status - Contains the Transaction status for the transaction (table below)
4	responseURL	M	Alphanum	Merchant Response URL

payLoad Detailed Field description

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Alphanum[20]	Payment ID - Unique Id generated by Payment Gateway, this is the same ID that Payment Gateway had provided along with the PaymentPage URL for the initial request
2	transId	M	Alphanum[19]	Transaction ID - Unique Transaction ID generated by Payment Gateway
3	ref	M	Alphanum[12]	RRN - The reference number of the transaction known to payment scheme. This number or series of letters is used for referential purposes by some acquiring/issuing bank/institutions and should be stored properly
4	paymentTimestamp	M	Alphanum	Timestamp - ARB request initiation Timestamp
5	trackId	M	Alphanum[255]	Merchant Track ID - Track ID value that was sent by merchant in the Purchase request

S. No	Fields	M/C/O	Field Type	Description
6	udf1	O	Alphanum[255]	User Defined Field 1 - Same udf value that merchant has sent in the initial transaction request to Payment Gateway
7	udf2	O	Alphanum[255]	User Defined Field 2 - Same udf value that merchant has sent in the initial transaction request to Payment Gateway
8	udf3	O	Alphanum[255]	User Defined Field 3 - Same udf value that merchant has sent in the initial transaction request to Payment Gateway
9	udf4	O	Alphanum[255]	User Defined Field 4 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
10	udf5	O	Alphanum255]	User Defined Field 5 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
11	udf6	O	Alphanum [255]	User Defined Field 6 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
12	udf7	O	Alphanum[255]	User Defined Field 7 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
13	udf8	O	Alphanum[255]	User Defined Field 8 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
14	udf9	O	Alphanum255]	User Defined Field 9 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway
15	udf10	O	Alphanum[255]	User Defined Field 10 -Same udf value that merchant has sent in the initial transaction request to Payment Gateway

S. No	Fields	M/C/O	Field Type	Description
16	amt	M	Numeric with decimal places - Max 12 digit with 2 digit decimal places	Transaction amount - Transaction Amount as sent by merchant in the request
17	authRespCode	O	Alphanum[3]	Authorization Response Code – Authorization code received from the issuer bank
18	authCode	O	Alphanum[6]	Authorization Code - The resulting authorization number of the transaction from the issuing bank. This number or series of letters is used for referential purposes by some acquiring/issuing bank/institutions and should be stored properly
19	actionCode	C	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)
20	origTransactionID	C	Alphanumeric	It is applicable only for Supporting Transactions action code like 2,3,5,9 and 14. It refers to the "transId" of the Source/Original transaction

result Parameter detailed description

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Alphanum[30]	Status of the Payment
2	error	M	Alphanum[255]	Error code of the transaction
3	errorText	M	Alphanum[255]	Error description

Notification Response from Merchant to PG for Webhook

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	Merchant Notification acknowledgment status

Webhook from PG:

PAYMENT

```
[{"result": [{"status": "CAPTURED"}], "responseURL":
"http://172.22.0.121:9207/MerchDemoREST/NotifyResponse.htm", "payload":
[{"date": "0415", "authRespCode": "00", "authCode": "623666",
"transId": "202110527755152", "trackId": "04d26d6c-ca9f-4ac3-b65d-
6c5bff7af1a6", "udf5": "Select",
"udf10": "null", "amt": 10, "udf3": "null", "udf4": "null", "udf1": "null", "udf2":
"null", "ref": "110533003557", "udf9": "null", "udf8": "null", "paymentTimes
tamp": "2021-04-15T02:58:30.448Z", "actionCode": "1"}], "type": "PAYMENT"}]
```

PAYMENT FAILURE

```
[{"result": [{"errorText": "00", "error": "IPAY00001", "status": "NOT
CAPTURED"}], "responseURL":
"http://172.22.0.121:9207/MerchDemoREST/NotifyResponse.htm", "payload":
[{"date": "0414", "ref": "110423000769", "authRespCode": "00", "authCode": "54
4377", "paymentId": "700202110427042639", "transId": "202110472940732", "track
Id": "e061c0ec-b32e-456b-9ac2-
8196fb8f16ec", "amt": 100, "paymentTimestamp": "2021-04-
13T22:48:21.082Z", "actionCode": "1"}], "type": "PAYMENT FAILURE"}]
```

Purchase HOST TIME OUT Webhook notification request (Action code will be Change in payload for each segments as per document.)

```
[{"result":[{"status":"HOST
TIMEOUT"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi
/PayNotification","payload":[{"date":"0527","authRespCode":"000","authC
ode":"163025","trackId":"ec4e181a-16e9-406e-a132-
bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Ma
da","udf6":"10018496416","udf10":"","amt":"5.0","udf3":"null","udf4":"n
ull","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","pay
mentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"1","pa
ymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

RISK FAILURE

```
[{"result":[{"errorText":"X","error":"IPAY00002","status":"IPAY0100045
- DENIED BY RISK"}],"responseURL":
"http://172.22.0.121:9207/MerchDemoREST/NotfiyRspnse.htm","payload":
"[{"authRespCode":"X","trackId":"605965227",
"transId":202101752650620,
"udf5":"UDF5","cardType":"Visa","udf6":"UDF6","udf10":"UDF10","amt":35.
0,"udf3":"UDF3","udf4":"UDF4","udf1":"UDF1","udf2":"UDF2","result":"IPA
Y0100045 - DENIED BY
RISK","udf9":"UDF9","paymentId":"600202101747330225",
"udf7":"UDF7","udf8":"UDF8","actionCode":"1"}],"type":"RISK FAILURE"}]
```

Refund Webhook notification request

```
[{"result":[{"status":"CAPTURED"}],"responseURL":"https://partner.se.co
m.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","au
thRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-
a132-
bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Ma
da","udf6":"10018496416","udf10":"FINALCAPTURE","amt":"5.0","udf3":"nul
l","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9
":"null","origTransactionID":"202214779988486","paymentId":700202214780
010325,"udf7":"","udf8":"null","actionCode":"2","paymentTimestamp":"202
2-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

VOIDED Webhook notification request

```
[{"result":[{"status":"VOIDED"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","authRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-a132-bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Mada","udf6":"10018496416","udf10":"","amt":"5.0","udf3":"null","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","origTransactionID":"202214779988486","paymentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"3","paymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

NOT VOIDED Webhook notification request

```
[{"result":[{"status":"NOT VOIDED"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","authRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-a132-bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Mada","udf6":"10018496416","udf10":"","amt":"5.0","udf3":"null","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","origTransactionID":"202214779988486","paymentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"3","paymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

Auth Webhook notification request

```
[{"result":[{"status":"APPROVED"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","authRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-a132-bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Mada","udf6":"10018496416","udf10":"FINALCAPTURE","amt":"5.0","udf3":"null","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","paymentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"4","paymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

Auth Webhook notification request

```
[{"result":[{"status":"NOT APPROVED"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","authRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-a132-bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Mada","udf6":"10018496416","udf10":"FINALCAPTURE","amt":"5.0","udf3":"null","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","paymentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"4","paymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

Capture Webhook notification request

```
[{"result":[{"status":"APPROVED"}],"responseURL":"https://partner.se.com.sa/RESTAdapter/AlRajhi/PayNotification","payload":[{"date":"0527","authRespCode":"000","authCode":"163025","trackId":"ec4e181a-16e9-406e-a132-bdfda42e4e91","transId":202214720027040,"udf5":"TrackID","cardType":"Mada","udf6":"10018496416","udf10":"FINALCAPTURE","amt":"5.0","udf3":"null","udf4":"null","udf1":"null","udf2":"null","ref":"214721008795","udf9":"null","origTransactionID":"202214779988486","paymentId":700202214780010325,"udf7":"","udf8":"null","actionCode":"5","paymentTimestamp":"2022-05-27T08:27:34.155Z"}],"type":"PAYMENT"}]
```

Note : "origTransactionID": "202101747330225" – Condition Parameter

Merchant should send acknowledgment to PG as below:

Merchant Response Sample:

```
[{  
  
  "status": "1"  
  
}]
```

If PG doesn't receive any acknowledgement from merchant , PG will keep sending the notification request till the acknowledgement is received .

Merchant Notification Flow for Bank Hosted Transactions

If Notification enabled at Merchant level,

1. Payment gateway will initiate REST call to send the transaction status notification request to merchant and wait for the acknowledgement.
2. If PG receives the acknowledgment from merchant, then Payment Gateway logs the response and provides the transaction update to Cortex system for processing the settlement and Payment Gateway sends the final response to merchant. The final response will be URL redirection.
3. Merchant displays transaction result to customer.
4. In case, if PG does not receive the acknowledgment from merchant, then PG will initiate the VOID transaction to respective scheme to reverse the transaction and there is no payment advise initiated to Cortex.

If Notification disabled at Merchant level,

1. Payment Gateway logs the response and provides the transaction update to Cortex system for processing the settlement and Payment Gateway sends final response to merchant and this will be URL redirection.
2. Merchant displays transaction result to customer.

Request - Notification Generation API

Notification request from PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by Payment gateway. Merchant can store the payment ID to match the final URL redirection response
2	trandata	M	Alphanumeric	All the below response parameters encrypted and send the encrypted value in trandata

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanumeric	Transaction status
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in

S. No	Fields	M/C/O	Field Type	Description
				the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authResp Code	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	actionCode	C	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Below is the sample merchant notification request and response,

Notification Request from ARB payment gateway to Merchant:

```
[{
  "paymentId":100201935044735860,
  "trandata":"<encrypted trandata>"
}]
```

Plain Trandata:

```
[{"paymentId":100201935166676976,
  "result":"CAPTURED",
  "ref":"935110000001",
  "transId":"201935166561122",
  "date":"1217",
```



```
"trackId":"1003383844",  
  
"udf1":"","  
  
"udf2":"","  
  
"udf3":"8870091137",  
  
"udf4":"FC",  
  
"udf5":"Tidal5",  
  
"amt":"70.0",  
  
"authRespCode","00",  
  
"authCode":"000000",  
  
"actionCode":"1"  
  
}]
```

Response - Acknowledgement

Notification response from Merchant to ARB Payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	Status should be '1'
2	result	M	Alphanumeric	Merchant response URL for which PG will provide the final URL redirection response

Acknowledgment response from merchant to PG:

```
[{  
  "status": "1",  
  "result": "https://merchant.com/finalresultURL"  
}]
```

Final response to merchant

If ARB payment gateway receives the acknowledgement from merchant, PG logs the response and provide the response back to merchant. This will be URL redirection.

Response from PG to merchant.

```
[{  
  "paymentId": "100201935166676976",  
  "trandata": "<encrypted trandata>",  
  "Error": "",  
  "ErrorText": ""  
}]
```

Plain Trandata:

```
[{  
  "paymentId": "100201935166676976",  
  "result": "CAPTURED",  
  "ref": "935110000001",  
  "transId": "201935166561122",  
  "date": "1217",  
  "trackId": "1003383844",  
  "udf1": "",  
  "udf2": "",  
  "udf3": "8870091137",  
  "udf4": "FC",  
}]
```

```
"udf5": "Tidal5",  
  
"amt": "70.0",  
  
"authRespCode": "00",  
  
"authCode": "000000",  
  
"actionCode": "1"  
}]
```

Response – No Acknowledgement

If there is no acknowledgement from Merchant After initiating the merchant notification, PG will wait for the response based on the configured time and no acknowledgement from merchant then PG will void the transaction and update the response back to merchant error URL received in the initial API request.

```
[{  
  "paymentId": "100201935166676976",  
  "Result": "Voided",  
  "error": "IPAY0200025",  
  "errorText": "!ERROR!-IPAY0200025 - Problem occurred while getting  
merchant acknowledgement & transaction reversed",  
  "trackId": "123456",  
  "amt": "12.0"  
}]
```

Issuer Country API (Card Bin Check)

Perform a BIN lookup using this API call. Merchant can send card bin number to PG and PG will respond with card issuer country details.

End Point: <https://securepayments.alrajhibank.com.sa/pg/payment/bincheck.htm>

Request from Merchant to PG

Detailed description of Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	Bin	M	Numeric	Card Bin First six digits of the card Number
2	Id	M	Alphanum	Tranportal ID. Merchant can download the same in merchant portal
3	Password	M	Alphanum	Tranportal password. Merchant can download the same in merchant portal.

Below is the sample request from Merchant to PG

```
{  
  "bin": "515735",  
  "id": "IPAYaQ03cVHs2q",  
  "password": "1$Q$VP73S2gycg@"  
}
```

Response from PG to Merchant

Detailed description of Response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	Bin	C	Numeric	Card Bin First six digits of the card Number
2	Bank	C	Alphanu m	Bank name for the card bin issuer
3	Country	C	Alphanu m	Country name for the card bin
4	Countryco de	C	Alphanu m	Country Code for the card bin
5	Card	C	Alphanu m	Card Type . Ex : MasterCard /Visa
6	errorText	C	Alphanu m with special chars	If validation failed, then payment gateway will provide the respective error description
7	Error	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error code
8	Status	M	Numeric	If the request validation is success, then status will be '1'. If the validation failed, then status will be '2'

Below is the sample response from PG to Merchant (Success Case)

```
{
  "country": "SAUDI ARABIA",
  "bank": "Al Bank Al Saudi Al Fransi",
  "bin": "446404",
  "countrycode": "SAU",
  "card": "MADA",
  "status": "1"
}
```

Below is the sample response from PG to Merchant (Error Case)

```
{
  "errorText": "!ERROR!-IPAY0100380-Bin number should be of 6 digits.",
  "error": "IPAY0100380",
  "status": "2"
}
```


Bank Hosted Recurring Transaction Flow (3D Secure)

Recurring Billing API methods enable merchant user (via payment gateway) to manage regular subscription payments.

Request - Payment Token Generation API

Request from Merchant to ARB Payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanum	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanum	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanum	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization : 4
3	password	M	Alphanum	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanum	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction

				response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanumeric	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.

Request from Merchant to ARB Payment gateway:

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata> ",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata:

Trandata will contain below parameters encrypted with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource Key

```
[{  
  
  //Mandatory Parameters  
  
  "amt": "12.00",  
  
  "action": "1",      // 1 - Purchase , 2 - Authorization  
  
  "password": "q@a680$27@JLkcK",  
  
  "id": "IPAYlCR6qZF7q6w",  
  
  "currencyCode": "682",  
  
  "trackId": "12345656789",  
  
  "responseURL": "https://merchantpage/PaymentResult.jsp",  
  "errorURL": "https://merchantpage/PaymentResult.jsp",  
  
  "udf1": "1", //To identify Recurring based transactions  
  
  "udf2": "SI",      //To identify Recurring based transactions  
  
  //Optional Parameters  
  
  "udf3": "udf3text",  
  
  "udf4": "udf4text",  
  
  "udf5": "udf5text",  
  
  "langid": "ar",
```

Split Payment or Payout.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanum	Bank Identification Code Min - 8 Max -12
3	iBanNum	C	Alphanum	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//Conditional if Merchant Opted for Split Payment or Payout.

"accountDetails": [

{"bankIdCode": "12345d6f", "iBanNum": "567896743281926354276254",

"beneficiaryName": "AlRajhi Bank Services",

"serviceAmount": "200.00", "valueDate": "20201231" },

{"bankIdCode": "1234ret3", "iBanNum": "987656743281926354276254",

"beneficiaryName": "DIGITAL CO",

"serviceAmount": "300.00", "valueDate": "20201231" }] ,
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15 Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant

    "billingDetails" :

    { "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE"
  },
```

Initial Response - Payment ID and Payment Page URL

Initial Response from PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

- ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response in case of successful validation, if failure then error code and description will be provided. The below response will be in plain format and there is no encryption for the below. Merchant can directly parse the response-based status and result fields as mentioned below.

Success:

```
[{
  "status": "1",
  "result": "100201931620827468:https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm", //Payment ID:Paymentpage URL
  "error": null,
  "errorText": null }]
Failure:
[{
  "status": "2",
  "error": " IPAY0100124",
  "errorText": "Problem occurred while validating transaction data",
```

```
"result": null  
}]
```

Framing Payment URL

- If success, Merchant needs to frame the payment page URL like the below sample

```
https://securepayments.alrajhibank.com.sa/pg/paymentpage.htm?PaymentID=100201931620827468
```

- If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

Final Response – Transaction Status

Final URL redirection response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique payment Id generated by PG and merchant can use this ID to match the response from PG
2	trandata	C	Alphanu m	All the below response parameters encrypted and send the encrypted value in trandata
3	error	C	Alphanu m	If any error, PG will send the error code
4	errorText	C	Alphanu m	If any error, PG will send the error description

Detailed description of Plain trandata parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanu m	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	custid	M	Alphanumeric	Customer ID needs to be sent in the subsequent Merchant Initiating recurring transactions
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Sample JSON Response - Final

```
//Redirection Parameters
```

- "paymentId": "100201935166676976",
- "trandata": "<encrypted trandata>",
- "error": "",
- "errorText": ""

Plain Trandata:

```
[{  
  "paymentId": "100201935166676976",  
  "result": "CAPTURED",  
  "transId": 201935166561122,  
  "ref": "935110000001",  
  "date": "1217",  
  "trackId": "1003383844",  
  "udf1": "",  
  "udf2": "",  
  "udf3": "8870091137",  
  "udf4": "FC",  
  "udf5": "Tidal5",  
  "amt": "70.0",  
  "authRespCode": "00",  
  "authCode": "000000",  
  "cardType": "Visa",  
  "custid": "202014785236784",  
  "actionCode": "1"  
}]
```

Merchant Hosted Recurring Transaction Flow (3D Secure)

Recurring Billing API methods enable merchant user to manage regular subscription payments.

Request - Payment Token Generation API

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain trandata request parameters

S.No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	password	M	Alphanumeric	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response.

S.No	Fields	M/C /O	Field Type	Description
				Merchant should ensure that field left blank when no data needs to be passed.
8	udf2	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
9	udf3	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field left blank when no data needs to be passed.
10	udf5	O	Alphanum	The user (merchant) defines these fields. The field data passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
12	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
13	expYear	M	Numeric	Expiry year of card
14	expMonth	M	Numeric	Expiry month of card
15	member	M	Alphanum	Card holder name
16	cvv2	M	Numeric	CVV of the card
17	cardNo	M	Numeric	Cardholders card number
18	cardType	M	Alphanum	Card type Ex : Credit card – C, Debit Card – D
19	browserLanguage	M	Alphanum	Value representing the browser language Returned from "navigator.language" property. Length 1 to 8 characters.
20	browserColorDepth	M	Alphanum	Value representing the bit depth of the colour palette for displaying images, in bits per pixel. Obtained from Cardholder browser using the "screen.colorDepth" property. Length 1 to 2 characters. Values Accepted :

S.No	Fields	M/C /O	Field Type	Description
				1 = 1 bit 4 = 4 bits 8 = 8 bits 15 = 15 bits 16 = 16 bits 24 = 24 bits 32 = 32 bits 48 = 48 bits
21	browserScreenHeight	M	Alphanumeric	Total height of the Cardholder's screen in pixels. Value is returned from the screen.height property. Length 1 to 6 characters.
22	browserScreenWidth	M	Alphanumeric	Total width of the cardholder's screen in pixels. Value is returned from the screen.width property. Length 1 to 6 characters.
23	browserJavaScriptEnabled	M	Alphanumeric	Value is returned from the navigator.javaEnabled property. Boolean value.
24	browserTZ	M	Alphanumeric	Time difference between UTC time and the Cardholder browser local time, in minutes. Value is returned from the getTimezoneOffset() method. Length 1 to 5 characters.
25	jsEnabled	M	Alphanumeric	Value whether the java script is enabled in browser or not.

Request from Merchant to ARB Payment gateway:

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>",
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata

Trandata will contain below parameters encrypted with AES algorithm with CBC Mode, PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource key.

```
[{  
  
  //Mandatory Parameters  
  
  "amt":"12.00",  
  
  "action":"1",      // 1 - Purchase , 4 - Authorization  
  
  "password":"q@a68O$27@JLkcK",  
  
  "id":"IPAYlCR6qZF7q6w",  
  
  "currencyCode":"682",  
  
  "trackId":"123456",  
  
  "expYear":"2022",  
  
  "expMonth":"12",  
  
  "member":"cardholdername",  
  
  "cvv2":"212",  
  
  "cardNo","5453*****5539",  
  
  "cardType":"C",  
  
  "responseURL":"https://merchantpage/PaymentResult.jsp",  
  
  "errorURL":"https://merchantpage/PaymentResult.jsp",  
  
  "browserJavaEnabled":"true",  
  
  "browserLanguage":"en",  
  
  "browserColorDepth":"48",  
  
  "browserScreenHeight":"400",  
  
  "browserScreenWidth":"600",  
  
  "browserTZ":"0",
```

```

"jsEnabled":"true",

"udf1":"1", //To identify Recurring based transactions

"udf2":"SI",      //To identify Recurring based transactions

//Optional Parameters

"udf3":"udf3text",

"udf4":"udf4text",

"udf5":"udf5text",

```

Credit Card Installment Transactions

Detailed description of Plain Trandata request parameters

S.No	Fields	M/C	Field Type	Description
1	udf6	C	Alphanum	Credit Card Installment Short Code
2	udf7	C	Alphanum	Credit Card Installment Type

Sample JSON request - Request from Merchant to ARB PG

```

//conditional - Credit Card Installment Transactions

"udf6":"12345",

"udf7":"C",

```

Payout Future

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	accountDetails	C	JSON Array	Conditional if Merchant Opted for Payout future. Split Payment or Payout Details.
2	bankIdCode	C	Alphanumeric	Bank Identification Code Min - 8 Max -12
3	iBanNum	C	Alphanumeric	IBAN Number Min - 24 Max - 35
4	beneficiaryName	C	Alphabetic	beneficiaryName (English Only) Max - 100
5	serviceAmount	C	Numeric	Service Amount
6	valueDate	C	Numeric	Value Date Format: YYYYMMDD

Sample JSON request - Request from Merchant to ARB PG

```
//conditional if Merchant opted for Payout Future.

"accountDetails": [

  { "bankIdCode": "12345d6f", "iBanNum": "567896743281926354276254", "beneficiaryName": "AlRajhi Bank Services", "serviceAmount": "200.00", "valueDate": "20201231" },

  { "bankIdCode": "1234ret3", "iBanNum": "987656743281926354276254", "beneficiaryName": "DIGITAL CO", "serviceAmount": "300.00", "valueDate": "20201231" } ],
```

SADAD

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	transactionType	C	Alphanumeric	Minimum length : 3 , Maximum length : 15

				Transaction Type Allowed Values 'PAY' 'PAY_SAVE' 'ADVANCE' 'PARTIAL_PAYMENT' 'OVER_PAYMENT'
2	billerID	C	Alphanumeric	Minimum length : 1 , Maximum length : 3 Biller ID
3	billAmount	C	Numeric	billAmount
4	billType	C	Alphanumeric	Minimum length : 7 , Maximum length : 8 Bill Type. The allowed values are: - PREPAID : Paid in advance - POSTPAID : Paid at the end
5	billNumber	C	Alphanumeric	Minimum length : 1 , Maximum length : 30 Bill Number as identified by SADAD
6	IDType	C	Alphanumeric	Minimum length : 2 , Maximum length : 2 ID Type 01 National ID Card 02 Iqama 03 Passport
7	IDNumber	C	Alphanumeric	Minimum : 1 , Maximum : 20 ID Number

Sample JSON request - Request from Merchant to ARB PG

```
//conditional -for SADAD merchant
```

```
"billingDetails" :
```

```
{ "IDType": "01", "IDNumber": "1072587916", "billNumber": "00100100018", "billType": "POSTPAID", "billerID": "169", "billAmount": 5, "transactionType": "ADVANCE" }
```

Initial Response - Payment ID and Payment Page URL

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response if the validation success. If failure then, Error code and description will be provided.

Initial Response from PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanu m	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanu m	If validation failed, then Payment gateway will provide the respective error description

Plain Response:

```

Success:

[ {
  "status": "1",
  "result": "700212030953264091:https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700212030953264091&id=r9Ht8R4U6g9dYtY",
  //Payment ID:Payment URL
  "error": null,
  "errorText": null
} ]

Failure:

[ {
  "status": "2",
  "error": " IPAY0100124",

```

```
"errorText": " Problem occurred while validating transaction data",  
"result": null  
}]
```

Framing Payment URL

Merchant needs to frame the payment page URL like the below sample

<https://securepayments.alrajhibank.com.sa/pg/TranportalVbv.htm?paymentId=700112030953264091&id=r9Ht8R4U6g9dYtYg>

Final Response – Transaction Status

Merchant needs to redirect the customer to ARB Payment gateway.

Customer browser will redirect to ACS page and will complete the authentication. PG then process for authorization with the respective schemes. Once payment response received from respective scheme, then ARB Payment gateway returns the response to merchant. This is URL redirection. Below is sample response from ARB PG to merchant,

Final Response from ARB payment gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by Payment gateway. Merchant can store the payment ID to match the final URL redirection response
2	trandata	C	Alphanumeric	All the below response parameters encrypted and send the encrypted value in trandata Ex: <pre>[{"paymentId":100201935166676976,"result":"CAPTURED","ref":"935110000001","transId":201935166561122,"date":1217,"trackId":"1003383844","udf1":"","udf2":"","udf3":"8870091137","udf4":"FC","udf5":"Tidal5","amt":"70.0","authRespCode":"","00"}]</pre>
3	Error	C	Numeric	If any error, PG will provide the error code
4	ErrorText	C	Alphanumeric	PG will provide the error description if any transaction declined.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
1	paymentId	M	Numeric	Unique ID generated by payment gateway. Based on this payment Id merchant can match the final URL redirection response
2	result	M	Alphanumeric	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for

S. No	Fields	M/C/O	Field Type	Description
				initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	custid	M	Alphanumeric	Customer ID needs to be sent in the subsequent Merchant Initiating recurring transactions
16	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization

S. No	Fields	M/C/O	Field Type	Description
				14-Authorization Extension(MADA)

Sample JSON Response - Final

Redirection Parameters

- "paymentId": "100201935044735860",
- "trandata": "<encrypted trandata>",
- "Error": "",
- "ErrorText": ""

Plain Trandata

```
[{"paymentId": "100201935044735860",
"result": "CAPTURED",
"amt": "10.55",
"date": 1221,
"ref": "935110000001", -
"udf1": "udf1text",
"udf2": "udf2text",
"udf3": "udf3text",
"udf4": "udf4text",
"udf5": "udf5text",
"trackId": "3423423",
"transId": "1242345345234",
"authRespCode": "00",
"authCode": "000000",
```

```
"cardType": "Visa",  
"custid": "202014785236784",  
"actionCode": "1"  
}]
```

Recurring Payment for MADA – Merchant Initiated Initial Response - Payment ID and Payment Page URL

Merchant forwards the API request to ARB Payment gateway, below is sample request.

Request

Request from Merchant to ARB payment gateway

S. No	Fields	M/C/O	Field Type	Description
1	Id	M	Alphanu m	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	Trandata	M	Alphanu m	All the below request parameters encrypted and pass the encrypted value in trandata.

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	Action	M	Numeric	It defines the transactions actions Purchase: 1 Authorization:4
3	Password	M	Alphanu m	Tranportal password. Merchant download the same in merchant portal.
4	id	M	Alphanu m	Tranportal ID. Merchant download the same in merchant portal
5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanu m	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
8	udf2	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	cardType	M	Alphanumeric	Card type Ex : Credit card – C, Debit Card – D
12	browserLanguage	M	Alphanumeric	Value representing the browser language Returned from "navigator.language" property. Length 1 to 8 characters.
13	browserColorDepth	M	Alphanumeric	Value representing the bit depth of the colour palette for displaying images, in bits per pixel. Obtained from Cardholder browser using the "screen.colorDepth" property. Length 1 to 2 characters. Values Accepted : 1 = 1 bit 4 = 4 bits 8 = 8 bits 15 = 15 bits 16 = 16 bits 24 = 24 bits 32 = 32 bits 48 = 48 bits
14	browserScreenHeight	M	Alphanumeric	Total height of the Cardholder's screen in pixels. Value is returned from the screen.height property. Length 1 to 6 characters.
15	browserScreenWidth	M	Alphanumeric	Total width of the cardholder's screen in pixels. Value is returned from the screen.width property. Length 1 to 6 characters.
16	browserJavaScriptEnabled	M	Alphanumeric	Value is returned from the navigator.javaEnabled property. Boolean value.

S. No	Fields	M/C/O	Field Type	Description
17	browserTZ	M	Alphanumeric	Time difference between UTC time and the Cardholder browser local time, in minutes. Value is returned from the getTimezoneOffset() method. Length 1 to 5 characters.
18	jsEnabled	M	Alphanumeric	Value whether the java script is enabled in browser or not.

Below is the sample encrypted request from Merchant to PG

```
[{
  //Mandatory Parameters
  "id": "IPAY1CR6qZF7q6w",
  "trandata": "<encrypted trandata>"
}]
```

Below are the plain Trandata request

```
[{
  //Mandatory Parameters
  "amt": "12.00",
  "action": "1",
  "password": "q@a68O$27@JLkcK",
  "id": "IPAY1CR6qZF7q6w",
  "currencyCode": "682",
  "trackId": "123456",
  "cardType": "C",
  "browserJavaEnabled": "true",
  "browserLanguage": "en",
  "browserColorDepth": "48",
}]
```

```

"browserScreenHeight":"400",

"browserScreenWidth":"600",

"browserTZ":"0",

"jsEnabled":"true",

"udf1":"1", //To identify Merchant Initiated Recurring transaction

"udf2":"SI", //To identify Merchant Initiated Recurring transaction


//Optional Parameters

"udf3":"udf3text",

"udf4":"udf4text",

"udf5":"udf5text",

```

Credit Card Installment Transactions

Detailed description of Plain Trandata request parameters

S.No	Fields	M/C	Field Type	Description
		/O		
1	udf6	C	Alphanum	Credit Card Installment Short Code
2	udf7	C	Alphanum	Credit Card Installment Type

Sample JSON request - Request from Merchant to ARB PG

```

//conditional - Credit Card Installment Transactions

"udf6":"12345",

"udf7":"C",

```

Response

Once ARB payment gateway verifies the transaction and returns the response to the same request.

Response from ARB Payment Gateway to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	AlphaNum	If any error during processing, PG will provide the error code
3	errorText	C	AlphaNum	If any error during processing, PG will provide the error description
4	status	M	AlphaNum	If transaction success 1. If transaction failure 2.

Detailed description of Plain trandata response parameters

S. No	Fields	M/C/O	Field Type	Description
2	result	M	AlphaNum	Transaction status . Value will be 'CAPTURED' for purchase successful and 'APPROVED' for authorization successful.
3	ref	M	Numeric	Transaction reference number (RRN)
4	transId	M	Numeric	Unique transaction Id generated by Payment gateway and merchant can use this id for initiating supported transactions (Void, refund and inquiry)
5	date	M	Numeric	Transaction date and time
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	AlphaNum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
8	udf2	O	AlphaNum	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.

S. No	Fields	M/C/O	Field Type	Description
9	udf3	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
10	udf5	O	Alphanumeric	The user (merchant) defines these fields. The field data is passed along with a transaction request and then returned in the transaction response. Merchant should ensure that field is left blank when no data needs to be passed.
11	amt	M	Numeric	Transaction amount
12	authRespCode	M	Numeric	Auth response code provided by PG
13	authCode	M	Numeric	6 digit authorization code received from switch
14	cardType	M	Alphabetic	Card Brand name . Value will be "Visa" or "MasterCard" or "Mada".
15	actionCode	M	Alphanumeric	Action code of transaction. Possible Values 1-Purchase 2-Credit 3-Void Purchase 4-Authorization 5-Capture 8-Inquiry 9-Void Authorization 14-Authorization Extension(MADA)

Below is the sample encrypted response from PG to Merchant

```
[{
  "tranid": "201931951332346",
  "trandata": "<encrypted trandata>",
  "status": "1", //1 for success transaction, 2 for failure transaction
  "error": null,
  "errorText": null
}]
```

Plain trandata in response

```
[{  
  "paymentId":-1,  
  "result":"CAPTURED",  
  "amt":"10.55",  
  "date":"1221",  
  "ref":"935110000001",  
  "udf1":"udf1text",  
  "udf2":"udf2text",  
  "udf3":"udf3text",  
  "udf4":"udf4text",  
  "udf5":"udf5text",  
  "trackId","3423423",  
  "transId":"1242345345234",  
  "authRespCode":"00",  
  "authCode":"000000",  
  "cardType":"Visa",  
  "actionCode":"1"  
}]
```


JS WIDGET Integration

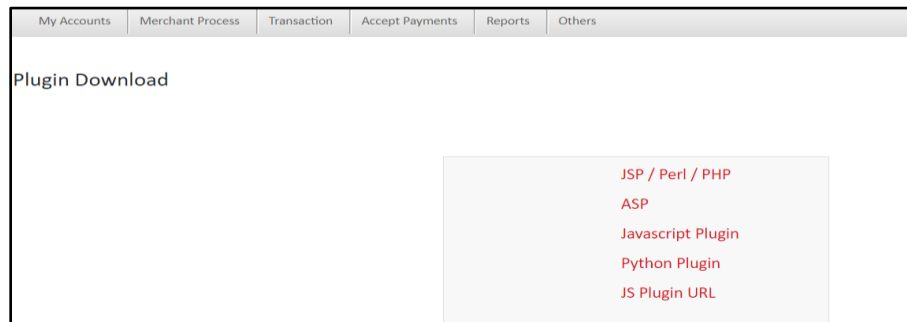
Integration Process

Steps to be followed by Merchant Integration Team:

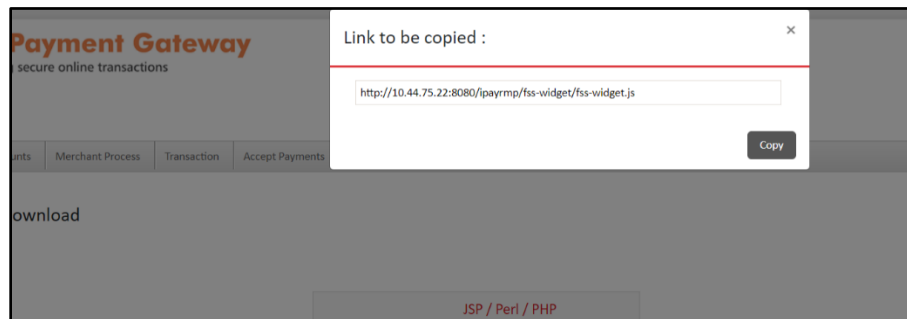
1. Merchant should login to certification environment with the user credential shared by the Acquirer Bank .
2. Ex. <https://www.securepayments.alrajhibank.com.sa/mrchptl/merchant.htm>
3. Merchant should copy the js plug-in url and domain url from the menu "Merchant Process -> Plug-in Download -> Js Plugin Url" inorder to connect Payment gateway and Key will be shared by bank prime team for Encryption and Decryption purpose.
4. Integrate the Js plug-in and plug-in resource details with merchant web page.
5. Construct the Request message as expected by Payment Gateway
6. Process the response message receiving the response from Payment Gateway.

JS Plug-in URL

1. Click **Merchant Process > Plug-in Download**.



2. From the **Plugin Download** page, click **Js Plugin URL** link.
The application displays the **URL** on pop-up:



3. Click the **Copy** button to copy the plug-in URL.
The application displays the Domain URL
4. Click the **Copy** button to copy domain URL.
Key will be shared manually by bank prime team for Encryption and Decryption purpose.

Integrate JS plug-in with merchant page

Code snippet for Integration

```
/** Request Processing**/
```

```
//Merchant can connect JS Plugin using below step:
```

1. Merchant should import the js plug-in url which is copied from Plugin download menu.

Merchant Process-> Plug-in Download

Ex. `<script src="https://securepayments.alrajhibank.com.sa/mrchptl/arb-widget/arb-widget.js"></script>`

2. Merchant should add the below mentioned code in their webpage.payment page will get loaded in this div component.

```
<div id="payment-widget-container" class="payment-widget-container">
```

```
</div>
```

```
//Set Values
```

3. Merchant should set payment values and customization values in merchant webpage as below mentioned format.

Detailed description of Plain Trandata request parameters

S.No	Parameter	Data Type	Data Input	Description
1	tranportalId	String	Mandatory	Terminal Unique ID
2	amount	String	Mandatory	Transaction Amount
3	merchantRefId	String	Optional	Merchant track id sent by merchant
4	Action	String	Mandatory	Transaction Identification
5	Language	String	Optional	Language Code
6	currencyCode	String	Mandatory	Terminal Currency Code

S.No	Parameter	Data Type	Data Input	Description
7	merchantResponseURL	String	Mandatory	Merchant call back success URL.
8	merchantErrorURL	String	Mandatory	Merchant call back Error URL
9	shipToFirstName	String	Conditional	Customer First Name. Applicable to Amex Transaction.
10	shipToCountryCd	String	Conditional	Customer Country Code. Applicable to Amex Transaction.
11	shipToPostalcd	String	Conditional	Customer Postal Code. Applicable to Amex Transaction.
12	shipToPhnNum	String	Conditional	Customer Phone Number. Applicable to Amex Transaction.
13	shipToLastName	String	Conditional	Customer Last Name. Applicable to Amex Transaction.
14	shipToAddress	String	Conditional	Customer Address. Applicable to Amex Transaction.
15	Udf1	String	Optional	User Defined Value
16	Udf2	String	Optional	User Defined Value
17	Udf3	String	Optional	User Defined Value
18	Udf4	String	Optional	User Defined Value
19	Udf5	String	Optional	User Defined Value
20	Udf6	String	Optional	User Defined Value
21	Udf7	String	Optional	User Defined Value
22	Udf8	String	Optional	User Defined Value
23	Udf9	String	Optional	User Defined Value
24	Udf10	String	Optional	User Defined Value
25	Udf11	String	Optional	User Defined Value
26	Udf12	String	Optional	User Defined Value
27	Udf13	String	Optional	User Defined Value

S.No	Parameter	Data Type	Data Input	Description
28	Udf14	String	Optional	User Defined Value
29	Udf15	String	Optional	User Defined Value
30	payorIDType	String	Conditional	ID type Applicable to MOF Transaction
31	PayorIDNumber	String	Conditional	ID number Applicable to MOF Transaction
32	billDetails	JSON Array	Conditional	Bill Details Applicable to MOF Transaction

Sample request

```
<script>
const reqBody={
    "shipToAddress":"","
    "udf15":"fifteen",
    "udf14":"fourteen",
    "udf13":"thirteen",
    "udf12":"twelve",
    "udf11":"eleven",
    "udf10":"ten",
    "language":"USA",
    "shipToPhnNum":"7548758433",
    "tranportalId":"200123",
    "udf9":"nine",
    "shipToCountryCd":"356",
    "udf7":"seven",
```

```
"udf8":"eight",

"customerId":"","

"action":"1",           //1-Purchase

"shipToLastName":"name",

"amount":"36.00",

"shipToFirstName":"tesdt",

"udf5":"6a8b21f7d9f9a4c5be6297d0f6a9b51e7461ce5b31a12861c655fc9ae300176a",

"shipToPostalcd":"4758845",

"udf6":"816",

"udf3":"9159277647",

"udf4":"","

"udf1":"","

"udf2":"","

"merchantResponseURL":"http://www.demomerchant.com/result.jsp",

"merchantRefId":"2009038955",    //Merchant Track ID

"currencyCode":"356",           //Transaction Currency

"merchantErrorURL":"http://www.demomerchant.com/error.jsp"

"contextPath":"https://products.arb.co.in/iPayRMP",

"payorIDType" : "C700",

"PayorIDNumber" : "9486576749",

"billDetails" :

"[{"issuerAgencyId":"123456789987654312","billingAccountId":"123456789",

"billingCycle":"789456132","dueAmount":"150","paidAmount":"18","billReferenceInfo":"bxcbbhgcfcgcfggf","agencyCode":"qwer"}, {"issuerAgencyId":
```

```

"123456789987654311", "billingAccountId": "1234567890", "billingCycle": "78
9465466", "dueAmount": "100", "paidAmount": "18", "billReferenceInfo": "xbdx
bvcgvfgh", "agencyCode": "qwtly"}] ]"

    }

styleReqBody = {

    "headerBgColor": "blue ",           //Header background color

    "payButtonBgColor": "blue",         //footer background color

    "payButtonColor": "blue",           //footer text color

    "textFontColor": "blue"             //font color

    "containerBgColor": "blue"          //container background color

    "formBgColor": "blue"               //Payment component background color

}

</script>

```

Domain url value which is copied from 'resource file download' menu should be set to reqBody.contextPath

Request from merchant after encryption:

```

{"tranportalId": "IPAYef5RJ0IA5fr", "trandata": "7A3694942E88633E572744674
7FF72F659364BD1A587001B5501722CF2D0EA303E7D2919C37AB4D7CA1908856CD05E11
4CFA0121DD7FFD576E7057D7EC2880EE7133077D692A20206DCEC14877CFDCE8733B982
CD731E0DB57DB0FD3B8A40534D7BCD488EE6979F05A1E7E922A81D047D62140179BAB82
D2DF17FED69F156124D3556436627BBA8C62C76DAEAF9A0A043CA5C71E24D67758DB8A
B481F37C620452D5982D580F8B978DC0206C6AB4F4A2FD77CFD71BFED233A0C5F7B33A9
C6989908466CD90DAC61BE9890F066BF6DFFFEAEB6D00DDFFFA8CED414242F5F4DD141F8
0AE4081C03B25CE08B4B62FFD04AF9C2F432E6617A56DFF610BB14D21BB23B49B410287
80183BB82B4E3E098F1303345E5830048AC51A2D8CA28F81A498ACFEA89B7984FCBAB03
6247D0E17B93A7F7BA0401D1E7FE81B3B19D2CE806E46F9393E61F68EA3ECA0FAEF66A3
09EF6F7FE1973B4145085C917CD8D72A02B86D200D913AEE2646FD3F2AF91E923C55E2C
8C29FB1C6A56829C4A95970578BA5437128E5D3C372DEE9A164FDA4CB63609D1655B47A
496565872238726271BC6D5AB70365F6D383FC38B08A7AF55070EF876D1B5B6C551E146
C25932294F472A1DCF46ABF89C16C10CFE2C968B1983CC906C5728942B2F9C115DFAEFE
2DB38844B4B23E88C0530412BE1094F65F3FD4F2C78F3CECAF0C436BCFD971B6A9CD6DD
DA30E36A9557120948C3915C45413AAEAC124CEC33CA273E4158661B49966A1108DFC2E
11718CB8CD5A545DBB25CDF7BBDE54A3E277D594235FADA6FDE03D0FB86E685C42AAA17
5E0666F0FC6E519119858E41D2F98344628B95AEDF8B43FCCF2B91BE61F8B0A8E7DAE53

```



```
118CAF81AA5B61A31E19D71E38200F422ADCAECAAA43C1148746710FCB65CE9152951B9C
067488549FA538A0962AD5A318A24A93B1F3CAC60B1A6E814D3C58E931C9D884F318D68
A0A29F74E884DD8603ABB79B5F72F0CF55FCC45F2977605084E10D695AD0E0CB4F1EDF1
6E797F686F881D63E104553CB7FC49E126FD5E964234A393A22A0057A55338D16CA07CB
07219A11351EAF2088DD686A64B00CCDB3DA30392A8AFC0956E711CB26171DD662E069B
4DBFACEFB0E0DE69C9B2C21F6019CD5B638A4883404C83C585404EFE6402FBF95F2FB1FE
0C674E8B4E9C4154896999CB3D80CEE8C70C635EC375E5D55F1F6EAB9F4C8236D306EBB
984EA37C99E57022481254804EDB07C7A3B7502C248254E945F34BCA6544B88FF91D369
DF13D535C0B3780ED0E42DED693880A4BCDD7BCECAD39AFF850B33A9A1127CC9C313D97
8BCE65D07C719EA69AD553252EA520A91FD17336D5A7F0BEE577F45BCA507541402B20C
08F0EA454A4B04A1DCC360FF09774C730C55143F33354EB4BAB236AE6C760513BCB7A04
7A9D3561DD191C3E0622AC8EE8E7368F538C91543DA232BA429F97CA67EF58620E502DE
E33ADB67723B0F026D4E5787E2D378860CEC452D2B6D55B408B72A3DBED3BFBE4062F0C
20863ABD10947D0273C01C67CE11A1ADE96E21942BFBAF1905678D4930C375CD1B52EC4
4E48DBC299B74B4511061FFC500FE39064F2FE426867EAD929783EE8EC34F2DEC1FFB0C
B76CC071E317B28B0FD984AF38315DDE5927ED570EC3C018EC1A94F38BC9D9F429DBE8D
1DDFA2BAFDCBFFD80DF8A56211EC4ED037E2FFA0D7EA614AB5E60FF5FBCAF88E9FA4296
05605D813826D43E8234B3D4F57F9BD8951E23CB7BF80B4D14A5C48191F724F9BC27E9B
95B075205CB94A65B34235F4F5E911B805DBCE83271D81ED08B1305AF98A520611729E8
C3EC1AD53ED8145286473BC52511B158725994F4F4B054CA4F96AC48FDAC2E87C3AFA01
F9212A5614A3F0EEA0A1A5ECA0DC602BD7B57304C28C19C84AC52232CDB454C4BB58ACF
4D66665548AE2963C9F06448379920BF18658582BD174B192569B0BC1064105E35D453F
D2161B9C277F50385C628E3B491BB790A7D4AC1DC911CA052F72E633D097B485044E598
36FF72B2D6C7A8CE86F1298C5EDE7ED73F7AB87D707D15B09FDB6CF823E707B2738698E
423AD9FB0786423DAB6427A1D5D8F92F67C7C1DB0491B489F6377A759190ED784E69224
9DF690C5A71E8559B0F402E5805EFC3855EA41EBFCD3FF163F348340EBC539C6B3BBB5A
5E77E3F60FE279F4AEF90D9BEF2A2894E694DA5166EB172CFFED1B3857E0C997F5CE834
8C74716F983B5EA744E81AC61CB2667F9550012C062542F01ADD971D91483BDB2D8CC46
025AC37C4750B4F8094E5E69BA37901DD61511C315F39FF3D337A21448C788F343E4796
CE70AEF9A9ECCABBB3353A616A9D7DDDC5F0CE9A73F8C8B51471F97A25BCE00E3DE9B77
98CFB400585A0F186C66C243A9557DDA28B8E206F54B30C1F739F640B8419655B5B995C
C3E7F36461653BFBE628C3B0424A6F1667BDDE9C91FE21A5017E6DD7ED3A75430D5DCA5
278DC004F79FFAE868B0A40B7CD519AE1A301D699EBC3BAF5470DBBD263BC6F54E285E8
51F4F31CC9ED56DDE4C35E48A261D2F0DC4CE00D8778A60D34C9D5019722BA69D3808A3
6EE9C5474900AB152328B0C0D8C10435E371709D80C209DA77D9AE6D3AE011421ADB5A0
996E63351A64A88F467D636BD339B4BD4786F509F681194282B59F6767DDA7DC9BA00A0
867DCE7D9D3A28F963967F659C74ED89F88B4A5B13129FE47D21A0A469D674A1233B60A
35F8DB233E9484CEE86DE99DED51CECE9A022B7010075B9525C62A0DCA46E3A7176297C
74E088BC3DEC4F092EF6EB18A808888D86765D4EF9FE7AC6B556D56BA5004E1176AF25B
14DEC0F429A2EFA4A26415BF4FC39358803E9A6713BD36E83511BD5D73D8F31", "conte
xtPath": "http://10.44.11.71:8085/iPayARBNET", "merchantResponseURL": "htt
p://10.44.11.71:8085/widgetdemo/jsp/arb/HostedPaymentResult.jsp", "merch
antErrorURL": "http://10.44.11.71:8085/widgetdemo/jsp/arb/HostedPaymentR
esult.jsp"}

/** End of Encryption **/
```

Loading payment page in merchant

```
// For Loading payment page from widget, the method to be called is
makePayment()

$("#pay").click(function(){

    $("#pay").hide();

    makePayment();

});

/** End of Request Processing**/
```

Final Response

S.No	Field	Type	Description
1.	paymentid	Numeric	Unique Id generated by Payment Gateway. Merchant MUST map this payment ID with merchant track Id and transaction amount for future transaction reference. (ex : 501201906013287030)
2.	result	Alpha	Describes the result of the transaction (ex: CAPTURED).
3.	auth	Alphanumeric	The resulting authorization number of the transaction from the issuing bank. This number or series of letters is used for referential purposes by some acquiring/issuing bank/institutions and should be stored properly. (ex: 099673)
4.	amt	Numeric	Transaction Amount as sent by merchant in the request
5.	ref	Alphanumeric	The resulting reference number of the transaction. This number or series of letters is used for referential purposes by some

S.No	Field	Type	Description
			acquiring/issuing bank/institutions and should be stored properly.
6.	postdate	Numeric	Transaction Date in the format of the authorization system, Post Date is business logic date of respective issuing bank and may not be same as the actual transaction date.
7.	trackid	Alphanumeric	Track ID value that was sent by merchant in the transaction request
8.	tranid	Numeric	Unique Transaction ID generated by Payment Gateway, this transaction ID is required for any future support/subsequent transaction (Refund/Void/Capture) to be performed by merchant on Payment Gateway, merchant should store this parameter properly
9.	avr	Alphanumeric	Address verification response.
10.	ErrorText	Alphanumeric	Describes the error with Error code. (ex. IPAY0100126-Brand not enabled)
11.	Error	Alphanumeric	Error code of the error. (ex. IPAY0100126)
12.	authRespCode	Alphanumeric	This is Response Code i.e. the reason code for the transaction. Merchant can use this code for additional validation of the transaction and can also use this response code field along with the reason code description list. (ex:00)
13.	udf1	Alphanumeric	Udf field sent in the request
14.	udf2	Alphanumeric	Udf field sent in the request
15.	udf3	Alphanumeric	Udf field sent in the request
16.	udf4	Alphanumeric	Udf field sent in the request

S.No	Field	Type	Description
17.	udf5	Alphanumeric	Udf field sent in the request
18.	Udf6	Alphanumeric	Udf field sent in the request
19.	Udf7	Alphanumeric	Udf field sent in the request
20.	Udf8	Alphanumeric	Udf field sent in the request
21.	Udf9	Alphanumeric	Udf field sent in the request
22.	Udf10	Alphanumeric	Udf field sent in the request
23.	Udf11	Alphanumeric	Udf field sent in the request
24.	udf12	Alphanumeric	Udf field sent in the request
25.	Udf13	Alphanumeric	Udf field sent in the request
26.	Udf14	Alphanumeric	Udf field sent in the request
27.	Udf15	Alphanumeric	Udf field sent in the request
28.	custid	Alphanumeric	Customer id for SI registration txn.

/** Response received from Payment Gateway to Merchant **/

```
{
  "trandata": "2B1E3C6E62299D5DC43D59BC864C7003577DE32CF9EBB5ED30C7CC86DA8BA324F1D88AFE9821E2378DDCEAF46206837B928A1BFF528C95DDBC8F06A24A7287BE1FE1726AB0D59870A8BC4DFB7BA6FD98F432051D2EDE899E42EEC157163B48DC2D217BD02DAC2542F25C3D0D042D387B4E53DA8C2BE66C56D292C58CF22709563A0FF7B1078147E4632D2E2BFC4C723905BE57106C4B4A4F4F4B8D010E2AD05331E8E051EBE61972B3DA02C3A1A712ACF9428A4E83D9D524BD4EF78AC286B64FFF32DD29086690E44666ADB13E4097B6DC706D09C7FE9689511C4B70A58544AE3D40D3808F451AFE15D2323C09CC42C5CBDB611F880EC46CF83CC146A0C314397A86835C3A383BBA51643D805533D3F332A46D14CE67E79B479A23F67882A1C889670BE5793A535EB2613D4948BFF5826A40B4B4FD9F7AE92FA1DA1DF945143661E1781D917F52796CE58278AE5CDB21C9AB843F515670B17BB7D654FB5846C2426A19C37B009464A505A22FE76C447B69F6CD6949D06AB1437D3820787863CA4278A1A0529DCA254017FFF315FB0D819D1B71495A37388CDF0992D5FECFD572EE7BC26D2C49D3003ACAF AAC365F0CDB6FCCD05F61CD79B24A4CEF7B7466F8E989B7EE3EFF2BFC5E0B4CDDA626D593A61AF4427A4683462302B2099063B50C634789E75EE7D3AC86444E7B998175596994026239BA00E27AEEB2FC64236E0F29FA638FF83424F45F53FBF59DCB78ACF2FB2A2E09E1825B6507062EC954C3D92B30EF3EAA86C911E636308DA1F115BD75830678E4391DA5F9977737BDDBC60D0A0F92"
}
```

```
3C381C0B0044942FEA7BA703545071E04FCC42589B2C93FA5D155BF9C055F31125A5178B08EEAD
9A8B424B7EB1EDF0B507EFB79F77E2FB465D2CDB98B4C59251A13530483CCDFA6128E8E18F20E6
F942D303A317FF716105C5A77355275EAEAE8D347BA605CAE3A7F97444A09537238232763732D9
36CB414B77DC0B4C77DAC7C049BB4830FD6A3D531015F9E340E165A2B3A0084C218BECE6667C11
18D122A5200110B94AEBFE8D91222925D2A80CC5BAA7EEB201241F8BD3750870AD04BAEFB79431
00A974EE4639FE9A4E2C50DAAC8331692DF283B7A81CF807665E6FBFBF6C80E3766EEDB50A310E
25699471B5EE4378FA4AD26732CD486722E9378FF0E65FABF6037E295FBD966EAFED826684DA88
C65044F376A6A8F44218AA5EB9D3EECE28AA3A577640712D5CBCA1D15BB58E91DE48FFD7A50F77
620AA118D65BC32BC60220EE97E466ECB0B78F8AD78C32E0E0BDE7EEB8894F198EEB58984A0B55
522FA27D8453DDC333A70147CE3D9E4FA36F36A465E3BA60B8A60A012E337E2CF1B3FDA28D69D1
4838364B1C003891972FA6EA7B73A124715DA4208C91D9D27F65DB6AC718AFC923655D4BA11369
1FB227EA1BA2432920635329E10C6B16F26CF4BFD5C1675AAD73C884CC0170B34BB3FB21BC621E
27406ACA2265E59728AAA3CD71B3EF0A647C6AE71EAFE2C0CB3EFB6918EC4475F7007372657186
9C6A511DE715A5EE4E3855AC359011EB424F7B02AA4CBD15A65DA01E9C794A57092D309712B433
AC83A3F76E7C727509399BE9765D4842A9E956AB4C2C81A43CC144273367002689079DD8F8F7F2
F6BD725A108FA6F57B3156439EEF477B9EAAE2177B65DB6B783672ACADC217F52AFEB9A920D1381
DEF2FC195DC96193ACABC6EABD56CDFD5B407F5AA42E85840BB7E6B266C8047538B3307FD97FAC
4B5D900226884CAA1793EC93D56F77CDCC3C3593A362C29BD25BC2B137C0A735CD4B305A8C8D31
F89C0EC433B913C37375451A75F577E77147B2A3B0D6873E82773762871E381DD05BEE88CAAFBF
0C457E201EE862A47", "currCD": "356", "paymentId": "100202128054842905", "trackid": "
2138298712", "tranAmt": 36, "custid": 202128045180741, "merchantResponseURL": "http:
//10.44.11.71:8085/widgetdemo/jsp/arb/HostedPaymentResult.jsp"}

```

After receiving the above response merchant need to decrypt the above trandata from response with the help of decryption code.

Response from payment gateway will be posted to merchantResponseURL.

Success Response Parameters	Description
result	CAPTURED for Purchase transaction. Other than above value considered the transaction as failure
date	Contains Postdate
Ref	Transaction reference no.
trackid	Containts merchant unique track id
tranid	PG Transactuaion ID
udf1 udf2 udf3 udf4 udf5	UDF Values sent during initial leg

udf6	
udf7	
udf8	
udf9	
udf10	
udf11	
udf12	
udf13	
udf14	
udf15	
paymentid	PG PaymentID
amt	Transaction amount
custid	Contains Customer ID for SI registration transaction.

Sample JSON Response - Final

If Merchant notification is disabled, then ARB Payment gateway will provide the final response in URL redirection. Below is the sample response from ARB PG to merchant

```
[{
  //Redirection Parameters
  "paymentId":"100201935166676976",
  "trandata":"<encrypted trandata>",
  "error":"","
  "errorText":""
}]
```

Plain Trandata:

Trandata will contain below parameters encrypted.

```
[{
  "paymentId":"100201935166676976",
  "result": "CAPTURED",
```

```

"transId":201935166561122,

    "ref":"935110000001",

    "date":"1217",

    "trackId":"1003383844",

    "udf1":"","

    "udf2":"","

    "udf3":"8870091137",

    "udf4":"FC",

    "udf5":"Tidal5",

    "amt":"70.0,

    "authRespCode","00",

    "authCode":"000000",

    "cardType":"Visa"

  }
}

```

Trandata after decryption :

```

[{"date":"1007","authRespCode":"00","authCode":"000000","agencyDetails":[{"bil
lReferenceInfo":"bxcbbhgcfcgcf", "dueAmount":"150","billingCycle":"789456132
","issuerAgencyId":"123456789987654312","billingAccountId":"123456789","agency
Code":"qwer","paidAmount":"18"}, {"billReferenceInfo":"xbdxcbvcgvfgh", "dueAmoun
t":"100","billingCycle":"789465466","issuerAgencyId":"123456789987654311","bil
lingAccountId":"1234567890","agencyCode":"qwt", "paidAmount":"18"}], "transId":
"202128045169265", "trackId":"2138298712", "udf5":"c3af040d7ce24256064aa421d5b2b
39898325614eafc2f79a3dfd82059906cf5", "cardType":"Visa", "udf6":"816", "udf10":"t
en", "amt":"36.0", "udf3":""," "udf4":""," "udf1":"1", "udf2":"SI", "result":"CAPTURED
", "ref":"128010000008", "udf9":"nine", "paymentId":"100202128054842905", "udf7":"
seven", "udf8":"eight", "fcCustId":"0", "paymentTimestamp":"2021-10-
07T07:06:01.554Z"}]

/** End of Response Processing**/

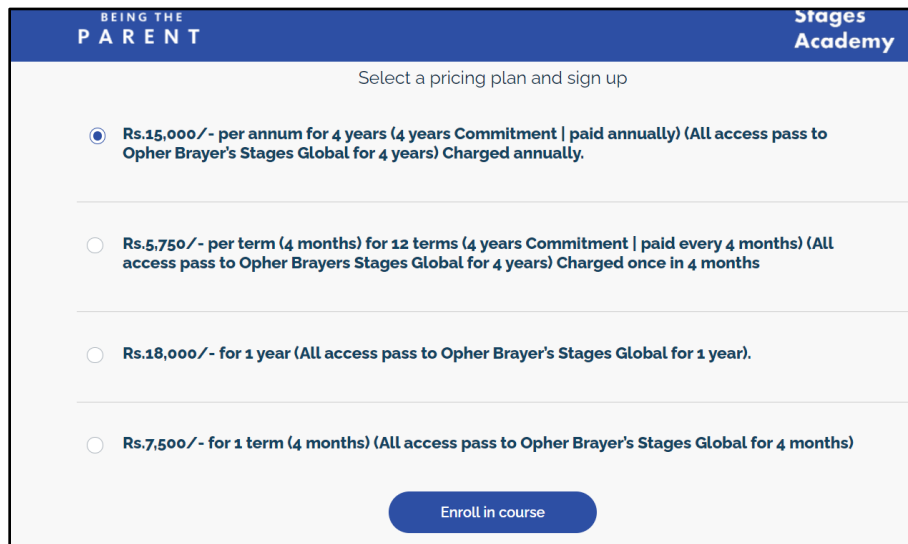
```

Sample Demo page Navigation

Hosted Payment Integration

Step 1: Merchant's page

When the Customer log on to the Merchant website, Merchant will show the plans page for online shopping.



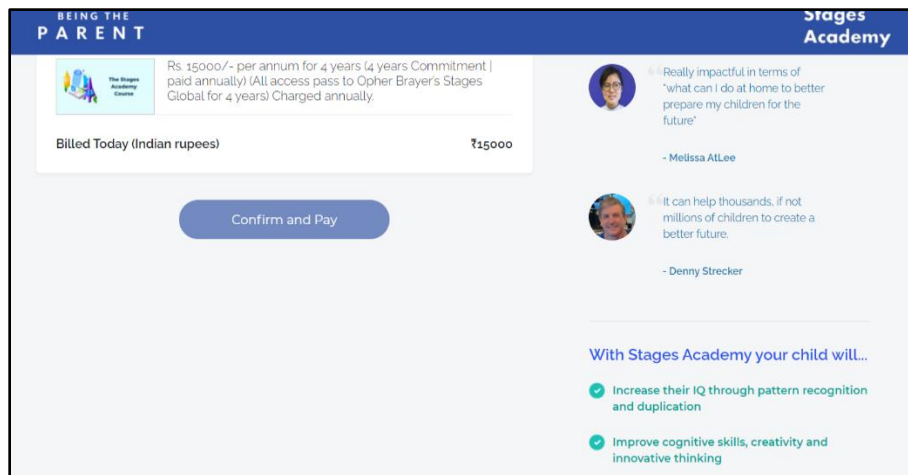
The screenshot shows a web page titled "BEING THE PARENT" by Stages Academy. The header includes the text "BEING THE PARENT" and "Stages Academy". Below the header, there is a sub-header "Select a pricing plan and sign up". The main content area lists four pricing plans, each with a radio button for selection:

- ☒ **Rs.15,000/- per annum for 4 years (4 years Commitment | paid annually) (All access pass to Opher Brayer's Stages Global for 4 years) Charged annually.**
- ☐ **Rs.5,750/- per term (4 months) for 12 terms (4 years Commitment | paid every 4 months) (All access pass to Opher Brayers Stages Global for 4 years) Charged once in 4 months**
- ☐ **Rs.18,000/- for 1 year (All access pass to Opher Brayer's Stages Global for 1 year).**
- ☐ **Rs.7,500/- for 1 term (4 months) (All access pass to Opher Brayer's Stages Global for 4 months)**

At the bottom of the page, there is a blue button labeled "Enroll in course".

Step 2: Plans selected by the customer for purchase:

If the customer clicks "Enroll in course", then the Order summary confirmation will be shown.



The screenshot shows the same web page as in Step 1, but now it displays the order summary confirmation. The header remains the same. Below the header, there is a sub-header "Select a pricing plan and sign up". The main content area shows the selected plan and the total amount:

Rs. 15000/- per annum for 4 years (4 years Commitment | paid annually) (All access pass to Opher Brayer's Stages Global for 4 years) Charged annually.

Billed Today (Indian rupees) ₹15000

Below this, there is a blue button labeled "Confirm and Pay".

To the right of the pricing plan, there are two testimonials:

- Really impactful in terms of "what can I do at home to better prepare my children for the future"**
- Melissa ALLee
- It can help thousands, if not millions of children to create a better future.**
- Denny Strecker

Below the testimonials, there is a section titled "With Stages Academy your child will..." with two bullet points:

- ✓ Increase their IQ through pattern recognition and duplication
- ✓ Improve cognitive skills, creativity and innovative thinking

Step 3: Payment Page:

Once the customer clicks “Confirm and Pay”, this page is shown by widget in merchant webpage. Here the customers have to give their card credentials.

//Payment page for Credit, Debit card instruments

The screenshot displays the FSS (Faster Payment System) payment interface. On the left, the 'Order Summary' section shows a transaction for Rs. 36.00/- per annum for 4 years, with a 'Today Amount (Indian rupees)' of Rs. 36.00/-. Below this, it prompts the user to 'Click on make Payment to continue' and lists 'Available offers' including 10% off on HDFC Bank credit card transactions and 10% off on HDFC Debit credit card transactions, both up to RS 500 on orders above RS 5000. On the right, the 'TEST MERCHANT' section shows 'ASDSDASDA SDA' with a 'Track ID : 540285530' and a total amount of 'INR 36.00'. The payment form includes fields for 'Card Number' (with MM/YY), 'CVV', 'Email', and 'Mobile Number'. There is a checkbox for 'Save for future payments' and a blue 'Pay INR 36.00' button at the bottom.

Step 4: Merchant Response Page:

After processing transaction, Payment Gateway shall send response to the Merchant.

Payment Information

Your transaction for amount INR 36.0 is successfull
please note transaction number 202117937707065 and payment ID 200202117962327595

Best Practices:

- a) The Merchant should mandatorily maintain logs for each transaction as mentioned below
 - a. The parameters before setting the values in the respective variable.
 - b. Request from the merchant server to Payment Gateway
 - c. Response that is received from the Payment Gateway in the Merchant Response URL
- b) The Merchant should maintain "OWASP" (Open Web Application Security Project) Top 10 recommendation in their web application. (These recommendations are available on www.owasp.org)
- c) The Merchant should have the latest SSL security certificate in the payment request and receive webpage, if any. Always ensure that the SSL certificate is valid and has not expired. Such certificates should be as per the approved list of the Acquiring Bank. Self signed certificates are not supported by Payment Gateway in Test and Production Environment.
- d) The Merchant should mandatory complete the UAT and ensure all results are in line with the recommended response prior to going LIVE.
- e) Any changes in the pages would need to be tested before moving to Production after proper communication to the Bank personnel and receipt of approval. If the pages have a change in logic or transaction flow particularly, the Acquiring Bank's consent is Mandatory.
- f) The transaction request and Response Handling: For ease in integration, "Sample/Demo pages "provided in the integration document are essentially for representation purposes only. The actual pages have to be necessarily developed and implemented by the Merchant's development team and used in both the Test and Production environment. The Sample demo pages are provided for the logical understanding and transaction flow only. An ideal logical flow for the merchant to process the customer input data is to collect the shopping details of the customer such as transaction amount, merchant track id and other parameters and stored in a secure storage location and validated immediately against the details of shopping cart module.
- g) Maintenance of Transaction Logs: It is essential for the transaction logs to be maintained in a secure storage location within the environment. This is crucial in order to trace transaction history in case of a dispute raised by a

customer or even internal audit purposes. These logs should ideally include the customer IP address as well apart from the other transaction details.

Private and Public key

- If Merchant opted for private and public key future. ARBPG provide two keys for encrypting and decrypting the request and response respectively.
- Purchase Transaction: Key 1 should be used for encrypting the Request. Key 2 should be used for decrypting the response from payment gateway.
- Supporting transaction: Inquiry, Refund and Reversal transactions should be done by key 2 only i.e Key 2 should be used for encrypting and decrypting the request and response respectively from Payment Gateway. [Toc31220960](#)

Setting Up Your Server

Overview

- All pages that include Apple Pay must be served over HTTPS.
- Your domain must have a valid SSL certificate.
- Your server must support the Transport Layer Security (TLS) 1.2 protocol and one of the cipher suites listed in the below.

```

TLS_RSA_WITH_AES_128_CBC_SHA256
TLS_RSA_WITH_AES_128_GCM_SHA256
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
  
```

Whitelist Apple Pay IP Addresses for Merchant Validation

- To enable merchant validation and receive a session object, your server must allow access over HTTPS (TCP over port 443) to the Apple Pay IP addresses and domains provided below

For production environment:

Domain	apple-pay-gateway.apple.com
17.171.78.7	apple-pay-gateway-nc-pod1.apple.com
17.171.78.71	apple-pay-gateway-nc-pod2.apple.com
17.171.78.135	apple-pay-gateway-nc-pod3.apple.com
17.171.78.199	apple-pay-gateway-nc-pod4.apple.com
17.171.79.12	apple-pay-gateway-nc-pod5.apple.com
17.141.128.7	apple-pay-gateway-pr-pod1.apple.com
17.141.128.71	apple-pay-gateway-pr-pod2.apple.com
17.141.128.135	apple-pay-gateway-pr-pod3.apple.com

17.141.128.199 apple-pay-gateway-pr-pod4.apple.com
17.141.129.12 apple-pay-gateway-pr-pod5.apple.com
17.171.78.9 apple-pay-gateway-nc-pod1-dr.apple.com
17.171.78.73 apple-pay-gateway-nc-pod2-dr.apple.com
17.171.78.137 apple-pay-gateway-nc-pod3-dr.apple.com
17.171.78.201 apple-pay-gateway-nc-pod4-dr.apple.com
17.171.79.13 apple-pay-gateway-nc-pod5-dr.apple.com
17.141.128.9 apple-pay-gateway-pr-pod1-dr.apple.com
17.141.128.73 apple-pay-gateway-pr-pod2-dr.apple.com
17.141.128.137 apple-pay-gateway-pr-pod3-dr.apple.com
17.141.128.201 apple-pay-gateway-pr-pod4-dr.apple.com
17.141.129.13 apple-pay-gateway-pr-pod5-dr.apple.com
101.230.204.232 cn-apple-pay-gateway-sh-pod1.apple.com
101.230.204.233 cn-apple-pay-gateway-sh-pod1-dr.apple.com
101.230.204.242 cn-apple-pay-gateway-sh-pod2.apple.com
101.230.204.243 cn-apple-pay-gateway-sh-pod2-dr.apple.com
101.230.204.240 cn-apple-pay-gateway-sh-pod3.apple.com
101.230.204.241 cn-apple-pay-gateway-sh-pod3-dr.apple.com
60.29.205.104 cn-apple-pay-gateway-tj-pod1.apple.com
60.29.205.105 cn-apple-pay-gateway-tj-pod1-dr.apple.com
60.29.205.106 cn-apple-pay-gateway-tj-pod2.apple.com
60.29.205.107 cn-apple-pay-gateway-tj-pod2-dr.apple.com
60.29.205.108 cn-apple-pay-gateway-tj-pod3.apple.com
60.29.205.109 cn-apple-pay-gateway-tj-pod3-dr.apple.com

For sandbox testing only:

17.171.85.7 apple-pay-gateway-cert.apple.com
101.230.204.235 cn-apple-pay-gateway-cert.apple.com

Whitelist Apple IP Addresses for Domain Verification

- Apple uses the following IP addresses when you register or verify your merchant domain. If your domain is protected from public access and you wish to complete domain verification, you should whitelist the following IP address ranges.

17.32.139.128/27
17.32.139.160/27

17.140.126.0/27
17.140.126.32/27
17.179.144.128/27
17.179.144.160/27
17.179.144.192/27
17.179.144.224/27
17.253.0.0/16

Apple Pay Process for merchant registration and certificates

Create a merchant identifier

- In Certificates, Identifiers & Profiles, select Identifiers from the sidebar, then click the Add button (+) in the upper-left corner.
- Select Merchant IDs, then click Continue.
- Enter the merchant description and identifier name, then click Continue.
- Review the settings, then click Register.

Create a payment processing certificate

- In Certificates, Identifiers & Profiles, select Identifiers from the sidebar.
- Under Identifiers, select Merchant IDs using the filter in the top-right.
- On the right, select your merchant identifier.
- Under Apple Pay Payment Processing Certificate, click Create Certificate.
- Click Choose File.
- Upload the Payment Processing CSR file i.e generated initially
- Click Continue.
- Click Download.
- Convert the downloaded .cer file to jks file and save to the .bin file for payment token decryption process.

Merchant Domain Verification

- In Certificates, Identifiers & Profiles, select Identifiers from the sidebar.
- Under Identifiers, select Merchant IDs using the filter in the top-right.
- On the right, select your merchant identifier.
- Under Merchant Domains, click Add Domain.
- Enter your domain name and click save.
- Download the apple-developer-merchantid-domain-association.txt file.
- Place the downloaded file in the virtual directory of the server to complete domain validation method using virtual directory.
- Apple will initiate the request in below format to check the file is present or not.

`https://www.yourdomain.com/.well-known/apple-developer-merchantid-domain-association.txt`

Create a merchant identity certificate (This is only required for Bank hosted integration)

- In Certificates, Identifiers & Profiles, select Identifiers from the sidebar, then select Merchant IDs from the pop-up menu on the top right.
- On the right, select your merchant identifier.
- Under Apple Pay Merchant Identity Certificate, click Create Certificate.
- Create a certificate signing request on your Mac, and click Continue.
- Click Choose File.
- Upload the Merchant CSR file i.e generated initially
- Click Continue.
- Click Download.

- Convert the downloaded .cer file to jks file and save to the trustore and keystore files to perform 2-way TLS handshake.

Commands to generate Merchant Identity Certificate and Payment Processing Certificate

Generating a Merchant Identity Certificate

- Generates a Certificate Signing Request (CSR) for a Merchant Identity
- Certificate per the following (and keeps the private key safely):
 - a) Command-line tools (either OpenSSL or Keytool can be used)
 - (i) OpenSSL
 - Generate key pair in a key file
 - `openssl req -new -newkey rsa:2048 -nodes -out rsacertreq.csr -keyout rsakey.key -subj /CN=www.mydomain.com`
 - (ii) Keytool
 - Generate key pair in a pkcs12 file
 - `keytool -genkeypair -keyalg RSA -keystore rsakeystore.p12 -storetype pkcs12 -keysize 2048 -alias rsakeyname -dname CN=www.mydomain.com`
 - Generate CSR from key pair in pkcs12 file
 - `keytool -certreq -alias rsakeyname -file rsacertreq.csr -keystore rsakeystore.p12 -storetype pkcs12`
 - Then uploads the Merchant Identity Certificate CSR in apple portal.
 - Download (and backup) the Apple signed Merchant Identity Certificate, `merchant_id.cer`
 - Import merchant certificate and private key to generate P12
 - a) Command-line tools
 - (i) OpenSSL
 - convert `merchant_id.cer` to PEM
 - `openssl x509 -inform DER -in merchant_id.cer -out merchant_id.pem`
 - Import merchant certificate and private key to generate P12
 - `openssl pkcs12 -export -out Certificates.p12 -inkey rsakey.key -in merchant_id.pem`
 - Convert `.p12` file to `jks` file and save into trustore and keystore file.

a) Command-line tools

(i) Keytool

```
keytool -importkeystore -srckeystore Certificates.p12 -srcstoretype
pkcs12 -destkeystore applepaytrustore.bin -deststoretype jks
```

- Ensure below mentioned Apple Root and Intermediate certificates are installed in your truststore to perform 2-way TLS handshake.

Download certificates using below links

<https://www.digicert.com/digicert-root-certificates.htm>

<https://www.apple.com/certificateauthority/>

(i) AppleRootCA-G3.cer

(ii) DigiCertGlobalCA-3G2.crt

(iii) DigiCertGlobalRootG2.crt

Generating a Payment Processing Certificate

- Generates a Certificate Signing Request (CSR) for a Payment Processing Certificate per the following (and keeps the private key safely):

a) Command-line tools

(i) OpenSSL

- Generate key pair in a key file

```
openssl ecparam -genkey -name prime256v1 -out ecckey.key
```

- Generate CSR from key pair in key file

```
openssl req -new -sha256 -key ecckey.key -out ecccertreq.csr -subj
/CN=www.mydomain.com
```

(ii) Keytool

- Generate key pair in a pkcs12 file

```
keytool -genkeypair -keyalg EC -keystore ecckeystore.p12 -
storetype
pkcs12 -keysize 256 -alias ecckeyname -dname
CN=www.mydomain.com
```

- Generate CSR from key pair in pkcs12 file

```
keytool -certreq -sigalg SHA256withECDSA -alias ecckeyname -file
ecccertreq.csr -keystore ecckeystore.p12 -storetype pkcs12
```

- Then uploads the Payment Processing Certificate CSR in apple portal.

- Download (and backup) the Apple signed Payment Processing Certificate, apple_pay.cer
- Import payment processing certificate and private key to generate P12
 - a) Command-line tools
 - (i) OpenSSL
 - convert apple_pay.cer to PEM
openssl x509 -inform DER -in apple_pay.cer -out apple_pay.pem
 - Import merchant certificate and private key to generate P12
openssl pkcs12 -export -out ecckystore.p12 -inkey ecckey.key -in apple_pay.pem
- Convert .p12 file to jks file and save into .bin file.
 - a) Command-line tools
 - (i) Keytool
keytool -importkeystore -srckeystore Certificates.p12 -srcstoretype pkcs12 -destkeystore applepaytrustore.bin -deststoretype jks

Merchant Hosted URPAY Integration

Merchant forwards the API request to ARB Payment gateway, below is sample request.

Attributes - Request from Merchant to ARB PG

S. No	Fields	M/C/O	Field Type	Description
1	id	M	Alphanumeric	Tranportal ID. Merchant can download the Tranportal id from Merchant portal
2	trandata	M	Alphanumeric	All the below request parameters encrypted and pass the encrypted value in trandata.
3	responseURL	M	Alphanumeric	The merchant success URL where Payment Gateway send the notification request.
4	errorURL	M	Alphanumeric	Merchant error URL

Detailed description of Plain Trandata request parameters

S. No	Fields	M/C/O	Field Type	Description
1	amt	M	Numeric	Transaction amount
2	action	M	Numeric	It defines the transactions actions Purchase: 1
3	password	M	Alphanumeric	Tranportal password. Merchant can download the same from the portal.
4	id	M	Alphanumeric	Tranportal ID. Merchant can download from the portal

5	currencyCode	M	Numeric	3-digit currency code of KSA. Ex:682
6	trackId	M	Numeric	Merchant unique reference no
7	udf1	O	Alphanum	The user (merchant) defines these fields. The
8	udf2	O	Alphanum	The user (merchant) defines these fields. The
9	udf3	O	Alphanum	The user (merchant) defines these fields. The
10	udf4	O	Alphanum	The user (merchant) defines these fields. The
11	udf5	O	Alphanum	The user (merchant) defines these fields. The
12	responseURL	M	Alphanum	The merchant success URL where Payment Gateway send the notification request.
13	errorURL	M	Alphanum	The merchant error URL where Payment Gateway send the response in case any error while Processing the transaction.
14	MobileNumber	M	Numeric	Contains 9 digit URPAY registered mobile number

Sample JSON request - Request from Merchant to ARB PG

```
[{
  //Mandatory Parameters
  "id": "IPAYlCR6qZF7q6w",
  "trandata": "<encrypted trandata>", (encrypted with AES algorithm with CBC Mode)
  "responseURL": "https://merchantpage/PaymentResult.jsp",
  "errorURL": "https://merchantpage/PaymentResult.jsp"
}]
```

Plain Trandata:

Trandata will contain below parameters

PKCS5Padding with initialization vector value **PGKEYENCDECIVSPC** under Resource key.

```
[{  
  
  //Mandatory Parameters  
  
  "amt": "12.00",  
  
  "action": "1",      // 1 - Purchase  
  
  "password": "q@a680$27@JLkcK",  
  
  "id": "IPAY1CR6qZF7q6w",  
  
  "currencyCode": "682",  
  
  "trackId": "123456",  
  
  "mobileNumber": "980765432",  
  
  "responseURL": "https://merchantpage/PaymentResult.jsp",  
  "errorURL": "https://merchantpage/PaymentResult.jsp",  
  
  //Optional Parameters  
  
  "udf1": "udf1text",  
  
  "udf2": "udf2text",  
  
  "udf3": "udf3text",  
  
  "udf4": "udf4text",  
  
  "udf5": "udf5text",  
  
}]
```

Initial Response from PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	status	M	Numeric	If the request validation success, then status will be '1'. If the validation failed, then status will be '2'
2	result	C	Alphanum	It contains payment ID and Payment URL if the validation success else this will be null
3	error	C	Alphanum	If validation failed, then Payment gateway will provide the respective error code
4	errorText	C	Alphanum	If validation failed, then Payment gateway will provide the respective error description

Plain Response:

ARB Payment gateway internally validates the request and gives payment ID and payment page URL in the response if the validation success. If failure then, Error code and description will be provided.

Success:

```
[{
  "status": "1",
  "result": "700212030953264091: https://securepayments.alrajhibank.com.sa/pg/URPaypage.htm", //Payment ID:Payment URL
  "error": null,
  "errorText": null
}]
```

```

Failure:

[ {

  "status": "2",

  "error": " IPAY0100124",

  "errorText": " Problem occurred while validating transaction data",

  "result": null

}]

```

Framing Payment URL :

Merchant needs to frame the URL like the below sample

<https://securepayments.alrajhibank.com.sa/pg/URPaypage.htm?paymentId=700112030953264091>

Final Response and Transaction Status

The ARB payment gateway verifies the transaction and returns the response to the same request.

Attribute - Final response from ARB PG to Merchant

S. No	Fields	M/C/O	Field Type	Description
1	trandata	M	AlphaNum	All the below response parameters will be provided in trandata field
2	error	C	Alphanum	If any error during processing, PG will provide the error code
3	errorText	C	Alphanum	If any error during processing, PG will provide the error
4	status	M	Alphanum	If transaction success 1. If transaction failure 2.

FAQ's on Integration Process

Q1. What are the pre requisities for integration process?

- Ans. 1. Tranportal ID
2. Tranportal Password
3. Resource Key
4. Payment gateway endpoint

Q2. Where to get tranportal ID , Password , Resource key and end point Url's?

Ans. Tranportal Id , Password , Resource Key and end point URL's will be shared to the merchant via mail to their registered E-mail Id.

Q3. What is resource key?

Ans. Resource key is unique for a terminal . It is required for encryption of request parameters and decryption of response paramters while connecting to ARB Payment Gateway. Bank user will share the resource key in a secured manner.

Q4. What is inquiry transaction?

Ans. ARB Payment Gateway allows merchants to do an inquiry of already completed transaction by passing certain details of the payment message, ARB Payment Gateway provides response to this request with appropriate fields in the response; merchant is expected to verify the relevant fields like Transaction amount, transaction status and other transaction fields.

Q5. What are the action codes for inquiry ,refunds and Void transactions?

Ans. action codes :

Action Codes

Action	Action code
Purchase	1
Authorization	4
Refund	2
Inquiry	8
Void	3
Capture	5
Auth Extension	14

Void Auth

9

Q6. Based on what parameters can transactions be inquired?

Ans. Transactions can be inquired based on transaction ID , Payment ID and Track ID of original transaction.

Q7.How to verify transactions and settlement ?

Ans. After the integration testing, transactions can be verified on merchant Portal To verify if a transaction has been settled or not merchant can always refer the transaction detail report in merchant Portal . Navigation to transaction detail report in merchant portal is as below :

Merchant Portal ->Reports->Transaction Reports->Transaction Detail Report

Q8 . What are the various result codes and their description.

Ans.

Result	Description
CAPTURED	Captured result will be considered as transaction success
NOT CAPTURED	This will be considered as transaction failure
APPROVED	This will be considered as transaction success for Authorization.
NOT APPROVED	This will be considered as transaction failure for Authorization.
VOIDED	Success for Void transaction
DENIED BY RISK	If the Risk validation failed, then PG will decline the transaction with this result
HOST TIMEOUT	If there is no response from respective interchange during authorization, then PG will provide the Host timeout result.

Sample Encryption and Decryption Code for JAVA

```
public static String encryptAES(String key,String encryptString) throws
Exception{

    String AES_IV = "PGKEYENCDECIVSPC";

    Byte [] encryptedText=null;

    IvParameterSpec ivspec=null;

    SecretKeySpec skeySpec=null;

    Cipher cipher=null;

    Byte [] text=null;

    String s=null;

    try {

        ivspec = new IvParameterSpec(AES_IV.getBytes("UTF-8"));

        skeySpec = new SecretKeySpec(key.getBytes("UTF-8"), "AES");

        cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");

        cipher.init(Cipher.ENCRYPT_MODE, skeySpec,ivspec);

        text = encryptString.getBytes("UTF-8");

        encryptedText = cipher.doFinal(text);

        s = byteArrayToHexString(encryptedText);

    } catch (Exception e) {

        e.printStackTrace();

    }

    finally

    {

        encryptedText=null;

        ivspec=null;
```



```
        sKeySpec=null;

        cipher=null;

        text=null;

    }

    return s.toUpperCase();

}
```

📌 **Note:** Before encrypting encryptString value. merchant needs to encode the value with URL Encoder.

```
public static String decryptAES(String key,String encryptedString) throws
Exception{

    String AES_IV = "PGKEYENCDECIVSPC";

    SecretKeySpec sKeySpec=null;

    IvParameterSpec ivspec=null;

    Cipher cipher =null;

    Byte [] textDecrypted=null;

    Try {

        Byte [] b = hexStringToByteArray(encryptedString);

        sKeySpec = new SecretKeySpec(key.getBytes("UTF-8"),"AES");

        ivspec = new IvParameterSpec(AES_IV.getBytes("UTF-8"));

        cipher = Cipher.getInstance("AES/CBC/PKCS5Padding");

        cipher.init(Cipher.DECRYPT_MODE, sKeySpec,ivspec);

        textDecrypted = cipher.doFinal(b);

    } catch (Exception e) {

        e.printStackTrace();

    }

    finally

    {
```

```
        sKeySpec=null;

        ivSpec=null;

        cipher =null;

    }

    return(new String(textDecrypted));

}
```

✍ **Note:** After decrypting encryptedString value. Merchant needs to decode the textDecrypted value with URL Decoder.

Sample Encryption and Decryption Code For JAVASCRIPT

```
function aesEncrypt(trandata,key)
{
    var iv = "PGKEYENCDECIVSPC";

    var rkEncryptionIv = aesjs.utils.utf8.toBytes(iv);

    var enckey= aesjs.utils.utf8.toBytes(key);

    var aesCtr = new aesjs.ModeOfOperation.cbc(enckey, rkEncryptionIv);

    var textBytes = aesjs.utils.utf8.toBytes(trandata);

    var encryptedBytes = aesCtr.encrypt(aesjs.padding.pkcs7.pad(textBytes));

    var encryptedHex = aesjs.utils.hex.fromBytes(encryptedBytes);

    return encryptedHex;
}
```

Note: Before encrypting trandata value. merchant needs to encode the value with URL Encoder.

```
function AESdecryption(encryptedHex,key)
{
    var iv = "PGKEYENCDECIVSPC";

    var enckey= aesjs.utils.utf8.toBytes(key);

    var rkEncryptionIv = aesjs.utils.utf8.toBytes(iv);

    var encryptedBytes = aesjs.utils.hex.toBytes(encryptedHex);

    var aesCbc = new aesjs.ModeOfOperation.cbc(enckey, rkEncryptionIv);

    var decryptedBytes = aesCbc.decrypt(encryptedBytes);

    var decryptedText = aesjs.utils.utf8.fromBytes(decryptedBytes);
}
```

```
    return decryptedText;  
}
```

- ☑ **Note:** After decrypting encryptedHex value. Merchant needs to decode the decryptedText value with URL Decoder.

Sample Encryption and Decryption Code For PHP

Encryption:

```
function encryptAES($str,$key)
{
    $str = $this->pkcs5_pad($str);
    $ivlen = openssl_cipher_iv_length($cipher="aes-256-cbc");
    $iv="PGKEYENCDECIVSPC";
    $encrypted = openssl_encrypt($str, "aes-256-cbc",$key, OPENSSL_ZERO_PADDING, $iv);
    $encrypted = base64_decode($encrypted);
    $encrypted = unpack('C*', ($encrypted));
    $encrypted=$this->byteArray2Hex($encrypted);
    $encrypted = urlencode($encrypted);
    return $encrypted;
}
```

☑ **Note:** Before encrypting transaction data, data needs to be encoded using URL-Encoder

Decryption:

```
function decryptAES ($codcode, $)
{
    $code = $this->hex2ByteArray(trim($code));
    $code=$this->byteArray2String($code);
    $iv = "PGKEYENCDECIVSPC";
```



```
$code = base64_encode($code);  
  
$decrypted = openssl_decrypt($code, 'AES-256-CBC', $key, OPENSSL_ZERO_PADDING,  
$iv);  
  
return $this->pkcs5_unpad($decrypted);  
}
```

📌 **Note:** After decrypting transaction data needs to be decoded using URL-Decoder

Chapter 4 TROUBLESHOOTING

Known Error Codes

The error codes are listed below:

The following table contains the known error codes and their descriptions:

Error Code	Error Code Description
IPAY0100114	Duplicate Record, transaction ID already exist
IPAY0100115	Transaction denied due to missing original transaction id.
IPAY0100116	Transaction denied due to invalid original transaction id.
IPAY0100118	Transaction denied due to card number length error
IPAY0100119	Transaction denied due to invalid card number
IPAY0100120	Transaction denied due to invalid payment instrument
IPAY0100121	Transaction denied due to invalid card holder name.
IPAY0100122	Transaction denied due to invalid address.
IPAY0100123	Transaction denied due to invalid postal code.
IPAY0100124	Problem occurred while validating transaction data
IPAY0100125	Payment instrument not enabled.
IPAY0100126	Brand not enabled.
IPAY0100127	Problem occurred while doing validate original transaction
IPAY0100128	Transaction denied due to Institution ID mismatch
IPAY0100129	Transaction denied due to Merchant ID mismatch
IPAY0100130	Transaction denied due to Terminal ID mismatch
IPAY0100131	Transaction denied due to Payment Instrument mismatch
IPAY0100132	Transaction denied due to Currency Code mismatch
IPAY0100133	Transaction denied due to Card Number mismatch
IPAY0100134	Transaction denied due to invalid Result Code
IPAY0100135	Problem occurred while doing perform action code reference id (Validate Original Transaction)
IPAY0200028	Problem occurred while loading default institution configuration (Validate Original Transaction)

Error Code	Error Code Description
IPAY0100136	Transaction denied due to previous capture check failure (Validate Original Transaction)
IPAY0100138	Transaction denied due to capture amount versus auth amount check failure (Validate Original)
IPAY0100139	Transaction denied due to void amount versus original amount check failure (Validate Original)
IPAY0100140	Transaction denied due to previous void check failure (Validate Original Transaction)
IPAY0100141	Transaction denied due to authorization already captured (Validate Original Transaction)
IPAY0100142	Problem occurred while validating original transaction
IPAY0200030	No external connection details for Extr Conn id :
IPAY0200031	Alternate external connection details not found for the alt Extr Conn id :
IPAY0100143	Transaction action is null
IPAY0100144	ISO MSG is null. See log for more details!
IPAY0100145	Problem occurred while loading default messages in ISO Formatter
IPAY0100147	Problem occurred while formatting purchase request in B24 ISO Message Formatter
IPAY0100150	Problem occurred while formatting Reverse purchase request in B24 ISO Message Formatter
IPAY0100152	Problem occurred while formatting authorization request in B24 ISO Message Formatter
IPAY0100153	Problem occurred while formatting Capture request in B24 ISO Message Formatter
IPAY0100155	Problem occurred while formatting reverse authorization request in B24 ISO Message Formatter
IPAY0100156	Problem occurred while formatting Reverse Capture request in B24 ISO Message Formatter
IPAY0100159	External message system error
IPAY0100160	Unable to process the transaction.
IPAY0100163	Problem occurred during transaction.
IPAY0100166	Transaction Not Processed due to Empty Authentication Status
IPAY0100167	Transaction Not Processed due to Invalid Authentication Status
IPAY0100168	Transaction Not Processed due to Empty Enrollment Status
IPAY0100169	Transaction Not Processed due to Invalid Enrollment Status
IPAY0100170	Transaction Not Processed due to invalid CAVV

Error Code	Error Code Description
IPAY0100171	Transaction Not Processed due to Empty CAVV
IPAY0100172	Problem occurred while converting amount.
IPAY0100173	Problem occurred while building refund request.
IPAY0100175	Problem occurred in refund process.
IPAY0100039	Invalid payment id .
IPAY0200009	Problem occurred while getting payment details.
IPAY0100041	Payment details missing.
IPAY0100042	Transaction time limit exceeds.
IPAY0200011	Problem occurred while getting IP block details.
IPAY0100043	IP address is blocked already
IPAY0100044	Problem occurred while loading payment page.
IPAY0100045	Denied by Risk
IPAY0200013	Problem occurred while updating description details in payment log.
IPAY0100047	Payment Page validation failed due to invalid Order Status:
IPAY0200015	Problem occurred while getting terminal details.
IPAY0100050	Invalid terminal key.
IPAY0100051	Missing terminal key.
IPAY0100053	Problem occurred while processing direct debit.
IPAY0100054	Payment details not available
IPAY0100056	Instrument not allowed in Terminal and Brand
IPAY0200016	Problem occurred while getting payment instrument.
IPAY0200018	Problem occurred while getting transaction details
IPAY0100057	Transaction denied due to invalid processing option action code
IPAY0100058	Transaction denied due to invalid instrument
IPAY0100059	Transaction denied due to invalid currency code.
IPAY0100060	Transaction denied due to missing amount.
IPAY0100061	Transaction denied due to invalid amount.
IPAY0100062	Transaction denied due to invalid Amount/Currency.
IPAY0100063	Transaction denied due to invalid track ID
IPAY0100064	Transaction denied due to invalid UDF1:
IPAY0100065	Transaction denied due to invalid UDF2:
IPAY0100066	Transaction denied due to invalid UDF3:
IPAY0100067	Transaction denied due to invalid UDF4:

Error Code	Error Code Description
IPAY0100068	Transaction denied due to invalid UDF5:
IPAY0100069	Missing payment instrument.
IPAY0100070	Transaction denied due to failed card check digit calculation.
IPAY0100071	Transaction denied due to missing CVD2.
IPAY0100072	Transaction denied due to invalid CVD2.
IPAY0100073	Transaction denied due to invalid CVV.
IPAY0100074	Transaction denied due to missing expiry year.
IPAY0100075	Transaction denied due to invalid expiry year.
IPAY0100076	Transaction denied due to missing expiry month.
IPAY0100077	Transaction denied due to invalid expiry month.
IPAY0100078	Transaction denied due to missing expiry day.
IPAY0100079	Transaction denied due to invalid expiry day.
IPAY0100080	Transaction denied due to expiration date.
IPAY0100081	Card holder name is not present
IPAY0100082	Card address is not present
IPAY0100083	Card postal code is not present
IPAY0100086	Transaction denied due to missing CVV.
IPAY0100095	Terminal inactive.
IPAY0100098	Terminal Action not enabled for Transaction request, Terminal "termid" ,Tran Action : "action"
IPAY0100099	Terminal Payment Instrument not enabled for Transaction request, Terminal "termid" , Tran Instrument:"PAYMENT_INSTRUMENT"
IPAY0100100	Problem occurred while authorize
IPAY0200019	Problem occurred while getting risk profile details
IPAY0100102	Denied by risk : Maximum Floor Limit Check - Fail
IPAY0100103	Transaction denied due to Risk : Maximum transaction count
IPAY0100104	Transaction denied due to Risk : Maximum processing amount
IPAY0200022	Problem occurred while getting currency.
IPAY0100106	Invalid payment instrument
IPAY0200024	Problem occurred while getting brand rules details.
IPAY0100107	Instrument not enabled.
IPAY0200025	Problem occurred while getting terminal details.
IPAY0100109	Invalid subsequent transaction, payment id is null or empty.

Error Code	Error Code Description
IPAY0200026	Problem occurred while getting transaction log details.
IPAY0200027	Missing encrypted card number.
IPAY0100111	Card decryption failed.
IPAY0100113	"transaction id" is a subsequent transaction, but original transaction id is invalid :
IPAY0100021	Missing currency.
IPAY0100024	Invalid amount.
IPAY0100027	Invalid track id.
IPAY0100030	Invalid user defined field3.
IPAY0200007	Problem occurred while validating payment details
IPAY0200008	Problem occurred while verifying payment details.
IPAY0100037	Payment id missing.
IPAY0100040	Transaction in progress in another tab/window.
IPAY0100052	Problem occurred during merchant response encryption.
IPAY0100055	Invalid Payment Status
IPAY0200017	Problem occurred while getting payment instrument list
IPAY0100094	Sorry, this instrument is not handled
IPAY0100101	Denied by risk : Risk Profile does not exist
IPAY0200020	Problem occurred while performing transaction risk check
IPAY0200021	Problem occurred while performing risk check
IPAY0200023	Problem occurred while determining payment instrument
IPAY0100108	Perform risk check : Failed
IPAY0100110	Invalid subsequent transaction, Tran Ref id is null or empty.
IPAY0100112	Problem occurred in method loading original transaction data(card number, exp month / year) for orig_tran_id
IPAY0100117	Transaction denied due to missing card number.
IPAY0100137	Transaction denied due to refund amount greater than auth amount check failure (Validate Original Transaction)
IPAY0200029	Problem occurred while getting external connection details.
IPAY0200032	Problem occurred while getting external connection details for Extr CConn id :
IPAY0100151	Problem occurred while formatting Refund request in B24 ISO Message Formatter
IPAY0100154	Problem occurred while formatting Reverse Refund request in B24 ISO Message Formatter

Error Code	Error Code Description
IPAY0100158	Host (SWITCH) timeout
IPAY0100161	Merchant is not allowed for encryption process.
IPAY0100176	Decrypting transaction data failed.
IPAY0100177	Invalid input data received.
IPAY0100178	Merchant encryption enabled.
IPAY0100179	IVR not enabled.
IPAY0100180	Authentication not available.
IPAY0100181	Card encryption failed.
IPAY0200037	Error occurred while getting Merchant ID
IPAY0100186	Encryption enabled.
IPAY0100189	Transaction denied due to brand directory unavailable
IPAY0100190	Transaction denied due to Risk : Maximum transaction count
IPAY0100191	Denied by risk : Negative Card check - Fail
IPAY0100192	Transaction Not Processed due to Empty XID
IPAY0100193	Transaction Not Processed due to invalid XID
IPAY0100202	Error occurred in Determine Payment Instrument
IPAY0100194	Transaction denied due to Risk : Minimum Transaction Amount processing
IPAY0100195	Transaction denied due to Risk : Maximum refund processing amount
IPAY0100196	Transaction denied due to Risk : Maximum processing amount
IPAY0100197	Transaction denied due to Risk : Maximum debit amount
IPAY0100198	Transaction denied due to Risk : Transaction count limit exceeded for the IP
IPAY0100199	Transaction denied due to previous refund check failure (Validate Original Transaction)
IPAY0100200	Denied by risk : Negative BIN check - Fail
IPAY0100201	Denied by risk : Declined Card check – Fail
IPAY0100203	Problem occurred while doing perform transaction
IPAY0100204	Missing payment details
IPAY0100206	Problem occurred while getting currency minor digits
IPAY0100207	Bin range not enabled
IPAY0100208	Action not enabled
IPAY0100209	Institution config not enabled
IPAY0100213	Problem occurred while processing the hosted transaction request

Error Code	Error Code Description
IPAY0100214	Problem occurred while verifying tranportal id
IPAY0100215	Invalid tranportal id
IPAY0100216	Invalid data received
IPAY0100217	Invalid payment detail
IPAY0100218	Invalid brand id
IPAY0100219	Missing card number
IPAY0100220	Invalid card number
IPAY0100221	Missing card holder name
IPAY0100222	Invalid card holder name
IPAY0100223	Missing cvv
IPAY0100224	Invalid cvv
IPAY0100225	Missing card expiry year
IPAY0100226	Invalid card expiry year
IPAY0100227	Missing card expiry month
IPAY0100228	Invalid card expiry month
IPAY0100229	Invalid card expiry day
IPAY0100230	Card expired
IPAY0100231	Invalid user defined field
IPAY0100232	Missing original transaction id
IPAY0100233	Invalid original transaction id
IPAY0100234	Problem occurred while formatting Reverse completion request in ISO Message Formatter
IPAY0100235	Problem occurred while formatting Reverse refund request in ISO Message Formatter
IPAY0100236	Problem occurred while formatting Reverse refund request in ISO Message Formatter
IPAY0100237	Problem occurred while formatting Reverse purchase request in ISO Message Formatter
IPAY0100238	Problem occurred while formatting Capture request in ISO Message Formatter
IPAY0100239	Problem occurred while formatting authorization request in ISO Message Formatter
IPAY0100240	Problem occurred while formatting refund request in ISO Message Formatter
IPAY0100241	Problem occurred while formatting purchase request in ISO Message Formatter

Error Code	Error Code Description
IPAY0100243	NOT SUPPORTED
IPAY0100244	Payment Instrument Not Configured
IPAY0100245	Problem occurred while sending/receiving ISO message
IPAY0100246	Problem occurred while doing perform ip risk check
IPAY0100249	Merchant response url is down
IPAY0100250	Payment details verification failed
IPAY0100251	Invalid payment data
IPAY0100253	Problem occurred while cancelling the transaction
IPAY0100254	Merchant not enabled
IPAY0100255	External connection not enabled
IPAY0100256	Payment encryption failed
IPAY0100257	Brand rules not enabled
IPAY0100260	Payment option(s) not enabled
IPAY0100261	Payment hashing failed
IPAY0100262	Problem occurred during VEREQ process
IPAY0100263	Transaction details not available
IPAY0100264	Signature validation failed
IPAY0100265	PARes status not successful
IPAY0100266	Brand directory unavailable
IPAY0100267	PARes status not successful
IPAY0100268	3d secure not enabled for the brand
IPAY0100269	Invalid card check digit
IPAY0100270	PARes status not successful
IPAY0100271	Problem occurred while formatting purchase request in MASTER ISO Message Formatter
IPAY0100272	Problem occurred while validating xml message format
IPAY0100273	Problem occurred while validation VERES message format
IPAY0100274	VERES message format is invalid
IPAY0100275	Problem occurred while formatting Credit request in MASTER ISO Message Formatter
IPAY0100276	Problem occurred while formatting Reverse purchase request in MASTER ISO Message Formatter
IPAY0100277	Problem occurred while formatting Reverse Credit request in MASTER ISO Message Formatter

Error Code	Error Code Description
IPAY0100278	Problem occurred while formatting reverse authorization request in MASTER ISO Message Formatter
IPAY0100279	Problem occurred while formatting Reverse Capture request in MASTER ISO Message Formatter
IPAY0100280	Problem occurred while formatting Capture request in MASTER ISO Message Formatter
IPAY0100281	Transaction Denied due to missing Master Brand
IPAY0100282	Transaction Denied due to missing Visa Brand
IPAY0100283	Problem occurred in determine payment instrument
IPAY0100284	Invalid subsequent transaction, track id is null or empty
IPAY0100285	Transaction denied due to invalid original transaction
IPAY0100289	Transaction denied due to Risk : Maximum credit amount
IPAY0100291	Problem occurred while validating original transaction
IPAY0100292	Transaction denied due to invalid PIN
IPAY0100293	Transaction denied due to duplicate Merchant trackid
IPAY0100294	Transaction denied due to missing Merchant trackid
IPAY0100295	Missing Merchant Track Id
IPAY0100327	Invalid Buyer Email ID
IPAY0100328	Invalid Buyer Mobile No.
IPAY0100329	Invalid Buyer Name
IPAY0100330	Invalid Minor digits length
IPAY0100331	Invalid Expiry Date
IPAY0100332	Invalid Invoice Id
IPAY0100333	Invalid Item Description
IPAY0100334	Invalid Udf1

Handling Final Response from Payment Gateway

Handling Transaction Response:

- By decrypting **"trandata"** Merchant can able to get the JSON Response. JSON response contains all the required success transaction data.

Handling Failure Transaction Response:

Step1:

From payment Gateway Response, Merchant needs to check the **"trandata"** first.

If the **"trandata"** not null.

By decrypting the trandata Merchant can able to get the failure transaction description by extracting the key **"result"**.

If **"result"** is null then merchant needs to use the key **"errorText"** inside **"trandata"** to get failure description.

Step 2:

If **"trandata"** is null or empty, then merchant needs to do follow below steps.

Step 3:

To get Error Description: Use Parameter Key name **"ErrorText"**.

Step 4:

To get Error Code: Use parameter key name **"Error"**.

Step 5:

To get Transaction Id: Use parameter key **"tranid"** this value can be null or empty.

Step 6:

To get Payment Id: Use parameter key **"paymentid"** this value can be null or empty.

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